
SPATIALAB: CAN VISION–LANGUAGE MODELS PERFORM SPATIAL REASONING IN THE WILD?

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ABSTRACT

Spatial reasoning is a foundational faculty of human cognition, and it remains the single most persistent obstacle confronting contemporary vision–language models (VLMs). Prevailing evaluations rest on synthetic or LLM-generated environments assembled from narrow task designs and puzzle-like setups, and they therefore fail to capture the real-world complexity, visual noise, and heterogeneous spatial relationships that VLMs must resolve in practice. We close this gap with **SPATIALAB**, a comprehensive benchmark that evaluates the spatial reasoning of VLMs directly in realistic, unconstrained visual contexts. SPATIALAB comprises 1,400 visual question–answer pairs spanning six principal categories, *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each decomposed into five subcategories to yield 30 distinct task types. With at least 25 questions per subcategory and at least 200 per principal category, the benchmark delivers balanced coverage and, through paired multiple-choice and open-ended protocols, isolates discriminative accuracy from generative spatial reasoning. Evaluating a broad spectrum of state-of-the-art VLMs, encompassing open- and closed-source systems, reasoning-focused models, and specialized spatial reasoning models, we establish a substantial and consistent gap between machine and human spatial competence that persists across model families, scales, and evaluation formats. In the multiple-choice setting, the strongest model, InternVL3.5-72B, reaches 54.93% accuracy against 87.57% for humans; in the open-ended setting, every model degrades by a further 10–25%, with GPT-5-mini leading at 40.93% against 64.93% for humans. These results localize the decisive limitations of current systems to complex spatial relationships, depth perception, navigation, and 3D geometry. By furnishing a diverse, real-world evaluation framework, SPATIALAB exposes at once the critical challenges and the concrete opportunities for advancing spatial reasoning in VLMs, and it supplies a rigorous instrument for steering future research toward robust, human-aligned spatial understanding. SPATIALAB will be released upon acceptance.

1 INTRODUCTION

Spatial reasoning is a foundational faculty of human cognition, subsuming the ability to apprehend spatial layouts, to imagine and manipulate objects, and to navigate environments. It underwrites an unusually broad application space, spanning robotics, autonomous driving, and augmented/virtual reality through to geospatial analytics, computer graphics, human–computer interaction, and education (Huang et al., 2024b; Qi et al., 2024; 2025). For artificial agents, autonomous systems that must perceive and act within dynamic environments, reliable spatial competence hinges on jointly interpreting geometric relations, depth cues, occlusion patterns, and agent-centric perspectives in scenes that are at once visually noisy and structurally complex (Bar-Anan et al., 2006; Cai et al., 2025; Trope & Liberman, 2010; Ramalho et al., 2018). Recent vision–language models have markedly advanced multimodal representation and language grounding; their spatial judgments, however, remain conspicuously fragile the moment realistic environments displace curated ones (Chen et al., 2025; Kosoy et al., 2025; Pothiraj et al., 2025).

Existing spatial benchmarks, however, are at once structurally narrow and perceptually simplified. The preponderance of them target elementary tasks, binary spatial relations, low-resolution depth categories, or schematic navigation cues, and they situate these tasks in synthetic or simplified settings

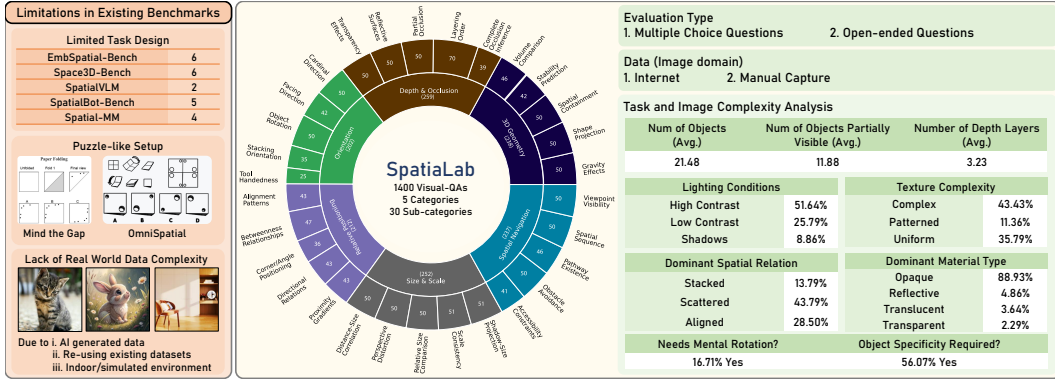


Figure 1: **Overview of SPATIALAB.** The benchmark redresses the limitations of prior datasets (left), contributes 1,400 visual QA pairs organized across six principal categories and 30 subcategories (center), and enables systematic evaluation through paired multiple-choice and open-ended tasks. Its imagery spans a wide range of task and scene complexity, with controlled variation in object counts, depth layers, lighting, textures, spatial relations, and materials (right).

that suppress visual clutter and homogenize object appearance (Du et al., 2024; Jia et al., 2025; Szymanska et al., 2024; Shiri et al., 2024; Kamath et al., 2023; Fu et al., 2024; Song et al., 2024; Yang et al., 2024; Qi et al., 2025; Cai et al., 2024). Such controlled conditions attenuate the perceptual and reasoning load, manufacturing an appearance of saturation that masks systematic failures under distribution shift. The very challenges most diagnostic of real-world competence, robust occlusion inference, scale-consistent judgments across viewpoints, and planning under partial observability, remain markedly under-sampled (Kosoy et al., 2025; Pothiraj et al., 2025; Chen et al., 2025; Cai et al., 2025). The consequence is direct and consequential: models that appear strong on existing benchmarks collapse on integrated, multi-step inference, and this discrepancy directly constrains safe deployment in real-world environments where spatial errors carry tangible consequences.

A meaningful evaluation must therefore span all core axes of spatial reasoning jointly: relative position, depth and occlusion, orientation, size and scale, navigation, and 3D geometry. Where prevailing task-isolated setups fracture these dimensions, human cognition integrates them seamlessly under noise and ambiguity (Driess et al., 2023; Brohan et al., 2023; Zitkovich et al., 2023; Collaboration et al., 2023; Yuan et al., 2024; Qi et al., 2025). A rigorous benchmark should accordingly probe selection and generation alike, calibrate model performance against human baselines, and draw its imagery from real-world scenes that surface precisely the failures simulator datasets conceal, principles that directly govern the design of SPATIALAB.

To meet these needs, we introduce **SPATIALAB**, a benchmark that assesses the spatial reasoning of vision–language models directly in realistic, unconstrained visual contexts, as outlined in Figure 1. **SPATIALAB** contributes a balanced suite of visual question–answer items that spans a broad taxonomy of spatial tasks and supports paired multiple-choice and open-ended evaluation. By construction, the benchmark foregrounds photographic diversity, complex relational queries, and tasks that demand the joint integration of depth, perspective, and 3D structure. We evaluate an extensive collection of state-of-the-art VLMs, spanning open- and closed-source models and reasoning-focused systems, benchmark them against human performance, and conduct in-depth error analysis. Our contributions are as follows:

1. **A large-scale, in-the-wild benchmark.** We introduce **SPATIALAB**, a benchmark of 1,400 visual question–answer pairs spanning six principal categories and thirty task types in realistic, unconstrained contexts. With at least 200 questions per category and 25 per subcategory, it guarantees balanced coverage and, through paired multiple-choice (SPATIALAB-MCQ) and open-ended (SPATIALAB-OPEN) protocols, cleanly isolates discriminative accuracy from generative reasoning.
2. **A comprehensive cross-family evaluation.** We evaluate 25+ state-of-the-art vision–language models, open- and closed-source, reasoning-oriented, and spatially specialized, against human baselines, and establish a consistent and substantial spatial reasoning

Table 1: Comparison of spatial reasoning benchmarks. SPATIALAB is the only benchmark to combine the broadest task taxonomy (30 categories), high-complexity in-the-wild imagery, manual annotation, and paired multiple-choice and open-ended evaluation; the correspondingly low best-baseline performance it induces exposes the headroom that prior benchmarks obscure.

Dataset	E.	T.C.	Data				QAs			Venue/ Year	Eval			A.L.	Md.	Best Baseline	B.B.P.
			Domain	Source	A.	Size	Count	Df.	Cmpl.		Type	Depth					
ScanQA	Y	7	Indoor	E.D.	M	800	41K	H	E	CVPR'22C	Close-ended	L	L	I		-	-
Visual Spatial	Y	7	R.W.	E.D.	T	10K	10K	E	E	TACL'23	MCQ,Bin.	L	M	I		LXMERT	70.10
What's up	N	6	Indoor	E.D.	T	5K	5K	E	E	EMNLP'23C	MCQ	S	Hi	I		XVLM-COCO	60.40
EmbSpatial-Bench	Y	6	Indoor	E.D.	T	2.2K	3.6K	H	E	ACL'24C	MCQ,%	M	S	I		Qwen-VL-Max	49.11
Space3D-Bench	N	6	Indoor	E.D.	M	211	1K	Mm	E	ECCV'24W	Mix	L	S	I		RAG3D-Chat	66.80
SpatialRGPT-Bench	N	12	Mix	E.D.	T	1.5K	1.5K	Mm	E	NeurIPS'24C	MCQ,Bin.	M	L	I		GPT-4o	57.83
BLINK-Spatial	Y	14	Mix	E.D.	M	7.3K	3.8K	H	E	ECCV'24C	MCQ,Bin.	M	M	I		GPT-4o	59.03
Spatial-MM	N	4	Internet	Internet	T	2.3K	2.3K	E	Mm	EMNLP'24C	Open	L	Hi	I		GPT-4 Vision	63.82
RoboSpatial	Y	4	Indoor	E.D.	T	1M	3M	H	E	CVPR'25C	Open,Bin.	L	S	I		LEO	71.90
SpatialVLM	N	2	Web	E.D.	T	10M	2B	H	E	CVPR'24C	Open	L	S	I		GPT-4V	68.00
VSL-Bench	Y	8	Indoor	M.E.D.	T	288	5K	Mm	Mm	CVPR'25C	MCQ,Nm.	M	M	V		Gemini-1.5 Pro	45.40
OmniSpatial	Y	50	Internet	Internet	M.P	1387	1.5K	H	Mm	2025	MCQ,Bin.	M	S	I		o3-2025-04-16	56.33
Mind the Gap	N	6	Mix	AI-Gen	M.P	1.8K	1.8K	E	Mm	2025	Mix	M	S	I		InternVL2.5-26B	48.83
VLM4D	N	4	Mix	Mix	M	1K	1.8K	E	H	ICCV'25	MCQ	Hi	M	Mix		Gemini-2.5 Pro	62.00
SPATIALAB-MCQ	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	H	2025	MCQ	Hi	H	Mix		InternVL3.5-72B	54.93
SPATIALAB-OPEN	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	VH	2025	Open	Hi	H	Mix		GPT-5-mini	40.93

gap that persists undiminished across model families, parameter scales, and evaluation formats.

3. **In-depth error analyses and diagnostic studies.** We localize the systematic failure modes of current systems in depth perception, navigation, occlusion, and 3D geometry, consolidating scattered observations into concrete insights and actionable directions for advancing robust, human-aligned spatial understanding.
4. **A systematic study of improvement strategies.** We investigate multiple strategies for strengthening spatial reasoning in VLMs, including **supervised fine-tuning (SFT)**, **chain-of-thought (CoT) prompting**, **CoT with self-reflection**, and **multi-agent architectures (SPATIOXOLVER) with multi-step reasoning**, and delineate precisely where each advances competence and where the benchmark's difficulty proves resilient across multiple-choice and open-ended tasks.

In the multiple-choice setting, the leading models reach roughly 50–55% accuracy while human annotators exceed 85%; in open-ended generation, model performance falls by a further 10–25 percentage points, the strongest systems settling near 40% against about 65% for humans. Error analysis concentrates these failures in occlusion inference, scale consistency, and multi-step navigational planning, pinpointing the limits of current compositional spatial representations. Beyond this diagnosis, our study establishes which interventions confer robust gains, which merely amplify existing biases, and how decisively perceptual grounding governs generalizable spatial reasoning. By supplying tasks, baseline results, and diagnostic analyses within a single framework, SPATIALAB lays a durable foundation for progress and for reproducible comparison in spatial reasoning evaluation.

2 PRELIMINARIES: SPATIAL REASONING IN VLMs

Visual-spatial reasoning, the capacity to perceive, represent, and manipulate the spatial relations among objects, occupies a central position in human cognition and is a prerequisite for embodied agents that navigate and act in the world (Palmer, 1999; Chabris et al., 2006). Formally, we cast a scene as a set of objects $O = \{o_1, \dots, o_n\}$, each endowed with attributes such as position, orientation, and size and situated within a world coordinate system W . Visual-spatial reasoning then reduces to estimating a mapping $f : (I, Q) \mapsto R$, in which I denotes an image (or image sequence), Q a linguistic query, and $R = \{r(o_i, o_j) \mid o_i, o_j \in O\}$ a structured representation of spatial relations encoding predicates such as *left-of*, *occludes*, or *supports*. For vision-language models (VLMs), estimating f is acutely challenging, since it demands the reconciliation of noisy visual perception with compositional linguistic abstraction under embodied constraints, among them perspective, gravity, and physical interaction (Dai et al., 2017; Alayrac et al., 2022).

Early approaches to spatial reasoning in VLMs relied predominantly on elementary spatial relations or narrow task designs (Cheng et al., 2024; Du et al., 2024; Szymanska et al., 2024), typically situated in synthetic datasets such as CLEVR (Johnson et al., 2016), structured question-answering benchmarks such as GQA (Hudson & Manning, 2019), or puzzle-like simulated setups such as MGT (Stogiannidis et al., 2025) and OmniSpatial (Stogiannidis et al., 2025), as summarized in Table

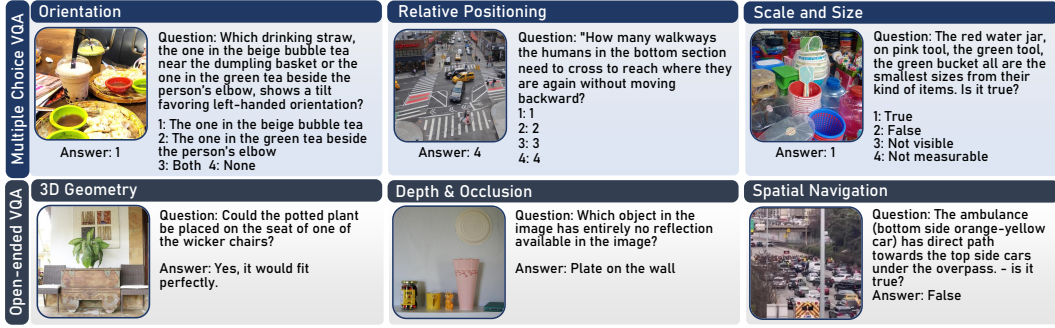


Figure 2: Representative examples drawn from the six principal categories, illustrating the paired open-ended and multiple-choice task formats and the visual complexity characteristic of SPATIALAB.

¹. Serviceable for controlled evaluation, these settings nonetheless oversimplify spatial structure, strip out environmental noise, and neglect dynamic factors such as temporal change and viewpoint shift. Overcoming these limitations requires that spatial reasoning be benchmarked in rich, real-world contexts that preserve scene complexity, temporal variation, and multiple perspectives; only such settings confront VLMs with the challenges intrinsic to embodied and interactive applications, robotics, AR/VR, and autonomous navigation foremost among them (Mu et al., 2023; Collaboration et al., 2023).

Accordingly, SPATIALAB organizes spatial reasoning into six principal categories, *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each aligned with a distinct cognitive faculty documented in psychology and spatial cognition (Baddeley, 1998; Previc, 1998). As shown in Figure 1 and Appendix B, every category is decomposed into five subcategories, yielding a taxonomy of thirty task types that isolate fine-grained reasoning skills such as directional alignment, occlusion inference, and stability prediction. This classification is engineered to span the full spectrum of visual-spatial reasoning challenges. By decomposing complex reasoning into measurable, individually testable units, the taxonomy allows us to assess not merely whether a model “solves” spatial tasks in aggregate, but precisely which dimensions remain fragile or underdeveloped (Gardner, 2011; Huang et al., 2022; Gupta & Kembhavi, 2023; Yang et al., 2023).

We further adopt two complementary evaluation formats: multiple-choice and open-ended question answering. The multiple-choice format probes recognition and fine-grained discrimination among distractors, whereas the open-ended format measures generative reasoning and compositional description (Driess et al., 2023; Liu et al., 2025; Zheng et al., 2022). Taken together, the two formats mirror the dual demands of real-world reasoning and expose complementary strengths and weaknesses that neither reveals in isolation. By uniting a principled taxonomy, diverse real-world imagery, and paired evaluation formats, SPATIALAB constitutes a structured framework for systematic benchmarking, one that enables fine-grained model comparison, calibration against human-level performance, and sharper insight into the development of robust spatial cognition in artificial agents.

3 SPATIALAB BENCHMARK

Spatial reasoning remains a persistent challenge for vision-language models (VLMs), most acutely under naturalistic conditions in which occlusion, perspective distortion, and noisy backgrounds co-occur. We meet this challenge with SPATIALAB, a benchmark that systematically evaluates the capacity of VLMs for complex spatial cognition under realistic visual conditions. The benchmark comprises 1,400 visual question-answer pairs organized into six principal categories, *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each subdivided into five subcategories to yield 30 distinct task types. Every subcategory contributes at least 25 QA pairs and each top-level category at least 200, guaranteeing balanced coverage across the taxonomy. SPATIALAB supports paired multiple-choice and open-ended evaluation,

¹ Here, E. → Embodied, R.W. → Real World, T.C. → Task Categories, E.D. → Existing Dataset, A. → Annotation, T → Template, M → Manual, P → Puzzles, Md. → Modality (I → Image, V → Video), Bin. → Bin., Nm. → Nm., Df. → Difficulty, Cmpl. → Complexity (E → Easy, Mm → Medium, H → Hard, S → Shallow, L → Low, Hi → Hi, VH/VHi → Very Hard/Very High), A.L. → Analysis Level, B.B.P. → Best Baseline Performance.

assessing discriminative accuracy alongside generative reasoning and explanatory competence; this dual protocol aligns with broader multimodal evaluation practice while foregrounding spatial-specific reasoning. High visual diversity, fine-grained annotation, and rigorous quality control jointly render **SPATIALAB** a reliable and scalable testbed, and Figure 2 presents representative examples from the six categories across both formats, conveying the richness and difficulty of real-world spatial reasoning.

3.1 BENCHMARK CONSTRUCTION

Image Collection. The image corpus was curated through three complementary acquisition strategies: (i) automated large-scale web crawling, (ii) targeted online retrieval of scene-specific images, and (iii) manual snapshots captured across diverse indoor and outdoor environments (Figure 3). This multi-source pipeline secures diversity across object categories, environmental conditions, and spatial configurations. To render natural variability explicit and controllable, every image was profiled along six meta-dimensions: *lighting condition* (high contrast, low contrast, shadows, and reflective settings), *texture complexity* (uniform, patterned, and complex), *edge complexity* (sharp and smooth boundaries), *dominant spatial relation* (stacked, scattered, aligned), *material type* (transparent, translucent, opaque, reflective), and *gravity constraints* (normal, floating, and unconstrained). Every dimension is systematically represented in the final corpus, so that the benchmark spans a broad spectrum of real-world, in-the-wild scenarios rather than simplified or synthetic setups. A quantitative complexity analysis of the corpus corroborates this richness: images contain 21.48 objects on average, of which 11.88 are partially visible, distributed across 3.23 distinct depth layers, while spatial reference chains (e.g., “object A is left of B, which is behind C”) average 2.07 links, attesting to genuine multi-step relational structure. Furthermore, 16.71% of the questions require mental rotation and 56.07% require object specificity (a particular color, pattern, or other distinguishing detail). These statistics establish decisively that **SPATIALAB** captures the cluttered, layered, and relationally rich scenes characteristic of real-world spatial reasoning and systematically underrepresented in synthetic datasets.

Annotation. Annotation proceeded in three phases. First, annotators completed targeted training (detailed in Appendix C.2) that enforced a consistent interpretation of the spatial categories and subcategories. Second, each image was paired with one or more spatial QA tasks spanning a range of reasoning types, among them proximity gradients, alignment patterns, occlusion inference, and stability prediction; questions were deliberately constructed to balance perceptual grounding against higher-order inference, such as predicting stability under gravity constraints. Third, each QA pair was encoded in both multiple-choice (four options) and open-ended formats, guaranteeing strict comparability across the two evaluation paradigms. This procedure produced approximately 1,500 spatial visual QA pairs across the six categories and 30 subcategories.

Review and Quality Control. To secure reliability, the annotation outputs passed through a three-tier review and validation pipeline. The first stage verified each QA item for semantic validity and alignment with its designated subcategory. The second stage tasked independent annotators with confirming the correctness of both the question phrasing and the designated answer, placing explicit emphasis on eliminating ambiguity and enforcing task clarity. The third stage conducted a gold-standard annotation round that established a set of high-confidence QA pairs for evaluation. All metadata was stripped from the images, and the dataset was carefully curated to avoid copyright concerns, though minor oversights may remain. This pipeline yielded a final benchmark of 1,400 rigorously validated QA items.

4 EXPERIMENTS AND EVALUATION

Models. We partition the evaluated systems into four groups that together span the current landscape of vision–language modeling. **Proprietary models** comprise GPT-4o-mini (OpenAI, 2024), GPT-5-nano, Gemini-2-Flash and Gemini-2.5-Flash (Team, 2025a), and Claude 3.5 Haiku (Anthropic, 2024). **Open-source models** span Intern-VL3 (1B, 2B, 4B) (Zhu et al., 2025), Qwen-VL2.5 (3B, 7B, 32B, 72B Instruct) (Bai et al., 2025), GLM 4.5V-3.06B and GLM-4.1V-9B-Thinking (Team et al., 2025), Gemma-3 (4B, 27B)-it (Gemma Team, Google DeepMind, 2025), Llama-3.2 (11B, 90B) Vision-Instruct (Grattafiori et al., 2024), and Step-3-321B-MoE (StepFun, 2025). **Reasoning models** include o4-mini (OpenAI, 2025), the Gemini-2-Flash and Gemini-2.5-Flash thinking variants (Team, 2025a), and Kimi-VL-A3B-Thinking-2506 (Team, 2025b). **Spatial reasoning specialists** comprise SpaceOm, SpaceThinker-Qwen2.5VL-3B, and SpaceQwen2.5-VL-

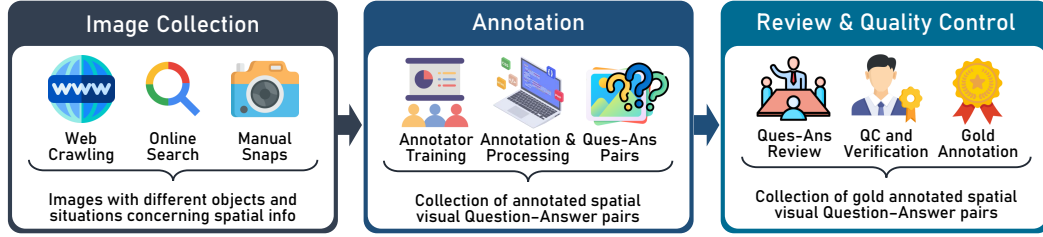


Figure 3: **Data creation pipeline for SPATIALAB.** Images are acquired through web crawling, targeted search, and manual snapshots; a structured annotation stage then derives the paired spatial question-answer items.

Table 2: Per-category multiple-choice accuracy (% , $\uparrow$ higher is better) on SPATIALAB-MCQ, reported for each spatial category alongside the overall score. Within each model group the strongest entry is rendered in **bold** and underlined. Even the best model, InternVL3.5-72B at 54.93% overall, trails the 87.57% human baseline by more than 32 points, and no model approaches human accuracy in any single category, evidencing a spatial reasoning gap that is uniform rather than localized.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Random Choice</i>	25.00	25.00	25.00	25.00	25.00	25.00	25.00
<i>Proprietary Models</i>							
GPT-4o-mini	47.06	39.00	47.03	47.17	49.60	49.79	46.50
GPT-5-mini	48.74	54.83	60.40	62.74	44.84	56.54	54.29
Gemini-2.0-Flash	47.06	55.21	53.96	58.02	54.37	46.84	52.50
Gemini-2.5-Flash	44.96	48.26	48.02	56.13	42.46	51.05	48.29
Gemini-2.5-Pro	47.48	50.19	49.50	58.49	43.65	52.32	50.07
Claude 3.5 Haiku	42.44	42.08	46.53	46.23	35.71	45.99	42.93
Mistral Medium 3.1	46.64	49.81	47.52	61.79	41.67	41.77	47.93
<i>Open-Source Models</i>							
InternVL3.5-1B	33.61	32.43	23.27	37.26	31.75	30.80	31.64
InternVL3.5-2B	34.03	31.66	31.68	40.57	32.54	32.49	33.71
Qwen-VL2.5-3B-Instruct	41.18	35.52	46.04	40.09	47.22	39.24	41.43
InternVL3.5-4B	42.86	42.86	42.08	54.72	36.51	42.19	43.29
Gemma-3-4B-it	43.70	34.36	46.53	45.75	37.30	37.97	40.57
Qwen-VL2.5-7B-Instruct	42.86	37.84	42.57	46.23	42.06	35.44	41.00
Llama-3.2-11B-Vision-Instruct	26.47	30.50	20.30	42.92	30.56	32.07	30.50
Gemma-3-27B-it	43.28	40.15	48.02	54.25	48.02	47.26	46.57
Qwen-VL2.5-32B-Instruct	41.18	40.15	46.53	45.28	45.24	41.77	43.21
InternVL3.5-72B	50.00	57.14	53.47	66.04	49.21	54.85	54.93
Qwen-VL2.5-72B-Instruct	47.06	48.65	51.98	54.25	43.65	48.95	48.86
Llama-3.2-90B-Vision-Instruct	46.22	52.12	50.50	58.96	46.83	48.52	50.36
<i>Reasoning Models</i>							
o3-mini	50.00	54.83	51.98	61.32	38.89	50.21	50.93
o4-mini-medium	51.26	58.30	54.95	64.15	40.87	51.48	53.21
Gemini-2-Flash-Thinking	37.82	41.31	41.58	45.75	50.40	43.04	43.36
Gemini-2.5-Flash-Thinking	45.80	53.67	52.97	56.60	55.16	53.59	52.93
Kimi-VL-A3B-Thinking-2506	42.86	41.31	40.59	51.42	39.68	41.35	42.71
<i>Spatial Reasoning Models</i>							
SpaceOm	42.44	38.61	48.02	37.74	42.86	39.24	41.36
SpaceThinker-Qwen2.5VL-3B	40.34	37.84	47.03	38.21	43.25	37.97	40.64
SpaceQwen2.5-VL-3B-Instruct	31.51	35.14	37.62	37.74	50.79	47.26	40.14
<i>Human Baseline</i>	93.70	74.13	91.58	91.51	88.89	87.76	87.57

3B-Instruct (Chen et al., 2024a). For GLM-4.5V-106B-MoE, GLM-4.1V-9B-Thinking, and Step-3-321B-MoE we report open-ended evaluation only: these models are not instruction-tuned and were trained predominantly on open-ended tasks, a regime that induced excessive refusals under the MCQ protocol. To streamline comparison across the results tables, we adopt a consistent color coding: **proprietary**, **open-source up to 7B**, **larger than 7B**, **proprietary reasoning**, **open-source reasoning**, **spatial specialists**, and **human baseline**.

Evaluation Process and Metrics. Accuracy is the primary metric under both the MCQ and the open-ended protocols. For **MCQ evaluation**, we employ direct prompting: each model returns the option index corresponding to its selected answer, and responses are checked automatically against the ground-truth option. The prompts and full protocol are documented in Appendix D.1. For **open-ended evaluation**, each model generates a free-form answer that is subsequently scored by a large-language-model judge, with prompts and protocol detailed in Appendix D.2. To certify

Table 3: Per-category open-ended (free-form generation) accuracy (% , $\uparrow$ higher is better) on SPATIALAB-OPEN, reported for each spatial category alongside the overall score. Within each model group the top result is rendered in **bold** and underlined. Withdrawing the multiple-choice scaffold widens the gap to humans (64.93% overall): the strongest model, GPT-5-mini, reaches only 40.93%, while small open-source and specialized spatial models collapse into the low teens, demonstrating that generative spatial reasoning is markedly harder than discriminative selection.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	23.53	16.60	23.27	30.66	17.86	21.94	26.00
GPT-5-mini	45.38	34.75	37.13	49.53	42.46	37.13	40.93
Gemini-2.0-Flash	31.93	24.32	27.23	31.13	26.19	24.47	27.43
Gemini-2.5-Flash	34.03	26.64	31.68	38.68	26.59	29.54	30.93
Gemini-2.5-Pro	37.14	45.45	36.36	37.14	23.91	24.44	33.61
Claude 3.5 Haiku	26.05	18.92	24.75	25.94	20.24	21.10	22.64
Mistral Medium 3.1	25.21	19.31	21.78	29.25	15.08	16.88	21.00
Open-Source Models							
InternVL3.5-1B	05.88	09.65	9.90	13.68	09.13	10.13	09.64
InternVL3.5-2B	12.18	11.20	10.89	23.58	11.90	18.14	14.50
Qwen-VL2.5-3B-Instruct	15.55	8.49	15.35	10.85	18.25	9.28	12.93
InternVL3.5-4B	19.33	17.76	15.84	19.81	16.27	18.99	18.00
Gemma-3-4B-it	20.17	13.13	14.85	23.58	15.08	19.83	17.64
Qwen-VL2.5-7B-Instruct	15.13	15.83	20.30	27.83	15.87	19.83	18.86
Llama-3.2-11B-Vision-Instruct	16.81	16.99	22.28	25.00	13.49	18.57	18.57
Gemma-3-27B-it	22.69	16.22	24.75	34.43	22.62	21.94	23.43
Qwen-VL2.5-32B-Instruct	16.39	09.65	14.85	16.98	09.92	13.50	13.36
InternVL3.5-72B	22.69	20.46	20.30	31.60	19.84	26.16	23.36
Qwen-VL2.5-72B-Instruct	26.89	20.85	25.25	30.66	24.60	20.68	24.64
Llama-3.2-90B-Vision-Instruct	22.69	23.17	21.29	28.30	21.83	27.00	24.00
GLM-4.5V-106B-MoE	31.09	20.46	25.25	26.42	24.21	24.47	25.21
Reasoning Models							
o3-mini	39.08	30.12	29.70	39.15	41.27	30.38	35.00
o4-mini-medium	40.76	32.82	32.18	42.92	44.05	34.18	37.86
Gemini-2-Flash-Thinking	31.09	27.41	31.19	34.43	29.37	29.54	30.36
Gemini-2.5-Flash-Thinking	37.14	45.45	36.36	37.14	21.74	22.22	32.77
Kimi-VL-A3B-Thinking-2506	13.45	14.29	11.88	17.92	12.70	18.57	14.79
GLM-4.1V-9B-Thinking	21.43	19.31	26.73	24.53	19.84	22.78	22.21
Step-3-321B-MoE	32.35	18.53	27.72	27.36	29.76	27.85	27.14
Spatial Reasoning Models							
SpaceOm	12.61	06.95	15.84	11.79	18.65	12.24	12.93
SpaceThinker-Qwen2.5VL-3B	13.45	09.27	17.82	10.38	19.44	10.13	13.36
SpaceQwen2.5-VL-3B-Instruct	12.61	03.86	13.86	09.43	11.90	11.39	10.36
Human Baseline							
	73.53	50.19	70.30	69.81	65.48	62.87	64.93

the judge against human evaluation, we conducted an LLM–human agreement analysis, reported in Appendix D.2. Table 5 establishes that the LLM judge (Gemini-2.5-Flash) attains substantial agreement with the human annotators, recording a Cohen’s kappa of 0.738 against the majority vote and 0.681–0.795 against individual annotators; raw accuracy against the majority vote reaches 0.880, and a Fleiss’ kappa of 0.774 among the human annotators confirms strong inter-annotator reliability. This agreement is high enough to license the LLM judge as a faithful, scalable proxy for human scoring across the open-ended benchmark. Additional details on the experimental setup, prompts, and evaluation procedures appear in Appendix D.

Improving Visual Reasoning Capabilities. To determine how far spatial reasoning can be advanced on this diverse benchmark, we systematically investigate a family of intervention strategies: the models’ inherent reasoning mechanisms (Appendix H.1), Chain-of-Thought (CoT) prompting (Appendix H.2), CoT with self-reflection (Appendix H.3), supervised fine-tuning (SFT) (Appendix H.4), and multi-agent systems (Appendix H.5). For SFT specifically, Qwen-VL2.5-3B-Instruct was fine-tuned on a stratified 40% split of the dataset, strengthening generalization across relational and geometric tasks; evaluation on the held-out 60% recorded reduced errors in navigation and orientation, with additive gains when CoT and SFT were combined. Agentic reasoning, by contrast, exhibited a sharply asymmetric profile: orientation benefits substantially, whereas other categories stagnate or degrade, which demonstrates that multi-step reasoning cannot substitute for the deeper perceptual and spatial grounding the benchmark demands.

5 FINDINGS AND ANALYSIS

5.1 OVERALL RESULTS

SPATIALAB-MCQ. Table 2 reports multiple-choice performance on SPATIALAB-MCQ, and the picture it paints is unambiguous: substantial variation across models, tasks, and families, with overall accuracies confined to roughly 30–55% (random choice = 25%) against a human baseline of 87.57%, a margin of headroom that no model comes close to closing. The strongest system is an open-source model, InternVL3.5-72B, at 54.93% accuracy. Reasoning- and instruction-tuned models (o4-mini at 53.21%, Gemini-2.5-Flash-Thinking at 52.93%, GPT-5-mini at 54.29%) and the largest open-source models (InternVL3.5-72B at 54.93%, Llama-3.2-90B at 50.36%) cluster near the top, yet scale alone is plainly not decisive: Llama-3.2-11B reaches only 30.50% despite its considerable size. By category, the strong models concentrate their competence in Orientation and 3D Geometry, several exceeding 60%, whereas Spatial Navigation, Depth & Occlusion, and Size & Scale are at once more variable and uniformly harder. Tellingly, the spatially specialized models (SpaceOm, SpaceThinker, SpaceQwen) settle in the low-to-mid 40s overall, so domain specialization does not by itself translate into higher MCQ accuracy on our diverse question set. Depth & Occlusion and Size & Scale yield mixed outcomes, some architectures handling occlusion competently while others fail outright, a divergence that points to differing inductive biases and training data. Taken together, these patterns establish that instruction-tuning and reasoning capability preferentially benefit geometric and orientation reasoning, while multi-step grounding tasks such as navigation remain an open and shared weakness.

SPATIALAB-OPEN. The open-ended scores in Table 3 fall substantially below the MCQ results and spread widely: overall accuracy ranges from about 9.6% (InternVL3.5-1B) to 40.9% (GPT-5-mini) against a human baseline of 64.9%, leaving a pronounced and uniform gap. Proprietary models lead the open-ended leaderboard (GPT-5-mini at 40.93%, GPT-4o-mini at 26.00%, Gemini-2.5-Flash at 30.93%), but the reasoning-tuned variants stand out most sharply: o4-mini (37.86%) and Gemini-2.5-Flash-Thinking (32.77%) surpass many open-source baselines, direct evidence that instruction- and CoT-style tuning materially improves generative outputs. Most small open-source models remain at the bottom, several in the teens (InternVL3.5-1B at 9.64%, Qwen-VL2.5-3B at 12.93%), while the largest open models climb only to the low 20s (InternVL3.5-72B at 23.36%, Qwen-VL2.5-72B at 24.64%), confirming that scale confers no more than modest gains. The spatial specialists (SpaceOm, SpaceThinker, SpaceQwen) reach only ~10–13% overall, so task specialization again fails to secure robust open-ended generation on our diverse question set. Across subtasks, the top models fare comparatively better on orientation and select relative-position items (GPT-5-mini reaches 49.53% on Relative Positioning and 37.13% on Orientation), whereas depth & occlusion, size & scale, and above all spatial navigation remain consistently hard, many models scoring below 30% on these subtasks. This pattern corroborates the earlier finding that multi-step grounding and perceptual reasoning are the decisive bottlenecks for open-ended generative answers.

5.2 PERFORMANCE DROP IN OPEN-ENDED EVALUATION COMPARED TO MCQ

Across 25 models, the average MCQ→open-ended gap in SPATIALAB is 23.0% ($\sigma = 5.5\%$), with subtask means confined to a narrow band from 22.89% (spatial navigation) to 24.57% (3D geometry); this uniformity marks the effect as systematic rather than incidental. Specialist spatial reasoning models incur the largest gaps (about 27%), driven by spatial navigation (up to 36.68%) and orientation (34.44%), whereas reasoning-oriented models hold the smallest gaps (around 19%) at lower subtask variance, reflecting the stabilizing influence of instruction-tuning and CoT decoding. Correlational analysis confirms that spatial navigation dominates the overall disparity (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) contributing substantially. The negative or near-zero gaps (Llama-3.2-11B at -1.98 in depth & occlusion; o4-mini at -3.18 in relative position) reveal that poorly designed MCQ distractors can misrepresent true competence in either direction. We trace this format sensitivity to four interacting factors, the structural advantage of the MCQ format, a specialization bias toward categorical tasks, the sequential-reasoning demands of spatio-navigation, and calibration differences in generative output, and we conclude that MCQ evaluation alone materially overestimates practical spatial reasoning ability. We therefore recommend complementing MCQ with open-ended evaluation, auditing distractors, adopting stepwise generation or instruction-tuning, and reporting per-subtask diagnostics, so that measured performance reflects genuine competence and format-dependent artifacts are suppressed. Full details appear in Appendix G and Table 18.

5.3 ERROR ANALYSIS

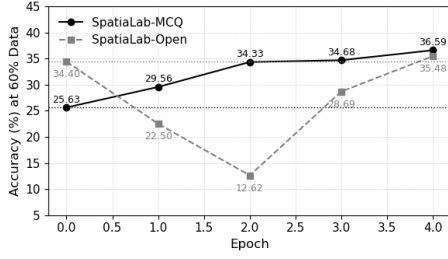
SPATIALAB-MCQ. Our analysis (Appendix E) establishes that VLMs attain selective peaks yet lack holistic spatial competence. Closed-source models reach the highest scores anywhere in the benchmark (85.71% in *Stacking Orientation*) even as they collapse to near-chance on *Relative Size Comparison*. Open-source scaling lifts ceilings (80.56% in *Corner/Angle Positioning*) without forestalling catastrophic failures (2.0% in *Object Rotation*). Reasoning-augmented and spatially specialized models contribute localized gains, in occlusion inference and navigation for instance, yet remain capped below 55% on many physical-abstraction tasks (*Gravity Effects*, *Stability Prediction*). The error distributions further localize systematic weaknesses in embodied reasoning (*Tool Handedness*), recursive relational chaining (*Pathway Existence*), and non-local cues (*Reflective Surfaces*), while the comparatively strong results in *Obstacle Avoidance* betray shortcut exploitation rather than robust planning. In aggregate, these patterns confirm that current VLMs lean heavily on surface-level correlations and lack stable internal encodings of orientation, physics, and compositional logic. Progress will accordingly demand both richer training distributions, embodied and physics-driven data among them, and architectural mechanisms that embed geometric and reference-frame grounding.

SPATIALAB-OPEN. Error analysis of SPATIALAB-OPEN (Appendix F) shows that even the top closed-source models (GPT-5-mini at 58.14% in *Directional Relations*) collapse on tasks such as *Proximity Gradients*, so their dominance is task-specific rather than general. Large open-source models post strong results in *Depth & Occlusion* (60.0%) yet still fall to 9.3% on *Proximity*, demonstrating that scale alone cannot secure robustness. Smaller models pair isolated strengths (26.0% in *Relative Size Comparison*) with outright breakdowns (0.0% in *Betweenness*), a signature of the structural limits of compact architectures. Reasoning-tuned systems are genuinely promising, Gemini-2.5-Flash-Thinking reaching 75.0% in *Tool Handedness*, yet comparable models fall to single digits, exposing fragile integration. The specialized spatial models perform worst overall, rarely exceeding 20% and frequently failing on core relational tasks. These failure patterns isolate systemic weaknesses in occlusion handling, orientation, and multi-step relational chaining, while the partial success on size-scale cues points to reliance on superficial correlations. The evidence converges on three root causes, missing representations for spatial analogies, architectural limits tied to scale and training diversity, and brittle reasoning pipelines, and thereby establishes the need for unified geometric encodings, physics-aware reasoning, and embodied data to advance from isolated peaks toward consistent spatial competence.

Qualitative Error Analysis. To characterize model weaknesses precisely, we conducted a qualitative error analysis (Appendix I) and found that failures condense into a small number of recurring classes rather than dispersing as random noise. The recurring modes are spatial mislocalization, in which models confuse referents in crowded scenes; perspective and scale mistakes, in which reliance on object-size priors overrides image-based cues; and occlusion or ordering failures, most acute with thin or partially hidden structures. We further identify attribute confusion, the conflation of perceptual with functional properties, and open-ended rationalization, the generation of fluent but visually ungrounded narratives. Because these errors recur invariantly across architectures and prompting styles, they reflect structural biases intrinsic to current VLMs rather than dataset-specific noise. Failure rates spike, moreover, precisely when multiple cues must be fused, as when depth ordering must be combined with relative size, and confidence calibration is especially poor in open-ended settings. Our diagnostic protocols confirm that models routinely ignore minimal but decisive visual features, fall back on brittle heuristics, and fail to update when counterfactual edits are applied. The root causes trace to insufficient object-centric binding, the absence of geometric supervision, and training objectives that reward plausibility over grounding. Collectively, these findings show that current VLMs command strong coarse perception yet struggle with multi-cue integration and grounded reasoning, establishing the need for geometry-aware supervision, multi-scale feature retention, and verification pipelines.

5.4 PERFORMANCE IMPROVEMENT APPROACHES

Inherent Reasoning Mechanisms. Reasoning-enabled models consistently outperform their non-reasoning baselines under both the MCQ and the open-ended format, the largest gains concentrating in relational and orientation tasks (up to +13.1% in *Relative Positioning* for MCQ). Reasoning does stabilize open-ended performance, yet its returns remain uneven across categories, declining most visibly in *Size & Scale*; this establishes that reasoning modules sharpen logical consistency without



(a) SFT performance trends over epochs.

Category	MCQ			Open-ended		
	Before	After	Gain	Before	After	Gain
3D Geometry	30.07	37.76	7.69	20.28	23.78	3.50
Depth & Occlusion	23.87	32.90	9.03	43.23	45.81	2.58
Orientation	23.33	36.67	13.33	39.67	42.15	2.48
Relative Positioning	26.19	27.78	1.59	51.18	50.39	-0.79
Size & Scale	27.63	44.74	17.11	18.42	14.47	-3.95
Spatial Navigation	22.38	38.46	16.08	36.62	39.44	2.82
Grand Total	25.63	36.59	10.97	34.40	35.48	1.07

(b) Per-category accuracy before and after SFT for the MCQ and open-ended formats. Fine-tuning improves every MCQ category (grand total +10.97%) but yields only a negligible open-ended gain (+1.07%), with regressions in Relative Positioning and Size & Scale, exposing a format-dependent transfer asymmetry.

Figure 4: SFT training results: (a) learning trajectories over epochs, and (b) final per-category accuracy. MCQ accuracy rises monotonically while open-ended accuracy remains essentially flat, evidencing that SFT strengthens discriminative selection without improving generative spatial grounding.

resolving the deeper problems of grounding and scale sensitivity. Detailed per-category comparisons appear in Appendix H.1.

Chain-of-Thought (CoT) Prompting. Across models, CoT prompting confers little benefit and frequently reduces accuracy (Table 19), orientation being the sole category to register consistent gains. Step-by-step reasoning thus aids directional alignment but falters wherever robust perceptual grounding is prerequisite, as in depth, scale, and spatial navigation. In sharp contrast to its effect on textual reasoning, CoT here amplifies flawed priors rather than correcting them, which demonstrates that solving SPATIALAB demands perceptual understanding beyond logical chaining (details in Appendix H.2).

Chain-of-Thought (CoT) with Self-Reflection. Augmenting CoT with an explicit self-reflection stage, in which the model is prompted to review its prior reasoning step-by-step and correct its mistakes (see Figure 8 in the Appendix for the prompt), yields modest benefits under the MCQ format, most notably for geometry and depth, yet fails to generalize to open-ended tasks (Appendix H.3). This asymmetry confirms that reflective reasoning cannot compensate for missing perceptual grounding, laying bare a core limitation of current VLMs.

Supervised Fine-Tuning (SFT). As shown in Figure 4, supervised fine-tuning of Qwen2.5-VL-3B-Instruct on a stratified 40% split of SPATIALAB (~560 samples) improves MCQ accuracy consistently across every spatial reasoning category, yet transfers little, and at times negatively, to open-ended tasks. This divergence signals overfitting to task-specific answer distributions, corroborating the concern that SFT often *teaches to the test* rather than fostering generalizable reasoning (Wang et al., 2025). The sharp decline in open-ended performance implicates biased internal representations, tuned for categorical discrimination but unstable in generative settings; the most likely mechanism is catastrophic forgetting of linguistic priors during fine-tuning (Huang et al., 2024a), compounded by objectives that privilege discrete-choice alignment over spatial reasoning expressed in language. In sum, while SFT strengthens performance under structured formats, it risks imposing an algorithmic straitjacket, boosting accuracy on constrained tasks while stalling progress on naturalistic spatial reasoning. Full details appear in Appendix H.4.

AI Agents for Spatial Reasoning (SPATIOXOLVER). To probe agentic capabilities, we employ SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025) to perform structured spatial reasoning over images. The framework decomposes complex visual reasoning into specialized sub-tasks assigned to dedicated agents, comprising modules for object segmentation, attribute extraction, spatial relation mapping, and transformation tracking, and consolidates their outputs into a unified structured representation through which low-level perceptual cues propagate into higher-order relational inferences. We instantiate the pipeline with Gemini-2.5-Flash at low temperature to enforce deterministic outputs across the multi-stage refinement process. The results are sharply asymmetric: the Orientation category improves substantially, by +8.00% in MCQ and by a striking +36.00% in open-ended evaluation, yet this success does not generalize, as Depth & Occlusion and Spatial Navigation decline severely under open evaluation (−24.00% and −12.00%, respectively). These drops show that where the inherent perceptual priors, depth and occlusion cues

in particular, are weak, multi-step reasoning reinforces misconceptions rather than correcting them. The finding is decisive: agentic workflows advance tasks reducible to sequential alignment but cannot compensate for absent perceptual grounding in complex 3D environments. Full details appear in Appendix H.5.

6 DISCUSSION

Spatial reasoning is central to embodied intelligence, yet most benchmarks rest on synthetic scenes, templated questions, or narrow task scopes that overlook real-world visual variability. We introduce SPATIALAB, a benchmark of 1,400 visual question-answer pairs spanning six principal spatial categories and 30 subcategories, among them 3D geometry, occlusion, and navigation. The benchmark pairs Multiple-Choice and Open-Ended formats to disentangle option-based pattern use from genuine generative reasoning, and its quality is underwritten by a multi-stage review process and complexity checks spanning lighting, texture, and spatial diversity. Evaluated against human baselines, more than 25 contemporary models establish SPATIALAB as a precise diagnostic instrument for exposing the weaknesses of current vision-language systems.

Detailed error analysis localizes systematic failures in handling transparency, multi-step navigation, and occlusion, demonstrating that models lean on surface-level texture cues in place of internalized 3D geometric representations. Supervised Fine-Tuning (SFT) delivered clear gains on structured MCQ tasks yet simultaneously induced instability in open-ended reasoning, a signature of the “catastrophic forgetting” of linguistic priors that accompanies optimization toward discrete answers. Chain-of-Thought reasoning proved effective for sequential tasks such as Orientation but failed to repair the underlying perceptual errors in Depth and Scale, amplifying hallucinations rather than resolving them. Agentic decompositions exhibited the same trade-off, sharpening specific alignment tasks while degrading holistic scene understanding. These convergent behaviors establish that scaling and reasoning scaffolds alone are insufficient: attaining human-level spatial intelligence will require architectural innovations that explicitly integrate geometric grounding and physics-aware representations.

7 CONCLUDING REMARKS

Spatial reasoning remains a core bottleneck for VLMs, most acutely in visually complex, real-world settings. SPATIALAB contributes 1,400 diverse QA pairs across six spatial categories and thirty subcategories, enabling fine-grained evaluation through paired MCQ and open-ended formats. Even after the application of several performance-improving methods, our experiments establish substantial gaps between state-of-the-art models and human performance, with open-ended generation incurring an additional 10–25% drop relative to multiple-choice accuracy. Error analysis localizes these persistent failures in depth, occlusion, 3D geometry, and multi-step navigation, all of which reflect reliance on superficial cues rather than grounded spatial representations. By exposing these limitations precisely, SPATIALAB furnishes a structured testbed for diagnosing spatial competence. The benchmark thus enables holistic evaluation and yields actionable guidance for developing VLMs with human-aligned spatial understanding. Continued progress will require richer training signals, embodied interaction and action-conditioned feedback among them, together with architectures capable of geometric grounding, relational chaining, and consistent multi-scale reasoning.

ETHICS STATEMENT

We adopted multiple measures to ensure ethical compliance throughout the development of this work. All metadata associated with the images was removed to protect privacy, and personally identifiable features such as human faces and license plates were minimized or excluded wherever possible. The dataset was carefully curated to avoid copyright concerns, with diligent attention to sourcing and licensing, though minor oversights may persist despite our best efforts. The benchmark construction process was designed for transparency and fairness, so that no demographic group was singled out or negatively represented. We adhered to the ICLR Code of Ethics and confirm that no human subjects or sensitive personal data were involved.

REPRODUCIBILITY STATEMENT

To facilitate reproducibility, we provide a detailed account of the dataset construction process, annotation pipeline, and evaluation protocol across the main text and the appendix. Every filtering, quality-control, and validation step is documented explicitly, together with precise definitions of the task categories and annotation criteria. Hyperparameters, model configurations, and training procedures are reported in full in the appendix to support faithful replication of our results.

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A RELATED WORK

Spatial reasoning in multimodal models. Spatial reasoning, the capacity to apprehend geometric structure and relational layout within visual scenes, is a foundational competence of embodied intelligence. The earliest multimodal benchmarks, exemplified by CLEVR (Johnson et al., 2016) and GQA (Hudson & Manning, 2019), interrogated compositional reasoning through synthetic renderings and curated templates; although these controlled regimes permitted precise diagnostic analysis, they conferred only limited ecological validity. The advent of large vision–language models (VLMs) (Li et al., 2023; Alayrac et al., 2022; OpenAI, 2023) has since motivated a thorough reappraisal of spatial reasoning in more naturalistic settings, and yet systematic deficiencies endure. Notwithstanding their commanding performance on captioning and visual question answering, these models fail with striking consistency on tasks that demand depth ordering, occlusion inference, or coherent reference-frame alignment. This dissociation lays bare a fundamental limitation: scale alone does not translate into progress on structured spatial understanding.

Spatial reasoning benchmarks. A rapid succession of benchmarks has recently sought to expose these gaps (Table 1). SCANQA (Azuma et al., 2022) ranks among the earliest embodied QA datasets, mandating reasoning over 3D indoor environments; its modest scale (800 scenes) and comparatively shallow complexity, however, sharply curtail broader generalization. Subsequent datasets such as VISUAL SPATIAL (Liu et al., 2023) and WHAT’S UP (Kamath et al., 2023) extended coverage to real-world and household contexts, yet their dependence on template-driven questions attenuated linguistic variability and overrepresented elementary binary or multiple-choice relations. More recent initiatives, notably EMBSPATIAL-BENCH (Du et al., 2024) and SPACE3D-BENCH (Szymanska et al., 2024), diversified the task repertoire but remained constrained both by scale (2.2K and 211 samples, respectively) and by annotation style, whether template-heavy or manual yet narrow. SPATIALRGPT-BENCH and BLINK-SPATIAL introduced mixed domains and as many as 14 categories, meaningfully advancing diversity, yet continued to privilege either static snapshots or limited multi-view simulation.

Embodiment, difficulty, and diversity. The trajectory toward embodiment is exemplified by datasets such as ROBOSPATIAL (Song et al., 2024), which scales to 1M scenes framed around robot-centric questions. Its template-based annotation, however, renders those questions less natural and leaves them susceptible to heuristic shortcuts. SPATIALVLM (Chen et al., 2024b) likewise attains unprecedented scale (billions of QA pairs), yet its two-category coverage forecloses fine-grained diagnostic evaluation. At the opposite pole, high-quality manual datasets such as VSI-BENCH (Yang et al., 2024) achieve superior annotation fidelity but, comprising only 288 scenes, remain inadequate for training or for robust cross-model comparison. Along the dimension of complexity, the majority of existing benchmarks plateau at “medium” (Mm in Table 1), only rarely attaining the “hard” or “extremely hard” regimes that demand multi-step causal inference. A substantial evaluation gap consequently persists for next-generation models, one that SPATIALAB is expressly designed to close.

Evaluation protocols. The evaluation protocol and its attendant analytic depth are of no less consequence. The preponderance of prior work commits to close-ended QA or binary judgments (e.g., VISUAL SPATIAL, EMBSPATIAL-BENCH), which expedite scoring at the cost of output expressivity. While a small number of datasets, among them ROBOSPATIAL, SPATIAL-MM, and SPATIALAB-OPEN, admit open-ended evaluation, only SPATIALAB-OPEN couples it with extreme complexity, manual puzzle-style annotation, and a comprehensive treatment of error patterns. This conjunction surfaces systematic weaknesses that lie well beyond raw accuracy, encompassing calibration and cross-model divergence, dimensions that existing datasets seldom capture. Absent such granularity, reported gains can prove deceptive, as models may overfit dataset-specific biases in place of acquiring robust, transferable spatial competence.

Positioning of SPATIALAB. SPATIALAB confronts these limitations head-on. Whereas prior datasets span between 2 and 14 task categories, SPATIALAB encompasses 30 distinct categories spanning geometry, occlusion, orientation, relative positioning, size and scale, and navigation, delivering a task diversity without precedent in the literature. In deliberate contrast to template-based benchmarks, every question is manually authored and puzzle-inspired, securing linguistic richness while categorically foreclosing shortcut exploitation. SPATIALAB is, furthermore, fully embodied and multi-modal, integrating static and dynamic visual contexts alike, and it thereby

bridges the longstanding divide between robotics-style datasets (ROBOSPATIAL) and web-scale corpora (SPATIALVLM). As Table 1 makes plain, even the strongest baselines, InternVL3.5-72B and GPT-5-mini, remain well below 60%, while the open-ended variant admits a mere 40.93% accuracy. This pronounced performance gap establishes at once the difficulty of the benchmark and its enduring value for driving future research.

Toward next-generation evaluation. By unifying high difficulty, broad coverage, manual annotation, and open-ended tasks within a single framework, SPATIALAB reorients the spatial reasoning landscape away from template-driven, narrow evaluations and toward comprehensive, embodied testing. This design supports not merely a binary verdict on whether a model answers correctly, but the systematic identification of error patterns, the rigorous interrogation of causal and counterfactual reasoning, and the quantification of calibration reliability. SPATIALAB thus furnishes a principled foundation for the next phase of multimodal research, complementing datasets such as MIND THE GAP and VLM4D that probe temporal or generative facets yet remain narrower in coverage. We anticipate that future work will adopt SPATIALAB as a high-bar diagnostic instrument, one that propels models beyond surface-level recognition toward genuine, grounded spatial intelligence.

A.1 DETAILED ANALYSIS OF LIMITATIONS IN PRIOR BENCHMARKS

To situate SPATIALAB precisely within the broader landscape, we undertake a systematic examination of existing spatial reasoning benchmarks. Prior efforts have laid important foundations, yet each is encumbered by structural and methodological constraints that diminish its value for evaluating contemporary VLMs. We organize these constraints along three principal axes: synthetic-domain bias, restricted task scope, and constrained evaluation formats. Table 4 consolidates the salient limitations we identify.

Synthetic and Template-Based Constraints. Benchmarks such as CLEVR and GQA rest on synthetic environments and template-generated linguistic structures. While these settings furnish controlled conditions for isolating compositional reasoning, their visual domain lacks real-world complexity and their linguistic space remains narrow, so that models may exploit template regularities in place of grounded spatial interpretation. Related simulated benchmarks such as ROBOSPATIAL manifest the identical failure mode, incurring a pronounced sim-to-real gap.

Domain and Scale Limitations. Datasets grounded in 3D reconstruction, including ScanQA and Space3D-Bench, provide depth supervision but confine themselves to indoor scenes, thereby precluding assessment in the outdoor and mixed environments that embodied agents must ultimately navigate. High-quality manual datasets such as VSI-Bench deliver richer annotation yet remain limited in scale, a constraint that undermines their utility for high-resolution subcategory analysis.

Taxonomic and Evaluation Narrowness. Large-scale datasets such as SpatialVLM cover broad tasks but lack fine-grained taxonomies, rendering them ill-suited to diagnosing specific weaknesses in orientation, occlusion, or layout reasoning. Many recent benchmarks, among them EMBSPATIAL-BENCH and WHAT’S UP, rely heavily on Binary or MCQ formats, which can inflate measured performance by rewarding option elimination rather than genuine generative spatial reasoning. SPATIALAB redresses these deficiencies decisively, uniting MCQ and Open-Ended formats across a detailed taxonomy of 30 spatial subcategories.

Table 4: Limitations of representative prior spatial benchmarks contrasted with SPATIALAB, organized by domain constraints, evaluation format, and task taxonomy. SPATIALAB alone combines in-the-wild imagery, a 30-subcategory taxonomy, and paired MCQ and open-ended evaluation, resolving each limitation the other benchmarks exhibit.

Benchmark	Domain	Eval. Format	Taxonomy	Primary Limitations Identified
CLEVR and GQA	Synthetic	Closed-Ended	Relational	Artificial visual domain and template-driven language actively invite shortcut learning.
ScanQA	Indoor 3D	Open-Ended	Navigation	Confined to indoor point clouds at limited scale ($< 1K$ scenes).
SpatialVLM	Web Data	Open-Ended	General	Covers only a handful of high-level spatial categories.
EmbSpatial-Bench	Indoor	MCQ / Binary	6 Categories	Relies on closed-ended formats with shallow reasoning chains.
RoboSpatial	Simulator	Hybrid	Embodied	Template-generated questions in simulated scenes yield a sim-to-real gap.
Space3D-Bench	Indoor	MCQ	6 Categories	Very small scale (~ 200 samples) precludes robust statistical evaluation.
OmniSpatial	Mix	MCQ / Binary	Metric	Prioritizes metric outputs, offering scant support for explanatory reasoning.
SpatialLab (Ours)	In-the-Wild	MCQ + Open	30 Sub-cats	Resolves each identified limitation through real-world imagery, a fine-grained taxonomy, and paired evaluation formats.

B TASK DESIGN AND DESCRIPTIONS

Spatial reasoning demands far more than the recognition of individual objects; it hinges fundamentally on how those objects relate, align, and interact within physical space. To capture this complexity in full, we introduce *SpatialLab*, a benchmark expressly engineered to probe the fine-grained facets of spatial understanding across a diverse taxonomy of categories. Its tasks are constructed to mirror the ways in which humans naturally describe and reason about their surroundings, ranging from relative positioning (e.g., *What’s to the left of the microwave?*) through to geometric inference (e.g., *Would this stack fall if another book is added?*). Each category decomposes into subcategories that isolate subtle yet indispensable dimensions of reasoning, among them occlusion, orientation, navigation, and physical effects such as gravity. By articulating every subtask in accessible, everyday terms, *SpatialLab* compels models to advance beyond mere object detection toward structured spatial interpretation. This design lets us rigorously assess how well systems generalize across the contexts of greatest consequence for robotics, AR/VR, autonomous driving, and human-computer interaction. The sections that follow furnish concise, task-level descriptions for every category and subcategory in *SpatialLab*.

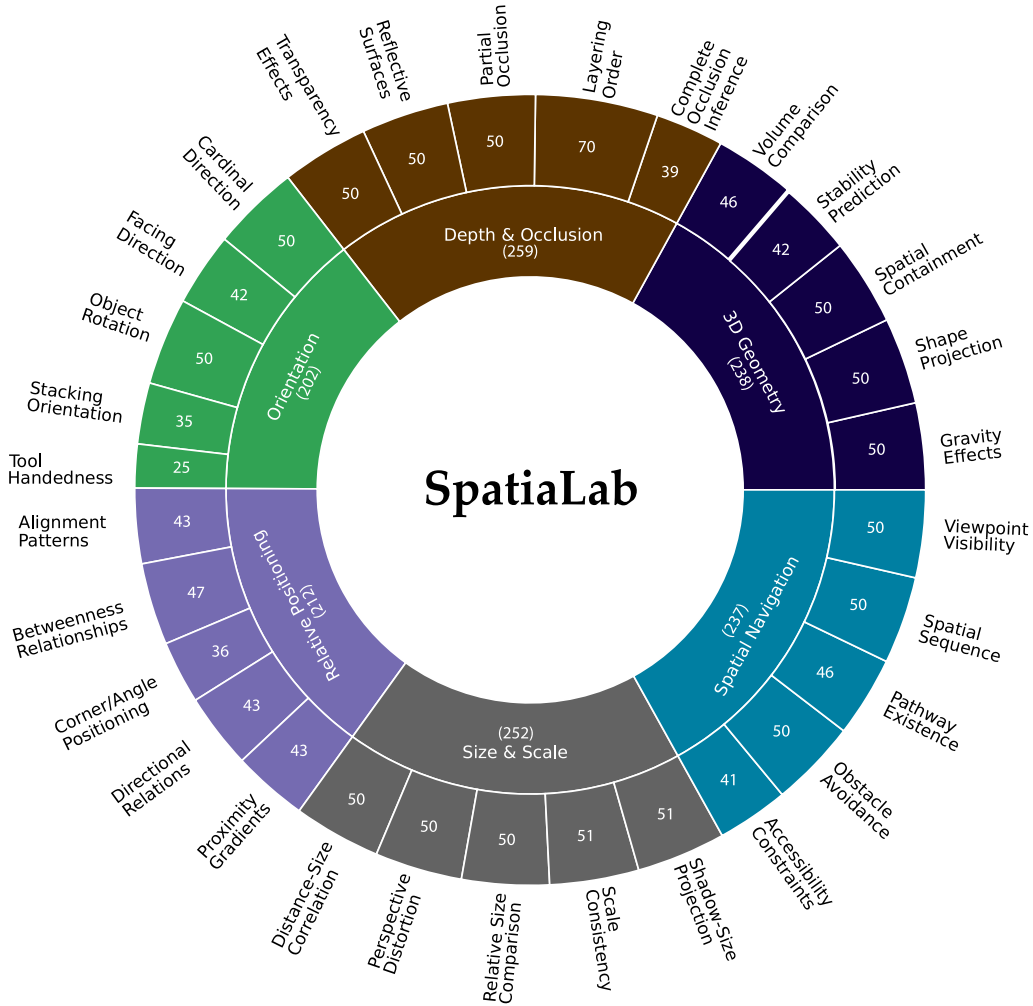


Figure 5: **Categories and subcategories of spatial reasoning in SPATIALAB.** Each of the six principal categories decomposes into five subcategories, yielding a taxonomy of thirty distinct task types in total.

We now articulate, category by category, the rationale and the real-world utility that jointly motivate each task design:

B.1 RELATIVE POSITIONING

Directional Relations. This subtask targets the directional predicates that situate objects relative to one another, namely left, right, above, and below. Command of these predicates enables a model to parse spatial layouts with the fluency of a human narrator, grounding its descriptions in a shared frame of reference. Resolving what lies immediately to the left of an appliance, for instance, is foundational to scene understanding and robot navigation alike.

Proximity Gradients. Proximity reasoning requires ordering objects by their closeness to a designated reference point, thereby isolating the relative-distance dimension of spatial cognition. It underlies queries such as “Which object is nearest to the doorway?” Models that estimate such gradients of closeness reliably furnish direct support for navigation tasks and assistive technologies.

Alignment Patterns. Alignment concerns whether a set of objects conforms to a common orientation or ordering, whether stacked vertically or arranged along a straight line. Detecting such regularities reveals how an environment is organized, from shelves and tables through to warehouses, and this capacity proves equally central to robotic sorting and automated inspection.

Betweenness Relationships. Betweenness isolates the relation of interposition, identifying the object that occupies the intermediate region between two references. Determining what sits between the lamp and the armchair, for example, demands reasoning about spatial continuity along an axis. Judgments of this kind feed directly into navigation, retrieval, and situational-awareness tasks.

Corner/Angle Positioning. Corners and angled locations function as salient reference anchors in spatial reasoning. Questions such as “What’s positioned in the back-right corner?” require the model to localize an object within a tightly constrained region of the scene. This form of reasoning is indispensable in robotics, where it underpins mapping and placement alike.

B.2 DEPTH & OCCLUSION

Layering Order. Layering demands the ordering of objects along the depth axis, distinguishing what stands in front from what recedes behind. Identifying the frontmost item on a table, for example, is a direct expression of depth reasoning. This skill is especially consequential in AR/VR applications, where visual stacking directly conditions the user experience.

Partial Occlusion. Partial occlusion arises when one object conceals a portion of another from view, obliging the model to infer what remains unseen. Recovering the identity of an item partly hidden behind a coffee cup, for instance, requires reconstructing the missing evidence. Such inferential robustness is vital for perception systems operating in cluttered environments.

Complete Occlusion Inference. When an object is fully occluded, its presence may nonetheless be betrayed by indirect cues such as cast shadows or telltale bulges. Inferring what lies beneath a blanket compels the model to reason from such indirect evidence rather than from direct observation, mirroring the confident inferences humans draw under uncertain visual conditions.

Transparency Effects. Transparent media, such as glass or water, permit partial visibility of whatever lies behind them. Determining what is discernible through a fish tank requires fusing occlusion reasoning with the perception of transparency. This interplay is pivotal for simulation, graphics, and embodied AI.

Reflective Surfaces. Reflective reasoning requires models to treat mirrors, polished metals, and glass reflections as bona fide sources of spatial information. Recognizing what is reflected in a mirror entails an indirect mapping between a virtual image and its physical origin. This competence is essential to safety in autonomous vehicles and to situational awareness in robotics.

B.3 ORIENTATION

Cardinal Direction. Cardinal reasoning grounds objects in compass directions such as north, south, east, and west. Establishing which way a front door faces requires integrating global orientation cues into a single consistent frame. This capability is critical for navigation and geospatial alignment.

1350 **Object Rotation.** Rotation reasoning quantifies the extent to which an object has been turned about
1351 an axis. Estimating that a chair has been rotated by a given angle is essential to accurate 3D modeling,
1352 and such estimates enable robots to align objects correctly prior to manipulation.

1353 **Facing Direction.** Facing direction pins down the orientation of an object’s front-facing surface.
1354 Deciding whether a monitor faces north, for example, requires mapping that surface orientation onto
1355 global references. This reasoning is important for surveillance and robotics alike.

1356 **Stacking Orientation.** Stacking orientation captures whether objects are piled vertically, horizontally,
1357 or in mixed configurations. Reading these arrangements correctly is precisely what secures stability
1358 in settings from warehouses to homes, and it is likewise critical to safe robotic stacking.

1359 **Tool Handedness.** Many tools, such as scissors or gloves, are engineered specifically for either
1360 left- or right-handed use. Discerning their handedness calls for fine-grained perception of shape and
1361 orientation, a precision that advances assistive robotics in human-centered environments.

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1365 B.4 SIZE & SCALE

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1367 **Relative Size Comparison.** Relative comparison determines whether one object is larger, taller,
1368 or smaller than another. Judging whether a vase stands taller than a statue exercises proportional
1369 reasoning, an ability crucial to packing, fitting, and recognition tasks.

1370 **Perspective Distortion.** Perspective can render distant objects apparently smaller than nearer ones
1371 even when their true dimensions dictate otherwise. Recognizing illusions such as a faraway car
1372 appearing smaller than a nearby bicycle demands explicit compensation for depth cues; mastering
1373 this renders models robust to the vagaries of camera perspective.

1374 **Distance-Size Correlation.** Here size judgments are coupled to distance estimation, so that apparent
1375 scale must be disentangled from true scale. A nearby bicycle looming larger than a distant car should
1376 not mislead reasoning about actual size, and resolving this correctly benefits navigation, AR/VR, and
1377 visual grounding.

1378 **Scale Consistency.** Scale consistency tests whether object sizes are plausible given their surrounding
1379 context. A mouse rendered larger than a chair signals a violation that models must detect, a vigilance
1380 that strengthens realism in simulated environments.

1381 **Shadow-Size Projection.** Shadows encode indirect cues about both object size and the position of
1382 the light source. Inferring how high a light must be from the length of a cast shadow requires explicit
1383 geometric reasoning, which proves valuable in graphics, architecture, and physics-based modeling.

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1387 B.5 SPATIAL NAVIGATION

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1389 **Pathway Existence.** Pathway reasoning determines whether an unobstructed route connects two
1390 points in a scene. Judging whether one can reach the desk without displacing the chairs probes
1391 navigability directly, a capacity that underpins indoor robot planning.

1392 **Obstacle Avoidance.** Avoidance requires selecting routes that minimize the risk of collision. Re-
1393 questing the clearest path from door to window maps directly onto robotic and human navigation
1394 alike, and it is essential in autonomous driving and indoor mobility aids.

1395 **Viewpoint Visibility.** From any given viewpoint, certain objects fall into view while others are
1396 occluded from it. Determining what is obscured when one is seated on the couch requires simulating
1397 that specific perspective, thereby strengthening AR/VR realism and robotic vision.

1398 **Spatial Sequence.** Sequence reasoning establishes the order in which objects are encountered along
1399 a trajectory: walking from the kitchen to the balcony, one would pass the dining table first. Ordering
1400 of this kind mirrors the navigational memory on which humans routinely rely.

1401 **Accessibility Constraints.** Constraint reasoning identifies which objects are rendered inaccessible
1402 by intervening obstacles. A cabinet, for instance, may be unreachable because a table stands directly
1403 in front of it. This reasoning is crucial for assistive AI and ergonomic design.

B.6 3D GEOMETRY

Volume Comparison. Volume comparison asks which object occupies the greater share of three-dimensional space. Judging whether a box exceeds a sphere in volume supports packing and containment tasks, and it is thus important for logistics and simulation.

Stability Prediction. Stability reasoning forecasts whether adding or displacing objects will topple a stack. Anticipating whether placing a book will collapse a pile mirrors human physical intuition and directly supports safety in robotics and construction.

Shape Projection. Projection tasks ask what silhouette an object would cast under a given rotation or illumination. A sphere, for instance, invariably projects a circle, whereas a cube may project a square or a hexagon. Reasoning of this kind strengthens genuine geometric understanding.

Spatial Containment. Containment evaluates whether one object can be accommodated inside another. Reasoning about whether a ball fits within a box supports packing and design and aligns closely with intuitive geometry.

Gravity Effects. Gravity reasoning predicts how objects move once physical forces act upon them. A rolling ball, for example, may come to rest beneath a table owing to friction and intervening barriers. Such reasoning strengthens real-world simulation capabilities.

C MORE DETAILS ON BENCHMARK CONSTRUCTION

C.1 ANNOTATOR RECRUITMENT

We drew our annotation cohort exclusively from engineering graduates aged 24 to 30, a deliberately curated pool whose technical training equipped them to reason precisely about spatial structure. Every annotator contributed on a strictly voluntary basis, worked independently of the others, and received no external incentives; this arrangement was no accident but a principled safeguard of the integrity and impartiality of their judgments. Before undertaking any annotation, each participant completed a structured training regimen calibrated to instill an unambiguous, shared understanding of the task and to enforce uniform consistency across the cohort. At every stage of data collection we adhered rigorously to established ethical and quality-control standards, holding each phase to a uniformly exacting bar. In combination, these measures yield annotations that are demonstrably reliable and tightly aligned with the design principles underpinning the dataset.

C.2 ANNOTATOR TRAINING AND PROTOCOL

Ahead of any annotation activity, we designed a comprehensive, structured training program whose purpose was to guarantee consistency and reliability across the full task taxonomy. Annotators were first oriented systematically to the six core categories (*Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*) and to the thirty subcategories they subsume. For each subcategory we supplied carefully curated exemplars that rendered the governing spatial relations explicit, among them directional relations, occlusion inference, and stability prediction, and we deliberately paired these with counter-examples chosen to expose the most common failure modes and interpretive pitfalls. Beyond the taxonomy proper, we trained annotators to reason systematically across six meta-dimensions of image variability, namely lighting, texture complexity, edge density, dominant relation, material type, and gravity constraints, so that the questions they authored would deliberately exercise the full breadth of visual diversity. We further demonstrated how these dimensions interact in authentic real-world settings, adjudicating occlusion under low-contrast lighting, for instance, or preserving scale consistency in the face of perspective distortion, thereby anchoring abstract principles in concrete perceptual challenges.

The training additionally encompassed the two complementary evaluation modes that jointly define the benchmark: multiple-choice and open-ended. Under the multiple-choice regime, annotators were schooled in the craft of engineering precise, targeted distractors capable of probing fine-grained perceptual discrimination; under the open-ended regime, they practiced formulating prompts calibrated to elicit compositional, multi-step reasoning. To operationalize and consolidate this instruction, we convened dedicated calibration sessions in which annotators authored both question types for a shared set of sample images and iteratively refined their designs in response to structured

feedback, progressively converging on a common standard of quality. To situate task design within the broader research landscape, we reviewed representative prior benchmarks with the annotators, among them SpatialBot-Bench, EmbSpatial, and SpatialMM, and subjected their limitations to critical analysis. Such resources frequently privilege simplified relations, synthetic or staged scenes, and a narrow repertoire of reasoning types, while systematically neglecting the natural complexity introduced by phenomena such as shadows, reflections, and gravity-sensitive stability. This deliberate gap analysis crystallized the imperative for spatial contexts that are richer, more genuinely ambiguous, and dynamically layered, and it directly motivated the design decisions our benchmark embodies.

Finally, we provided exhaustive demonstrations of the annotation protocol itself. These demonstrations codified precise criteria for selecting images of sufficient spatial complexity, for mapping each question unambiguously onto its governing subcategory, and for scrupulously avoiding trivial or under-constrained formulations. Annotators were explicitly directed to favor tasks demanding genuine multi-step reasoning, such as depth layering, relational chains, and cross-view scale inference, over reductive binary labels. Through this systematic training, coupled with sustained iterative feedback, we ensured that the resulting dataset strikes a principled equilibrium between comprehensive taxonomic coverage and authentic real-world relevance.

D MORE DETAILS ON EXPERIMENTS AND EVALUATION

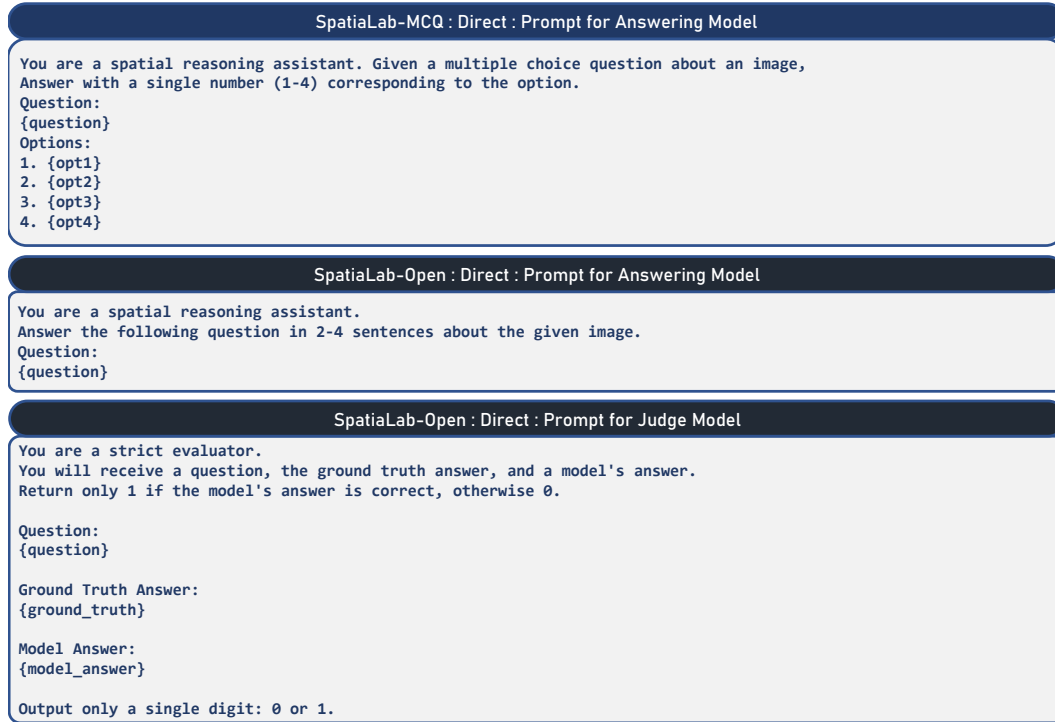


Figure 6: Prompts used for the multiple-choice answer model, the open-ended answer model, and the LLM judge.

D.1 MULTIPLE CHOICE EVALUATION

Figure 6 reproduces the prompts supplied to the answer model. To render decoding fully deterministic, and thereby guarantee exactly reproducible predictions, we fix `temperature` to 0 and cap `max_tokens` at 10, holding all remaining generation parameters at their default settings.

D.2 OPEN-ENDED EVALUATION

Figure 6 documents the prompts that govern both the answer model and the judge model. To accommodate the free-form character of open-ended responses, we set `temperature` to 0.7 and raise `max_tokens` to 1200, holding every other generation parameter at its default value.

D.2.1 EVALUATION OF LLM-AS-A-JUDGE FOR OPEN-ENDED EVALUATION

To establish the trustworthiness of *Gemini-2.5-Flash* as an automatic judge, we conducted a controlled validation study on a balanced set of 240 samples spanning every category. Each sample constituted a complete evaluation instance, namely the question, its ground truth, and the model output, and was scored independently by three human annotators. To eliminate ambiguity from the annotation protocol, judgments were recorded as binary labels, with 1 denoting a correct response and 0 an incorrect one. We then benchmarked the judge’s decisions against these human annotations through a suite of complementary agreement metrics, computed both in aggregate and at the granularity of individual categories. Specifically, we adopt **Cohen’s Kappa**, **Accuracy with Majority Vote**, and **Fleiss’ Kappa**, each defined precisely below.

Cohen’s Kappa. Cohen’s kappa (κ) quantifies the pairwise agreement between two raters after discounting the concordance expected under chance alone. For a binary task it is defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, in which p_o denotes the observed agreement and p_e the agreement anticipated by chance, $p_e = \sum_{c \in \{0,1\}} p_c^{(1)} p_c^{(2)}$. Here $p_c^{(1)}$ and $p_c^{(2)}$ are the marginal probabilities with which each rater assigns class c . The coefficient admits a directly interpretable scale: $\kappa = 1$ signals perfect agreement, $\kappa = 0$ indicates agreement no better than chance, and $\kappa < 0$ reflects systematic disagreement. We computed Cohen’s kappa between the judge and each annotator individually (κ_{Human1} , κ_{Human2} , κ_{Human3}), and, in parallel, between the judge and the annotators’ majority vote (κ_{majority}).

Accuracy with Majority Vote. As a complementary, uncorrected measure, we additionally quantify the raw agreement between the judge and the annotator majority. Accuracy is defined as $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i^{\text{LLM}} = y_i^{\text{maj}})$, where y_i^{LLM} is the judge’s decision for sample i and y_i^{maj} is the majority vote among the annotators. Unlike Cohen’s kappa, accuracy incorporates no correction for chance; it reports directly the proportion of labels on which the judge and the majority coincide.

Fleiss’ Kappa. To gauge the reliability of the human reference itself, we computed Fleiss’ kappa (κ_F), which generalizes Cohen’s kappa from a pair of raters to an arbitrary number of them. Let n_{ij} denote the number of annotators assigning category j to item i , and define $P_i = \frac{1}{n(n-1)} \sum_j n_{ij}(n_{ij} - 1)$, in which n is the number of annotators. The observed agreement is the mean of these per-item quantities, $\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i$, whereas the chance-expected agreement is $\bar{P}_e = \sum_j p_j^2$, with $p_j = \frac{1}{Nn} \sum_i n_{ij}$. Fleiss’ kappa then follows as $\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$. Larger values attest to stronger inter-annotator reliability beyond what chance alone would yield.

Table 5: Agreement between Gemini-2.5-Flash as an automatic judge and three human annotators, reported both overall and per category. Each row lists Cohen’s kappa against the majority vote and against each annotator individually, raw accuracy relative to the majority, and Fleiss’ kappa quantifying inter-annotator reliability. Bold values mark the strongest agreement attained within each category.

Category	κ_{Majority}	κ_{Human1}	κ_{Human2}	κ_{Human3}	Acc. (majority)	Fleiss κ (all)
Overall	0.738	0.738	0.681	0.795	0.880	0.774
3D Geometry	1.000	1.000	0.875	1.000	1.000	0.916
Depth & Occlusion	0.805	0.805	0.899	0.697	0.909	0.871
Orientation	0.500	0.500	0.426	0.588	0.810	0.571
Relative Positioning	0.833	0.657	0.676	0.833	0.917	0.778
Size & Scale	0.417	0.571	0.417	0.696	0.714	0.704
Spatial Navigation	0.815	0.815	0.667	1.000	0.933	0.773

Interpretation Table 5 establishes that the LLM judge (Gemini-2.5-Flash) agrees strongly with the human annotators, attaining a Cohen’s kappa of 0.738 against the majority vote and spanning 0.681–0.795 against the annotators taken individually. Overall raw accuracy relative to the majority reaches 0.880, and a Fleiss’ kappa of 0.774 among the human annotators attests to substantial consistency in the reference itself. Several categories are especially compelling: 3D Geometry attains perfect agreement, with $\kappa = 1.000$ and an accuracy of 1.000, and Depth & Occlusion likewise records high agreement ($\kappa_{\text{majority}} = 0.805$, accuracy = 0.909). Orientation ($\kappa = 0.500$, accuracy = 0.810) and Size & Scale ($\kappa = 0.417$, accuracy = 0.714), by contrast, exhibit only moderate kappa agreement, yet both remain within acceptable reliability ranges.

Taken together, these results demonstrate that Gemini-2.5-Flash scores correctness on a par with human evaluators: its Cohen’s kappa values fall predominantly within the substantial range (0.61–0.80), and several categories ascend into the almost-perfect range (above 0.81). The Fleiss’ kappa of 0.774 further confirms that the annotators are highly consistent among themselves, which in turn validates the reliability of the reference scores against which the judge is measured. The judge secures perfect or near-perfect agreement in well-specified categories such as 3D Geometry ($\kappa = 1.000$, accuracy = 1.000) and Spatial Navigation ($\kappa = 0.815$, accuracy = 0.933), evidence of dependable recognition of the intended patterns. Where agreement is merely moderate, as in Orientation and Size & Scale, the residual gap is most plausibly attributable to the intrinsic difficulty and subjectivity of those tasks rather than to any deficiency of the judge. On balance, the validation confirms that the LLM-as-judge constitutes a robust and reliable scoring instrument, one that closely mirrors human evaluation and can be deployed with confidence for automated scoring across every category of the benchmark.

E ERROR ANALYSIS OF SPATIALAB-MCQ

E.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models. Closed-source models (Table 6) define the upper envelope of the benchmark, establishing the highest ceilings attained anywhere in SPATIALAB-MCQ. That envelope is anchored by **GPT-4o-mini**, whose 85.71% in *Stacking Orientation* stands as the single highest score recorded across every model and every subcategory. This apex is no aberration: **Gemini-2.0-Flash** exhibits assured command of occlusion and projective inference, posting 64.0% in *Partial Occlusion* and 64.71% in *Shadow-Size Projection*, while **GPT-5-mini** carries the same tendency into embodied and layering judgments with 68.0% in *Tool Handedness* and 55.71% in *Layering Order*. These selective triumphs are nonetheless inseparable from a pronounced fragility, most starkly embodied by **Claude 3.5 Haiku** and its meagre 32.0% in *Relative Size Comparison*. Aggregated across categories, closed-source systems outstrip their open-source counterparts by a decisive 15–25 percentage points; within any single model, however, the dispersion routinely exceeds 50 percentage points. The diagnosis admits no ambiguity: proprietary training pipelines elevate peak attainment yet withhold the uniform spatial competence on which robust reasoning depends.

Open-source Large Models. Scaling to large open-source models (Table 7) yields demonstrable gains in peak attainment while, in the same stroke, laying bare severe representational fragilities. At the summit, **InternVL3.5-72B** secures 80.56% in *Corner/Angle Positioning*—among the strongest results any open-source system attains—and it converges with **llama-3.2-90b-vision-instruct** at 66.67% in *Complete Occlusion Inference*. Set against these advances is an arresting collapse: **llama-3.2-11b-vision-instruct** musters a mere 2.0% in *Object Rotation*. The same signature of localized strength recurs in *Stacking Orientation*, where **llama-3.2-90b-vision-instruct** reaches 71.43%; such peaks are evidently purchased at the expense of neighbouring categories, however, and gravity reasoning is held to a modest 56.0%. With the within-model spread routinely surpassing 70 percentage points, the evidence indicts brittle specialization rather than broad mastery. We conclude that parameter scaling amplifies selective strengths and lifts ceilings while leaving the foundational representational gaps structurally intact.

Open-source Small Models. Open-source small models (Table 8) manifest the sharpest internal contrasts of any family, traversing the full dynamic range of the taxonomy. The divergence is severe: **qwen-2.5-vl-7b-instruct** attains 74.29% in *Stacking Orientation*, whereas **InternVL3.5-1B** collapses to 12.0% in *Cardinal Direction*. Between these extremes, competence thins wherever abstraction

Table 6: Per-subcategory multiple-choice accuracy (%) on SPATIALAB-MCQ for the closed-source models, resolved across all 30 task types. Within each model group the strongest entry is rendered in **bold** and underlined.

Category	Sub-Category	GPT-4o-mini	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	52.0	50.0	46.0	52.0	52.0	46.0
	Shape Projection	54.0	48.0	50.0	50.0	46.0	60.0
	Spatial Containment	36.0	46.0	44.0	38.0	34.0	38.0
	Stability Prediction	52.38	52.38	59.52	50.0	45.24	52.38
	Volume Comparison	41.3	47.83	39.13	34.78	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	61.54	69.23	61.54	61.54	61.54	46.15
	Layering Order	32.86	55.71	50.0	42.86	32.86	45.71
	Partial Occlusion	42.0	54.0	64.0	50.0	44.0	52.0
	Reflective Surfaces	34.0	54.0	46.0	48.0	44.0	44.0
	Transparency Effects	32.0	44.0	60.0	44.0	36.0	62.0
Orientation	Cardinal Direction	26.0	48.0	44.0	30.0	44.0	38.0
	Facing Direction	45.24	57.14	57.14	47.62	35.71	45.24
	Object Rotation	40.0	60.0	50.0	46.0	38.0	44.0
	Stacking Orientation	85.71	77.14	62.86	65.71	68.57	62.86
	Tool Handedness	52.0	68.0	64.0	64.0	56.0	56.0
Relative Positioning	Alignment Patterns	44.19	58.14	55.81	53.49	46.51	58.14
	Betweenness Relationships	51.06	63.83	70.21	63.83	44.68	61.7
	Corner/Angle Positioning	47.22	63.89	50.0	58.33	50.0	77.78
	Directional Relations	48.84	79.07	67.44	69.77	53.49	69.77
	Proximity Gradients	44.19	48.84	44.19	34.88	37.21	44.19
Size and Scale	Distance-Size Correlation	56.0	56.0	42.0	34.0	34.0	42.0
	Perspective Distortion	50.0	40.0	46.0	36.0	38.0	34.0
	Relative Size Comparison	42.0	44.0	58.0	46.0	32.0	38.0
	Scale Consistency	47.06	45.1	47.06	39.22	33.33	41.18
	Shadow-Size Projection	52.94	39.22	64.71	49.02	41.18	52.94
Spatial Navigation	Accessibility Constraints	56.1	60.98	65.85	65.85	48.78	56.1
	Obstacle Avoidance	38.0	48.0	34.0	34.0	46.0	40.0
	Pathway Existence	54.35	41.3	43.48	45.65	39.13	36.96
	Spatial Sequence	66.0	78.0	48.0	62.0	56.0	44.0
	Viewpoint Visibility	36.0	54.0	48.0	50.0	40.0	34.0

Table 7: Per-subcategory multiple-choice accuracy (%) on SPATIALAB-MCQ for the large open-source models, resolved across all 30 task types. Within each model group the strongest entry is rendered in **bold** and underlined.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-72b-instruct	InternVL3-5-72B	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct
3D Geometry	Gravity Effects	22.0	48.0	54.0	52.0	52.0	56.0
	Shape Projection	32.0	42.0	44.0	50.0	46.0	50.0
	Spatial Containment	32.0	34.0	36.0	44.0	46.0	36.0
	Stability Prediction	33.33	54.76	38.1	52.38	47.62	52.38
	Volume Comparison	13.04	39.13	32.61	39.13	43.48	36.96
Depth and Occlusion	Complete Occlusion Inference	30.77	61.54	58.97	66.67	48.72	66.67
	Layering Order	34.29	34.29	37.14	50.0	50.0	38.57
	Partial Occlusion	32.0	34.0	44.0	60.0	48.0	58.0
	Reflective Surfaces	28.0	38.0	26.0	54.0	50.0	44.0
	Transparency Effects	26.0	40.0	40.0	60.0	46.0	62.0
Orientation	Cardinal Direction	22.0	48.0	34.0	46.0	48.0	40.0
	Facing Direction	21.43	42.86	52.38	54.76	50.0	50.0
	Object Rotation	2.0	40.0	38.0	48.0	48.0	44.0
	Stacking Orientation	28.57	65.71	65.71	68.57	62.86	71.43
	Tool Handedness	40.0	48.0	52.0	56.0	56.0	56.0
Relative Positioning	Alignment Patterns	27.91	55.81	48.84	53.49	55.81	53.49
	Betweenness Relationships	42.55	55.32	55.32	65.96	61.7	61.7
	Corner/Angle Positioning	58.33	55.56	36.11	80.56	50.0	75.0
	Directional Relations	60.47	62.79	51.16	83.72	62.79	65.12
	Proximity Gradients	27.91	41.86	32.56	48.84	39.53	41.86
Size and Scale	Distance-Size Correlation	36.0	50.0	54.0	52.0	44.0	46.0
	Perspective Distortion	22.0	42.0	36.0	46.0	34.0	52.0
	Relative Size Comparison	28.0	48.0	40.0	48.0	52.0	46.0
	Scale Consistency	29.41	47.06	41.18	43.14	37.25	37.25
	Shadow-Size Projection	37.25	52.94	54.9	56.86	50.98	52.94
Spatial Navigation	Accessibility Constraints	29.27	58.54	43.9	75.61	53.66	43.9
	Obstacle Avoidance	24.0	38.0	38.0	46.0	42.0	40.0
	Pathway Existence	28.26	54.35	36.96	39.13	54.35	47.83
	Spatial Sequence	34.0	52.0	56.0	66.0	54.0	62.0
	Viewpoint Visibility	44.0	36.0	34.0	50.0	42.0	48.0

is demanded—the best models are tied at a scant 34.78% in *Pathway Existence*, and gravity-related reasoning is capped at 50.0% for **InternVL3.5-4B** and **Gemma-3-4B-it**. A within-model spread that frequently exceeds 60 percentage points is itself diagnostic of brittle generalization: these systems are acutely cue-dependent, flourishing where strong visual templates are available and faltering wherever a task demands deeper spatial abstraction. We read this inconsistency as direct evidence that representation learning proceeds unevenly across the subcategory space. The composite portrait is one of localized competence that never consolidates into robustness across the full spectrum of spatial reasoning.

Reasoning Models. Reasoning-enhanced models (Table 9) confer tangible advantages on select tasks but do not convert this depth into consistent breadth. **o4-mini** outperforms most open baselines with 65.12% in *Alignment Patterns* and 56.0% in *Gravity Effects*, while **Gemini-2.0-Flash-Thinking** evinces sharper geometric reasoning at 62.75% in *Shadow-Size Projection* and 52.17% in *Volume*

Table 8: Per-subcategory multiple-choice accuracy (%) on SPATIALAB-MCQ for the small open-source models, resolved across all 30 task types. Within each model group the strongest entry is rendered in **bold** and underlined.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	46.0	38.0	48.0	50.0	50.0	50.0
	Shape Projection	30.0	42.0	38.0	42.0	40.0	44.0
	Spatial Containment	30.0	28.0	34.0	34.0	36.0	34.0
	Stability Prediction	30.95	35.71	50.0	52.38	45.24	52.38
	Volume Comparison	30.43	26.09	36.96	36.96	47.83	34.78
Depth and Occlusion	Complete Occlusion Inference	30.77	30.77	41.03	64.1	33.33	41.03
	Layering Order	30.0	24.29	30.0	32.86	22.86	27.14
	Partial Occlusion	38.0	30.0	38.0	54.0	46.0	42.0
	Reflective Surfaces	38.0	34.0	36.0	32.0	42.0	40.0
	Transparency Effects	26.0	42.0	36.0	40.0	32.0	44.0
Orientation	Cardinal Direction	12.0	18.0	36.0	24.0	38.0	30.0
	Facing Direction	21.43	28.57	52.38	40.48	42.86	38.1
	Object Rotation	12.0	28.0	42.0	34.0	42.0	34.0
	Stacking Orientation	45.71	57.14	62.86	68.57	68.57	74.29
	Tool Handedness	40.0	36.0	40.0	60.0	48.0	48.0
Relative Positioning	Alignment Patterns	27.91	30.23	39.53	44.19	41.86	41.86
	Betweenness Relationships	38.3	38.3	46.81	65.96	46.81	53.19
	Corner/Angle Positioning	50.0	55.56	47.22	69.44	55.56	50.0
	Directional Relations	51.16	58.14	27.91	60.47	46.51	48.84
	Proximity Gradients	20.93	23.26	39.53	34.88	39.53	37.21
Size and Scale	Distance-Size Correlation	32.0	20.0	40.0	36.0	38.0	36.0
	Perspective Distortion	26.0	40.0	42.0	38.0	44.0	44.0
	Relative Size Comparison	34.0	32.0	46.0	38.0	30.0	42.0
	Scale Consistency	27.45	35.29	41.18	31.37	27.45	41.18
	Shadow-Size Projection	39.22	35.29	66.67	39.22	47.06	47.06
Spatial Navigation	Accessibility Constraints	36.59	29.27	51.22	41.46	39.02	43.9
	Obstacle Avoidance	24.0	26.0	38.0	38.0	38.0	36.0
	Pathway Existence	34.78	30.43	32.61	34.78	34.78	28.26
	Spatial Sequence	24.0	42.0	40.0	56.0	42.0	44.0
	Viewpoint Visibility	36.0	34.0	36.0	40.0	36.0	26.0

Table 9: Per-subcategory multiple-choice accuracy (%) on SPATIALAB-MCQ for the reasoning models, resolved across all 30 task types. Within each model group the strongest entry is rendered in **bold** and underlined.

Category	Sub-Category	o4-mini	Gemini-2.0-Flash-Thinking	Gemini-2.5-Flash-Thinking	Kimi-VL-A3B-Thinking-2506
3D Geometry	Gravity Effects	56.0	38.0	46.0	50.0
	Shape Projection	50.0	36.0	40.0	34.0
	Spatial Containment	56.0	28.0	48.0	42.0
	Stability Prediction	50.0	35.71	50.0	50.0
	Volume Comparison	43.48	52.17	45.65	39.13
Depth and Occlusion	Complete Occlusion Inference	69.23	51.28	71.79	51.28
	Layering Order	57.14	30.0	41.43	35.71
	Partial Occlusion	50.0	42.0	58.0	40.0
	Reflective Surfaces	54.0	40.0	52.0	42.0
	Transparency Effects	64.0	50.0	54.0	42.0
Orientation	Cardinal Direction	46.0	34.0	50.0	28.0
	Facing Direction	50.0	42.86	54.76	45.24
	Object Rotation	56.0	36.0	48.0	30.0
	Stacking Orientation	71.43	57.14	57.14	65.71
	Tool Handedness	56.0	44.0	60.0	44.0
Relative Positioning	Alignment Patterns	65.12	41.86	44.19	34.88
	Betweenness Relationships	65.96	55.32	70.21	63.83
	Corner/Angle Positioning	75.0	55.56	61.11	47.22
	Directional Relations	65.12	44.19	65.12	60.47
	Proximity Gradients	51.16	32.56	41.86	48.84
Size and Scale	Distance-Size Correlation	44.0	42.0	56.0	44.0
	Perspective Distortion	40.0	44.0	50.0	36.0
	Relative Size Comparison	46.0	46.0	58.0	32.0
	Scale Consistency	39.22	56.86	52.94	43.14
	Shadow-Size Projection	35.29	62.75	58.82	43.14
Spatial Navigation	Accessibility Constraints	56.1	41.46	53.66	31.71
	Obstacle Avoidance	32.0	44.0	44.0	36.0
	Pathway Existence	45.65	41.3	47.83	34.78
	Spatial Sequence	64.0	48.0	68.0	54.0
	Viewpoint Visibility	60.0	40.0	54.0	48.0

Comparison. The family’s most commanding showing belongs to **Gemini-2.5-Flash-Thinking**, which records 71.79% in *Complete Occlusion Inference* and 70.21% in *Betweenness Relationships*. Tellingly, that very model cedes ground elsewhere, subsiding to 34.0% in *Cardinal Direction* and 34.0% in *Shape Projection*. This juxtaposition of deep peaks and shallow troughs, with within-model spreads frequently exceeding 35 percentage points, demonstrates that explicit reasoning mechanisms deepen performance within particular subcategories without extending coverage across them. We conclude that reasoning augmentation furnishes localized capacity while leaving the structural gaps in spatial understanding unaddressed: these systems evidence genuine promise, though not yet generalizable competence.

Table 10: Per-subcategory multiple-choice accuracy (%) on SPATIALAB-MCQ for the specialized spatial reasoning models, resolved across all 30 task types. Within each model group the strongest entry is rendered in **bold** and underlined.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	48.0	46.0	32.0
	Shape Projection	38.0	36.0	36.0
	Spatial Containment	40.0	36.0	24.0
	Stability Prediction	50.0	50.0	28.57
	Volume Comparison	36.96	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	38.46	35.9	38.46
	Layering Order	30.0	32.86	24.29
	Partial Occlusion	44.0	46.0	38.0
	Reflective Surfaces	44.0	44.0	38.0
	Transparency Effects	40.0	32.0	42.0
Orientation	Cardinal Direction	40.0	38.0	30.0
	Facing Direction	59.52	54.76	50.0
	Object Rotation	40.0	42.0	28.0
	Stacking Orientation	62.86	62.86	40.0
	Tool Handedness	40.0	40.0	48.0
Relative Positioning	Alignment Patterns	39.53	44.19	39.53
	Betweenness Relationships	44.68	42.55	53.19
	Corner/Angle Positioning	41.67	41.67	38.89
	Directional Relations	25.58	25.58	20.93
	Proximity Gradients	37.21	37.21	34.88
Size and Scale	Distance-Size Correlation	40.0	44.0	48.0
	Perspective Distortion	40.0	38.0	42.0
	Relative Size Comparison	42.0	46.0	48.0
	Scale Consistency	33.33	33.33	49.02
	Shadow-Size Projection	58.82	54.9	66.67
Spatial Navigation	Accessibility Constraints	48.78	46.34	53.66
	Obstacle Avoidance	36.0	32.0	44.0
	Pathway Existence	32.61	34.78	41.3
	Spatial Sequence	48.0	44.0	50.0
	Viewpoint Visibility	32.0	34.0	48.0

Spatial Reasoning Models. Spatially specialized models (Table 10) deliver precisely targeted improvements yet conspicuously want for generality. **SpaceQwen2.5-VL-3B-Instruct** distinguishes itself in size-scale reasoning, clearing many open-source baselines with 48.0% in *Distance-Size Correlation* and 49.02% in *Scale Consistency*, and it further leads the navigation subcategories at 53.66% in *Accessibility Constraints* and 44.0% in *Obstacle Avoidance*. In keeping with their domain-specific mandate, **SpaceOm** and **SpaceThinker-Qwen2.5VL-3B** each reach 50.0% in *Stability Prediction*. These gains, however, refuse to generalize: performance falls sharply on adjacent tasks such as *Spatial Containment*, where accuracy is only 24.0%, and even the strongest areas seldom exceed 55.0%, the subcategory spread routinely surpassing 30 percentage points. The evidence indicates that domain targeting sharpens narrow peaks but stops short of holistic competence; specialization amplifies isolated abilities without integrating them into the robust, general framework that spatial reasoning in the wild demands.

Overall Discussion. Viewed across the full taxonomy and every model family (Table 8, Table 7, Table 6, Table 10, Table 9), the benchmark yields one unmistakable and pervasive signature: fragmentation. Small open-source models peak at 74.29% in *Stacking Orientation* only to collapse to 12.0% in *Cardinal Direction*; larger models raise the ceiling to 80.56% in *Corner/Angle Positioning* yet still fail catastrophically at 2.0% in *Object Rotation*. Closed-source systems establish the benchmark’s high-water mark, **GPT-4o-mini** reaching 85.71% in *Stacking Orientation*, and nonetheless subside to 32.0% in *Relative Size Comparison*. Specialized models translate their domain focus into navigation (53.66%) and stability (50.0%) strengths while remaining capped below 55% overall, and reasoning models post gains such as 71.79% in *Complete Occlusion Inference* even as other subcategories languish at 34.0%. No approach escapes the pattern: within-model spreads routinely exceed 50 percentage points, and the subcategories that demand physical abstraction—*Gravity Effects*, *Stability Prediction*, and *Spatial Containment*—score below 35–40% regardless of family. We trace this fragility to two root causes: the absence of stable internal encodings of orientation and physics, and an entrenched over-reliance on surface-level correlations. The sheer variance, spanning 85.71% highs and 2.0% lows, betrays pattern-matching masquerading as robust spatial reasoning. Scaling, propri-

Table 11: Item-difficulty distribution on SPATIALAB-MCQ, tabulating how many questions were solved by a given number of models (0–5), broken down by category and sub-category. Here “0” indicates that no model answered the item correctly, whereas “5” indicates that only five of the models tested succeeded on it; lower bins thus isolate the questions that are hardest across the model pool.

Category	Sub-Category	Number of Models Successfully Answered						
		0	1	2	3	4	5	Total
3D Geometry	Gravity Effects	9	7	3	2	3	1	25
	Shape Projection	4	6	0	9	5	7	31
	Spatial Containment	6	9	6	3	6	1	31
	Stability Prediction	5	6	4	2	2	3	22
	Volume Comparison	6	5	6	5	4	2	28
3D Geometry Total		30	33	19	21	20	14	137
Depth & Occlusion	Complete Occlusion Inference	0	4	0	6	3	4	17
	Layering Order	9	8	7	8	6	5	43
	Partial Occlusion	2	2	5	7	5	2	23
	Reflective Surfaces	6	5	5	7	4	1	28
	Transparency Effects	4	6	3	2	5	7	27
Depth & Occlusion Total		21	25	20	30	23	19	138
Orientation	Cardinal Direction	1	7	10	7	3	4	32
	Facing Direction	2	3	8	5	3	1	22
	Object Rotation	4	6	7	3	5	4	29
	Stacking Orientation	2	3	0	2	1	1	9
	Tool Handedness	2	1	4	3	1	1	12
Orientation Total		11	20	29	20	13	11	104
Relative Positioning	Alignment Patterns	3	4	4	4	4	1	20
	Betweenness Relationships	3	1	2	2	5	7	20
	Corner/Angle Positioning	0	1	3	2	3	3	12
	Directional Relations	1	4	1	2	3	2	13
	Proximity Gradients	4	6	6	5	2	3	26
Relative Positioning Total		11	16	16	15	17	16	91
Size & Scale	Distance-Size Correlation	6	5	6	1	1	7	26
	Perspective Distortion	8	7	3	3	6	4	31
	Relative Size Comparison	5	5	7	4	4	4	29
	Scale Consistency	7	6	7	4	5	2	31
	Shadow-Size Projection	5	7	2	1	1	4	20
Size & Scale Total		31	30	25	13	17	21	137
Spatial Navigation	Accessibility Constraints	3	2	1	5	4	3	18
	Obstacle Avoidance	2	6	9	6	10	2	35
	Pathway Existence	2	10	4	1	4	3	24
	Spatial Sequence	3	2	3	2	7	8	25
	Viewpoint Visibility	2	2	5	9	8	5	31
Spatial Navigation Total		12	22	22	23	33	21	133
Grand Total		116	146	131	122	123	102	740

etary training, and reasoning augmentation each confer real but strictly partial benefits, producing isolated peaks that coexist with glaring blind spots rather than any semblance of general coverage. The precision of this diagnosis is itself the contribution: SPATIALAB-MCQ localizes competence to specific axes and pinpoints exactly where it fails. Progress toward holistic spatial intelligence will accordingly demand architectural change directed at embedding consistent geometric grounding and physics-aware mechanisms; our results document at once the remarkable selective gains and the persistence of fundamental, systematically identifiable gaps.

E.2 MODEL FAILURE ANALYSIS

The per-item distribution of correct answers across categories and subcategories in Table 11 sharpens the diagnostic picture, isolating where competence concentrates and where it dissolves. 3D Geometry and Size & Scale present comparatively balanced distributions, yet both retain a sizeable fraction of items that no model answered correctly, an indication that even elementary geometric invariances are not reliably internalized. Depth & Occlusion, by contrast, carries a heavier mid-range tail (2–4 correct)—the signature of partial but inconsistent reasoning about layered scenes. Orientation emerges as the weakest category overall, its especially low totals in Stacking Orientation and Tool Handedness betraying that subtle directional and embodied affordances go largely uncaptured. Relative Positioning proves more heterogeneous still: tasks such as Proximity Gradients attract moderate success, whereas Corner/Angle Positioning accumulates near-zero correct counts, exposing acute brittleness in compositional reasoning over reference frames. Spatial Navigation registers comparatively stronger performance, notably in Obstacle Avoidance and Viewpoint Visibility, yet its persistent errors in Pathway Existence reveal that relational chaining remains a bottleneck. In aggregate, no category attains uniformly high accuracy, and the Grand Total disperses evenly across the 0–5 bins, confirming that success is acutely task-dependent rather than the expression of general spatial competence. The consolidated evidence establishes that current models command isolated strengths while lacking systematic reliability across the spatial dimensions the benchmark probes.

These distributions resolve into a small set of identifiable root causes, each anchored to a specific representational or reasoning bottleneck. The elevated error rates in Stacking Orientation and Tool Handedness point to shallow grounding in physics-based and embodied reasoning, a deficiency consistent with training on static internet imagery in place of interactive or task-driven data. The difficulty in Corner/Angle Positioning and Pathway Existence isolates a distinct failure mode—inadequate multi-step relational chaining—that curtails the models’ capacity for recursive spatial logic. The inconsistency in Transparency Effects and Reflective Surfaces implicates a third mechanism: the handling of visual phenomena that hinge on non-local cues such as secondary reflections or occluded geometry. Against these deficits, the comparatively stronger showing in Obstacle Avoidance is best construed not as genuine planning ability but as a learned shortcut that exploits dataset biases, notably the frequent co-occurrence of walkable paths. Taken together, these failure modes converge on a single characterization of current training regimes: they overweight correlation-driven vision–language alignment while systematically underrepresenting embodied, counterfactual, and physics-based reasoning. Remedying them will plausibly require both richer training distributions—embodied simulation data and physics-augmented learning among them—and architectural innovations that explicitly model reference frames and relational constraints. Absent such advances, models will persist in the brittle competence this analysis pinpoints, one that predictably fractures under compositional or physically grounded demands.

F ERROR ANALYSIS OF SPATIALAB-OPEN

F.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models (Table 12). The open-ended results establish closed-source systems as the strongest family overall, yet the analysis makes equally plain that this advantage is apportioned unevenly across the taxonomy. GPT-5-mini emerges as the decisive leader, consistently commanding the best scores in **3D Geometry** and **Relative Positioning** and peaking at 58.14% in Directional Relations and 55.81% in Alignment Patterns. The same family, however, lays bare pronounced brittleness elsewhere: Gemini-2.5-Flash falls to 16.28% in Proximity Gradients and Claude 3.5 Haiku to 12.0% in Perspective Distortion. This breadth of outcome—authoritative on discrete relational and geometric judgments, fragile on subtle, continuous visual cues—demonstrates that even frontier proprietary models lack uniform spatial competence. We attribute the disparity to the differential sophistication of these models’ spatial reasoning pipelines, in which certain architectures encode relational and geometric structure far more faithfully than others. Proximity Gradients are diagnostic precisely because they demand analogical, graded inference rather than discrete classification, and GPT-5-mini’s capacity to prevail where its peers collapse indicates that scaling and proprietary training corpora confer measurable, if partial, advantages. Crucially, the leader’s own vulnerabilities establish that closed-source dominance is relative rather than absolute: the family still falls well short of robust, human-like spatial reasoning.

Table 12: Per-subcategory open-ended accuracy (%) on SPATIALAB-OPEN for the closed-source models, resolved across all 30 task types. Within each model group the strongest entry is rendered in **bold** and underlined.

Category	Sub-Category	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	48.0	36.0	30.0	24.0	20.0
	Shape Projection	40.0	18.0	22.0	16.0	16.0
	Spatial Containment	50.0	38.0	50.0	36.0	36.0
	Stability Prediction	50.0	26.19	35.71	30.95	26.19
	Volume Comparison	39.13	41.3	32.61	23.91	28.26
Depth and Occlusion	Complete Occlusion Inference	46.15	35.9	38.46	28.21	25.64
	Layering Order	30.0	22.86	21.43	17.14	15.71
	Partial Occlusion	38.0	20.0	28.0	14.0	14.0
	Reflective Surfaces	32.0	28.0	32.0	22.0	22.0
	Transparency Effects	32.0	18.0	18.0	16.0	22.0
Orientation	Cardinal Direction	34.0	40.0	34.0	26.0	30.0
	Facing Direction	28.57	28.57	26.19	19.05	14.29
	Object Rotation	42.0	14.0	28.0	30.0	14.0
	Stacking Orientation	37.14	20.0	31.43	22.86	28.57
	Tool Handedness	48.0	36.0	44.0	24.0	24.0
Relative Positioning	Alignment Patterns	55.81	34.88	37.21	37.21	34.88
	Betweenness Relationships	48.94	23.4	31.91	23.4	21.28
	Corner/Angle Positioning	44.44	30.56	41.67	16.67	36.11
	Directional Relations	58.14	46.51	58.14	34.88	37.21
	Proximity Gradients	39.53	20.93	25.58	16.28	18.6
Size and Scale	Distance-Size Correlation	48.0	32.0	36.0	20.0	22.0
	Perspective Distortion	48.0	24.0	22.0	20.0	12.0
	Relative Size Comparison	36.0	26.0	28.0	18.0	16.0
	Scale Consistency	43.14	21.57	23.53	19.61	9.8
	Shadow-Size Projection	37.25	27.45	23.53	23.53	15.69
Spatial Navigation	Accessibility Constraints	41.46	31.71	39.02	24.39	21.95
	Obstacle Avoidance	40.0	22.0	22.0	26.0	14.0
	Pathway Existence	23.91	15.22	21.74	10.87	19.57
	Spatial Sequence	52.0	32.0	34.0	24.0	20.0
	Viewpoint Visibility	28.0	22.0	32.0	20.0	10.0

Open-source Large Models (Table 13). Among large open-source models the analysis discloses a markedly more distributed landscape, one in which distinct systems dominate isolated sub-categories rather than any single model leading throughout. llama-3.2-90b-vision-instruct attains 60.0% in Depth and Occlusion, while GLM-4.5V-106B-MoE leads Spatial Containment at 38.0%. These localized peaks notwithstanding, several systems collapse on Proximity Gradients—qwen-2.5-vl-32b-instruct and qwen-2.5-vl-72b-instruct each registering only 9.3%—which establishes that architectural scaling broadens coverage without ever guaranteeing it across demanding continuous-reasoning tasks. We attribute much of this variation to heterogeneous training data: models exposed to richer depth and geometry corpora fare better on occlusion and containment, whereas others generalize poorly. The persistent deficit on Proximity Gradients indicates that these systems are simply not optimized for subtle spatial analogies and distance scaling. The group likewise exhibits higher inter-model variance than the closed-source family, a signature of open-source development heterogeneity. In sum, large open-source models present encouraging peaks alongside inconsistent coverage, confirming that scale alone is insufficient absent targeted pre-training and fine-tuning.

Open-source Small Models (Table 14). Small open-source models predictably trail their larger counterparts, yet the analysis surfaces noteworthy localized strengths that temper any blanket dismissal. qwen-2.5-vl-3b-instruct achieves 18.0% in Perspective Distortion and 26.0% in Relative Size Comparison, while Gemma-3-4B-it leads Spatial Containment (24.0%) and Stability Prediction (21.43%). These pockets of competence are offset by wholesale categorical breakdowns: InternVL3.5-2B scores 0.0% in Betweenness Relationships, and InternVL3.5-1B manages a mere 2.0% in Shape Projection. We attribute this extreme variability to limited capacity coupled with restricted training diversity, which jointly prevent compact models from acquiring robust representations of geometric and relational principles. Sub-categories that demand precise reasoning over orientation and hierarchy precipitate complete breakdowns, as minor misinterpretations propagate into total failure. That these models nonetheless occasionally overtake larger systems on narrow tasks suggests that even small-scale training can instill isolated competencies—evidence for the value of specialization, but in equal measure a confirmation of the structural ceiling imposed by compact architectures. In short, small open-source models exhibit pockets of promise embedded within pervasive brittleness, their overall accuracy bounded by size and data constraints.

Table 13: Per-subcategory open-ended accuracy (%) on SPATIALAB-OPEN for the large open-source models, resolved across all 30 task types. Within each model group the strongest entry is rendered in **bold** and underlined.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct	GLM-4.5V-106B-MoE
3D Geometry	Gravity Effects	14.0	22.0	12.0	30.0	24.0	28.0
	Shape Projection	10.0	20.0	10.0	20.0	16.0	20.0
	Spatial Containment	14.0	30.0	18.0	22.0	20.0	38.0
	Stability Prediction	23.81	30.95	16.67	21.43	26.19	38.1
	Volume Comparison	23.91	10.87	26.09	41.3	28.26	32.61
Depth and Occlusion	Complete Occlusion Inference	25.64	20.51	12.82	25.64	30.77	33.33
	Layering Order	8.57	15.71	10.0	14.29	15.71	10.0
	Partial Occlusion	12.0	16.0	8.0	14.0	20.0	16.0
	Reflective Surfaces	26.0	16.0	8.0	34.0	32.0	24.0
	Transparency Effects	18.0	14.0	10.0	20.0	22.0	26.0
Orientation	Cardinal Direction	32.0	34.0	20.0	38.0	28.0	30.0
	Facing Direction	11.9	26.19	14.29	21.43	11.9	16.67
	Object Rotation	26.0	16.0	10.0	18.0	28.0	20.0
	Stacking Orientation	17.14	25.71	17.14	25.71	5.71	37.14
	Tool Handedness	20.0	20.0	12.0	20.0	32.0	24.0
Relative Positioning	Alignment Patterns	23.26	46.51	20.93	46.51	30.23	30.23
	Betweenness Relationships	34.04	31.91	17.02	25.53	38.3	29.79
	Corner/Angle Positioning	27.78	36.11	11.11	27.78	27.78	16.67
	Directional Relations	30.23	39.53	25.58	34.88	25.58	32.56
	Proximity Gradients	9.3	18.6	9.3	18.6	18.6	20.93
Size and Scale	Distance-Size Correlation	20.0	26.0	12.0	26.0	30.0	26.0
	Perspective Distortion	10.0	18.0	10.0	18.0	12.0	20.0
	Relative Size Comparison	8.0	26.0	14.0	32.0	22.0	24.0
	Scale Consistency	11.76	21.57	3.92	19.61	21.57	23.53
	Shadow-Size Projection	17.65	21.57	9.8	27.45	23.53	27.45
Spatial Navigation	Accessibility Constraints	21.95	26.83	26.83	34.15	39.02	24.39
	Obstacle Avoidance	12.0	18.0	12.0	20.0	30.0	22.0
	Pathway Existence	19.57	21.74	6.52	17.39	15.22	17.39
	Spatial Sequence	22.0	28.0	12.0	22.0	28.0	34.0
	Viewpoint Visibility	18.0	16.0	12.0	12.0	24.0	24.0

Table 14: Per-subcategory open-ended accuracy (%) on SPATIALAB-OPEN for the small open-source models, resolved across all 30 task types. Within each model group the strongest entry is rendered in **bold** and underlined.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	4.0	6.0	14.0	24.0	16.0	22.0
	Shape Projection	2.0	8.0	10.0	4.0	10.0	0.0
	Spatial Containment	6.0	4.0	12.0	20.0	24.0	8.0
	Stability Prediction	2.38	11.9	14.29	19.05	21.43	19.05
	Volume Comparison	15.22	32.61	28.26	30.43	30.43	28.26
Depth and Occlusion	Complete Occlusion Inference	2.56	5.13	10.26	17.95	20.51	23.08
	Layering Order	5.71	7.14	10.0	14.29	11.43	14.29
	Partial Occlusion	14.0	8.0	12.0	16.0	4.0	10.0
	Reflective Surfaces	16.0	16.0	6.0	14.0	18.0	26.0
	Transparency Effects	10.0	20.0	4.0	28.0	14.0	8.0
Orientation	Cardinal Direction	12.0	16.0	16.0	26.0	20.0	24.0
	Facing Direction	9.52	4.76	28.57	14.29	7.14	19.05
	Object Rotation	4.0	10.0	12.0	8.0	14.0	16.0
	Stacking Orientation	17.14	11.43	8.57	17.14	20.0	28.57
	Tool Handedness	8.0	12.0	8.0	12.0	12.0	12.0
Relative Positioning	Alignment Patterns	13.95	27.91	32.56	11.63	41.86	34.88
	Betweenness Relationships	12.77	17.02	0.0	19.15	17.02	25.53
	Corner/Angle Positioning	22.22	30.56	2.78	22.22	19.44	22.22
	Directional Relations	11.63	25.58	9.3	34.88	23.26	37.21
	Proximity Gradients	9.3	18.6	9.3	11.63	16.28	18.6
Size and Scale	Distance-Size Correlation	6.0	10.0	14.0	16.0	20.0	12.0
	Perspective Distortion	8.0	12.0	18.0	16.0	12.0	16.0
	Relative Size Comparison	10.0	16.0	26.0	24.0	10.0	16.0
	Scale Consistency	7.84	9.8	11.76	11.76	7.84	11.76
	Shadow-Size Projection	13.73	11.76	21.57	13.73	25.49	23.53
Spatial Navigation	Accessibility Constraints	4.88	26.83	7.32	17.07	19.51	19.51
	Obstacle Avoidance	16.0	16.0	12.0	18.0	28.0	26.0
	Pathway Existence	0.0	8.7	2.17	8.7	17.39	8.7
	Spatial Sequence	12.0	22.0	8.0	26.0	20.0	28.0
	Viewpoint Visibility	16.0	18.0	16.0	24.0	14.0	16.0

Reasoning Models (Table 15). Reasoning-oriented models exhibit a sharper internal divide, with Gemini-2.5-flash-thinking standing decisively above its peers. It records the best scores in nearly every sub-category, including Tool Handedness at 75.0% and Complete Occlusion Inference at 57.14%. Kimi-VL-A3B-Thinking-2506, by contrast, struggles dramatically, scoring just 6.0% in Spatial Containment and trailing across the board. We trace this gap to divergent strategies for integrating multi-modal streams into coherent reasoning frameworks: Gemini-2.5-flash-thinking appears able to construct and query internal world models, enabling more effective inference over object interactions and physical properties, whereas weaker systems retreat to surface-level pattern matching inadequate for compositional or causal reasoning. The consistently high performance of Gemini-2.5 across categories establishes that explicit reasoning mechanisms confer tangible benefits for spatial understanding. Its failure to dominate Shape Projection and Betweenness Relationships, however, demonstrates that even advanced reasoning modules are not yet universally effective. Reasoning models therefore hold genuine promise, though the category itself remains heterogeneous and far from solved.

Table 15: Per-subcategory open-ended accuracy (%) on SPATIALAB-OPEN for the reasoning models, resolved across all 30 task types. Within each model group the strongest entry is rendered in **bold** and underlined.

Category	Sub-Category	gemini-2.0-flash-thinking	gemini-2.5-flash-thinking	Kimi-VL-A3B-Thinking-2506	Step3
3D Geometry	Gravity Effects	32.0	40.0	14.0	38.0
	Shape Projection	24.0	0.0	10.0	30.0
	Spatial Containment	36.0	50.0	6.0	30.0
	Stability Prediction	26.19	42.86	14.29	33.33
	Volume Comparison	36.96	44.44	23.91	30.43
Depth and Occlusion	Complete Occlusion Inference	33.33	57.14	17.95	33.33
	Layering Order	20.0	20.0	14.29	14.29
	Partial Occlusion	28.0	50.0	8.0	14.0
	Reflective Surfaces	34.0	63.64	10.0	16.0
	Transparency Effects	26.0	57.14	22.0	20.0
Orientation	Cardinal Direction	40.0	20.0	18.0	26.0
	Facing Direction	35.71	37.5	14.29	28.57
	Object Rotation	18.0	40.0	4.0	22.0
	Stacking Orientation	22.86	0.0	5.71	34.29
	Tool Handedness	44.0	75.0	20.0	32.0
Relative Positioning	Alignment Patterns	46.51	66.67	25.58	41.86
	Betweenness Relationships	27.66	20.0	12.77	25.53
	Corner/Angle Positioning	36.11	50.0	11.11	11.11
	Directional Relations	39.53	44.44	27.91	37.21
	Proximity Gradients	23.26	16.67	11.63	18.6
Size and Scale	Distance-Size Correlation	32.0	57.14	14.0	22.0
	Perspective Distortion	18.0	11.11	10.0	30.0
	Relative Size Comparison	28.0	11.11	14.0	36.0
	Scale Consistency	33.33	20.0	11.76	31.37
	Shadow-Size Projection	35.29	18.18	13.73	29.41
Spatial Navigation	Accessibility Constraints	31.71	45.45	17.07	41.46
	Obstacle Avoidance	26.0	10.0	16.0	18.0
	Pathway Existence	23.91	25.0	13.04	19.57
	Spatial Sequence	34.0	12.5	30.0	36.0
	Viewpoint Visibility	32.0	12.5	16.0	26.0

Spatial Reasoning Models (Table 16). Spatial reasoning models, though purpose-built for the task, record the lowest results overall. Their performance is acutely fragmented: several models tie at 16.0% in Gravity Effects, 8.0% in Shape Projection, and 14.29% in Stability Prediction. SpaceQwen2.5-VL-3B-Instruct stands marginally apart with 23.91% in Volume Comparison and 13.73% in Scale Consistency, yet even these figures remain far below competitive thresholds. At the opposite extreme, Betweenness Relationships and Corner/Angle Positioning collapse outright, recording 0.0% and 2.78%, respectively. Such outcomes indicate that the models’ smaller capacity and narrowly focused training severely constrict their representational flexibility. We read these failures as direct evidence that narrow specialization, absent sufficient capacity or data breadth, cannot yield generalizable spatial reasoning: on geometric and positional tasks in particular, even slight representational errors cascade into total breakdown. Consequently, despite their explicit focus, spatial reasoning models prove the least effective family, which underscores that scale and diversity alike are prerequisites for robust performance.

Overall Discussion. Viewed across the full taxonomy and every model family—closed-source (Table 12), open-source large (Table 13), open-source small (Table 14), reasoning (Table 15), and spatial reasoning (Table 16)—the evaluation returns a single, consistent verdict: spatial competence in the wild is fragmented and systemically incomplete. Closed-source models lead overall, GPT-5-mini reaching 58.14% in Directional Relations and 55.81% in Alignment Patterns, yet even they collapse on Proximity Gradients. Large open-source models post impressive peaks such as 60.0% in Depth and Occlusion but fall to just 9.3% in Proximity Gradients, demonstrating that scale without specialization remains insufficient. Small open-source models reinforce the identical conclusion, pairing rare localized strengths (e.g., 26.0% in Relative Size Comparison) with catastrophic lows such as 0.0% in Betweenness Relationships. Reasoning models show that explicit reasoning mechanisms can materially improve results—Gemini-2.5-flash-thinking reaching 75.0% in Tool Handedness—even as peers like Kimi-VL-A3B descend to single digits, exposing the fragility of less-integrated reasoning frameworks. Spatial reasoning models, their explicit specialization notwithstanding, underperform the most, with scores frequently below 15% and outright failures on Betweenness Relationships. Taken together, these patterns isolate three root causes: (1) the absence of robust representations for continuous and analogical spatial cues, (2) structural limitations tied to scale and training diversity, and (3) brittle reasoning pipelines that foreclose generalization across tasks. We conclude decisively

Table 16: Per-subcategory open-ended accuracy (%) on SPATIALAB-OPEN for the specialized spatial reasoning models, resolved across all 30 task types. Within each model group the strongest entry is rendered in **bold** and underlined.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	14.0	12.0	16.0
	Shape Projection	8.0	8.0	6.0
	Spatial Containment	8.0	10.0	10.0
	Stability Prediction	14.29	14.29	7.14
	Volume Comparison	19.57	23.91	23.91
Depth and Occlusion	Complete Occlusion Inference	7.69	10.26	7.69
	Layering Order	8.57	11.43	4.29
	Partial Occlusion	10.0	12.0	4.0
	Reflective Surfaces	4.0	4.0	2.0
	Transparency Effects	4.0	8.0	2.0
Orientation	Cardinal Direction	18.0	24.0	8.0
	Facing Direction	23.81	23.81	16.67
	Object Rotation	20.0	20.0	18.0
	Stacking Orientation	8.57	8.57	17.14
	Tool Handedness	0.0	4.0	8.0
Relative Positioning	Alignment Patterns	32.56	32.56	23.26
	Betweenness Relationships	4.26	2.13	0.0
	Corner/Angle Positioning	2.78	2.78	2.78
	Directional Relations	11.63	6.98	11.63
	Proximity Gradients	6.98	6.98	9.3
Size and Scale	Distance-Size Correlation	18.0	18.0	16.0
	Perspective Distortion	16.0	20.0	2.0
	Relative Size Comparison	22.0	20.0	8.0
	Scale Consistency	9.8	11.76	13.73
	Shadow-Size Projection	27.45	27.45	19.61
Spatial Navigation	Accessibility Constraints	7.32	7.32	9.76
	Obstacle Avoidance	10.0	8.0	14.0
	Pathway Existence	6.52	6.52	8.7
	Spatial Sequence	16.0	12.0	16.0
	Viewpoint Visibility	20.0	16.0	8.0

that no current approach—proprietary scaling, open-source expansion, reasoning integration, or task specialization—delivers comprehensive coverage. The recurrence of performance gaps exceeding 50 percentage points within a single model confirms that progress remains piecemeal. Future work must therefore unify robust geometric encodings, physics-informed reasoning, and cross-task transfer, advancing beyond isolated gains toward consistent spatial intelligence.

F.2 MODEL FAILURE ANALYSIS

The distribution in Table 17 lays bare persistent, category-spanning weaknesses, with a pronounced skew toward the lower bins (0–2 models correct). 3D Geometry proves broadly forbidding: Shape Projection (19 items with zero correct) and Stability Prediction (10 items with zero correct) yield the poorest outcomes, signalling that models lack reliable physical and geometric reasoning. Depth and Occlusion is the hardest category overall, amassing the highest zero-correct count (72), concentrated in Layering Order (20 zero) and Transparency Effects (16 zero)—both of which demand nuanced handling of occlusion and non-local cues. Orientation exhibits comparable difficulty: although Cardinal Direction attains some mid-range success, Stacking Orientation (11 zero) and Tool Handedness (5 zero, none with perfect success) underscore the brittleness of embodied directional reasoning. Relative Positioning reveals pervasive inconsistency, with Betweenness Relationships and Proximity Gradients seldom solved by more than two or three models and Corner/Angle Positioning showing low but comparatively balanced performance. Size and Scale stands relatively stronger: several subcategories (e.g., Scale Consistency with 6 full successes) attest that some models internalize coarse perspective and scaling cues, even as failures remain high in Shadow-Size Projection (17 zero). Spatial Navigation is bifurcated—Spatial Sequence and Obstacle Avoidance register occasional full-model success, whereas Pathway Existence (16 zero) and Accessibility Constraints (10 zero) expose enduring difficulty with multi-step relational chaining. Aggregating across 356 items, the grand total corroborates the diagnosis: zero-correct dominates (356), and only 74 items are solved by all five models, which establishes that consistent spatial competence remains elusive.

Table 17: Item-difficulty distribution on SPATIALAB-OPEN, tabulating how many questions were solved by a given number of models (0–5), broken down by category and sub-category. Here “0” indicates that no model answered the item correctly, whereas “5” indicates that only five of the models tested succeeded on it; lower bins thus isolate the questions that are hardest across the model pool.

Category	Sub-Category	Number of Models Successfully Answered					
		0	1	2	3	4	5
3D Geometry	Gravity Effects	13	7	3	4	1	2
	Shape Projection	19	5	6	2	5	1
	Spatial Containment	10	4	3	6	2	3
	Stability Prediction	10	3	7	3	0	2
	Volume Comparison	11	3	4	3	4	2
3D Geometry Total		63	22	23	18	12	10
Depth & Occlusion	Complete Occlusion Inference	7	5	4	2	5	2
	Layering Order	20	14	10	6	3	3
	Partial Occlusion	14	8	5	5	4	2
	Reflective Surfaces	15	4	1	5	2	5
	Transparency Effects	16	7	7	3	1	2
Depth & Occlusion Total		72	38	27	21	15	14
Orientation	Cardinal Direction	8	10	0	1	5	2
	Facing Direction	13	5	5	1	3	2
	Object Rotation	12	10	6	4	1	2
	Stacking Orientation	11	5	1	2	1	1
	Tool Handedness	5	4	1	4	2	0
Orientation Total		49	34	13	12	12	7
Relative Positioning	Alignment Patterns	7	5	3	1	1	1
	Betweenness Relationships	11	5	7	2	1	3
	Corner/Angle Positioning	7	6	2	1	3	2
	Directional Relations	5	4	4	4	2	0
	Proximity Gradients	10	10	3	3	2	4
Relative Positioning Total		40	30	19	11	9	10
Size & Scale	Distance-Size Correlation	7	5	9	4	1	5
	Perspective Distortion	12	6	6	8	5	2
	Relative Size Comparison	14	7	4	2	2	4
	Scale Consistency	14	7	5	3	3	6
	Shadow-Size Projection	17	4	4	4	3	2
Size & Scale Total		64	29	28	21	14	19
Spatial Navigation	Accessibility Constraints	10	5	4	0	1	2
	Obstacle Avoidance	15	5	8	2	2	5
	Pathway Existence	16	6	7	3	1	1
	Spatial Sequence	14	3	3	3	3	5
	Viewpoint Visibility	13	3	9	5	3	1
Spatial Navigation Total		68	22	31	13	10	14
Grand Total		356	175	141	96	72	74

These findings locate the root causes of failure not in dataset design but in the representational and reasoning limitations of current models. First, the concentration of errors in Depth and Occlusion reflects a reliance on superficial appearance features in preference to structured scene representations: models fasten onto surface textures instead of reasoning about hidden layers, producing systematic errors in Layering Order. Second, failures in Orientation tasks such as Tool Handedness expose the absence of embodied affordance modeling, since pretrained internet imagery seldom carries explicit annotations of left/right handedness or stacking physics. Third, the brittleness observed in Relative Positioning and Spatial Navigation points to deficiencies in chaining multi-hop spatial relations—Pathway Existence, for instance, requires integrating local connectivity with global layout, which attention-based co-occurrence priors capture poorly. By contrast, the partial success in Size and Scale indicates that models exploit statistical regularities in perspective distortion and scale consistency

Table 18: **MCQ-to-open-ended accuracy gap (%)** ($\downarrow$ lower is better) across spatial categories on SPATIALAB, defined as the multiple-choice score minus the open-ended score. A value near zero indicates consistent performance across both answer formats, whereas a larger value marks pronounced degradation once the multiple-choice scaffold is withdrawn; negative values identify the atypical regime in which open-ended generation is the more reliable probe. Within each model group, the largest gap is rendered in **bold** and underlined.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Proprietary Models</i>							
GPT-4o-mini	22.40	23.76	16.51	31.74	27.85	20.50	23.53
GPT-5-mini	20.08	23.27	13.21	2.38	19.41	13.36	3.36
Gemini-2.0-Flash	30.89	26.73	26.89	28.18	22.37	25.07	15.13
Gemini-2.5-Flash	21.62	16.34	17.45	15.87	21.51	17.36	10.93
Claude 3.5 Haiku	23.16	21.78	20.29	15.47	24.89	20.29	16.39
Mistral Medium 3.1	30.50	25.74	32.54	26.59	24.89	26.93	21.43
<i>Open-Source Models (Small)</i>							
InternVL3.5-1B	22.78	13.37	23.58	22.62	20.67	22.00	27.73
InternVL3.5-2B	20.46	20.79	16.99	20.64	14.35	19.21	21.85
Qwen-VL2.5-3B-Instruct	27.03	30.69	29.24	28.97	29.96	28.50	25.63
InternVL3.5-4B	25.10	26.24	34.91	20.24	23.20	25.29	23.53
Gemma-3-4B-it	21.23	31.68	22.17	22.22	18.14	22.93	23.53
Qwen-VL2.5-7B-Instruct	22.01	22.27	18.40	26.19	15.61	22.14	27.73
Llama-3.2-11B-Vision-Instruct	13.51	-1.98	17.92	17.07	13.50	11.93	9.66
Gemma-3-27B-it	23.93	23.27	19.82	25.40	25.32	23.14	20.59
Qwen-VL2.5-32B-Instruct	30.50	31.68	28.30	35.32	28.27	29.85	24.79
InternVL3.5-72B	36.68	33.17	34.44	29.37	28.69	31.57	27.31
Qwen-VL2.5-72B-Instruct	27.80	26.73	23.59	19.05	28.27	24.22	20.17
Llama-3.2-90B-Vision-Instruct	28.95	29.21	30.66	25.00	21.52	26.36	23.53
<i>Reasoning Models</i>							
o4-mini	25.48	22.77	21.23	-3.18	17.30	15.35	10.50
Gemini-2-Flash-Thinking	13.90	10.39	11.32	21.03	13.50	13.00	6.73
Gemini-2.5-Flash-Thinking	8.22	16.61	19.46	33.42	31.37	20.16	8.66
Kimi-VL-A3B-Thinking-2506	27.02	28.71	33.50	26.98	22.78	27.92	29.41
<i>Spatial Reasoning Models</i>							
SpaceOm	31.66	32.18	25.95	24.21	27.00	28.43	29.83
SpaceThinker-Qwen2.5VL-3B	28.57	29.21	27.83	23.81	27.84	27.28	26.89
SpaceQwen2.5-VL-3B-Instruct	31.28	23.76	28.31	38.89	35.87	29.78	18.90

yet buckle under less frequent cases such as shadows. Collectively, these patterns crystallize into a coherent diagnosis: current models succeed when cues are salient and correlate with training priors but fail the moment reasoning demands counterfactual, compositional, or embodied inference. Redressing these limitations will require architectural advances that explicitly model reference frames, occlusion layers, and physical dynamics, complemented by training data enriched with embodied simulations and counterfactual augmentations. Absent such advances, progress in spatial reasoning will remain fragmented—assured on surface-level cues yet breaking down on structured, real-world spatial logic.

G ANALYSIS OF MCQ VS. OPEN-ENDED PERFORMANCE GAPS

Through a systematic decomposition of the divergence between the multiple-choice (MCQ) and open-ended evaluation formats, this section demonstrates that MCQ accuracy, taken in isolation, materially overstates the practical spatial competence of vision-language models. The decomposition ranges over every spatial reasoning category in SPATIALAB—reported in full in Table 18—and issues in an unequivocal methodological conclusion: faithful assessment requires reporting the two formats jointly and pairing them with per-subtask diagnostics. This dual-format, subtask-resolved protocol constitutes the central methodological contribution of the present analysis.

G.1 QUANTITATIVE SNAPSHOT

The disparity is large, pervasive, and remarkably stable across the entire evaluation suite. Averaged over 25 models, the MCQ-to-open-ended gap reaches **23.0%** against a standard deviation of merely 5.5%; the effect is therefore neither an artifact of a handful of outliers nor localized to particular capabilities, but rather a systematic property of the two formats. Its uniformity across subtasks is equally striking: the dataset-wide means—24.57 for 3D geometry, 23.45 for depth & occlusion, 23.70

for orientation, 23.11 for relative positioning, 23.42 for size & scale, and 22.89 for spatial navigation—converge tightly on an overall average of 23.01. Individual models span a considerable range, from a low of 11.93 to a high of 31.57 at a median near 23.34, and yet the central tendency remains conspicuously consistent. Partitioning by model family sharpens the picture. Spatial reasoning specialists incur the widest average gap at 27.03, whereas reasoning-oriented models prove the most format-robust at 19.11; proprietary systems (21.09) outperform the broader open-source field (24.51) on average, and even the human baseline registers a comparable 22.64. The extremes are equally instructive: the largest gaps accrue to InternVL3.5-72B (31.57%), Qwen-VL2.5-32B (29.85%), and SpaceQwen2.5-VL-3B-Instruct (29.78%), while the smallest belong to Llama-3.2-11B (11.93%), Gemini-2-Flash-Thinking (13.00%), and GPT-5-mini (13.36). Correlational analysis pinpoints where the aggregate disparity concentrates: spatial navigation is its dominant driver (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) likewise tracking the overall gap closely.

G.2 PATTERNS AND OBSERVATIONS

Four regularities organize these results. First, specialist spatial reasoning models exhibit the widest MCQ→open-ended gaps, a pattern consistent with optimization for classification-style supervision at the expense of robustness under free-form generation; reasoning-focused models, by contrast, remain comparatively narrow, reflecting the stabilizing influence of chain-of-thought-style training on generative outputs. Second, spatial navigation stands out as the single strongest predictor of overall disparity: the models that falter on open-ended navigation reasoning are precisely those that inflate the aggregate gap, with orientation and 3D geometry contributing substantially while relative position and size/scale vary more idiosyncratically across models. Third, robustness is decidedly not a function of scale: the compact Llama-3.2-11B ranks among the most format-stable models, whereas the far larger InternVL3.5-72B posts the widest gap of all. Fourth, a small number of negative or near-zero gaps—depth & occlusion for Llama-3.2-11B at -1.98 , relative position for o4-mini at -3.18 —identify cases in which open-ended evaluation is the more reliable probe, an inversion we attribute to misleading distractors that depress MCQ scores.

G.3 HYPOTHESES ON ROOT CAUSES

We hypothesize that several interacting factors jointly underlie the MCQ→open-ended performance gaps observed in SPATIALAB. Each hypothesis below is grounded in the statistical evidence; we emphasize, nonetheless, that these remain proposed explanations rather than established causes:

1. **MCQ structural advantage.** By confining the output to a closed set of options, the MCQ format permits a model to exploit surface cues or eliminate distractors rather than construct an answer de novo. The consequence is systematic: mean MCQ scores exceed open-ended scores across the board, at an average gap of 23.0% and $\sigma = 5.5\%$. The exceptions are no less telling—the negative gaps for Llama-3.2-11B in depth & occlusion (-1.98) and o4-mini in relative position (-3.18) demonstrate that open-ended generation can surpass MCQ when distractors mislead, reinforcing the conclusion that the closed format can inflate or otherwise distort measured performance.

Recommendation: We recommend complementing MCQ evaluations with open-ended tasks and subtask-level diagnostics, which capture model competence more faithfully and curtail the overestimation induced by option-based cues.

2. **Specialization bias in spatial reasoning models.** That the models engineered expressly for spatial tasks post the largest average gaps (27.03%) is a revealing paradox: we hypothesize that their training relies disproportionately on categorical or synthetic selection objectives rather than free-form generation. InternVL3.5-72B exemplifies the pattern acutely—this 72B-parameter spatial reasoning model records the widest overall gap (31.57%), propelled by spatial navigation (36.68%) and orientation (34.44%), the very signature of classification-like supervision. The concentration of these format-dependent failures in sequential and geometric reasoning is corroborated by the correlations between subtask and overall gaps (SpNav: $r = 0.99$, Orientation: $r = 0.83$).

Recommendation: To counteract this bias, we recommend fine-tuning or instruction-tuning on open-ended spatial generation tasks, which strengthens robustness to generative formats and attenuates reliance on categorical supervision.

3. **Instruction-tuning and stepwise decoding.** At the opposite pole, reasoning-oriented models such as Gemini-2-Flash-Thinking and o4-mini combine smaller average gaps (around 19.11%) with lower cross-subtask variance. Gemini-2-Flash-Thinking is emblematic, confining its overall gap to 13.0% at uniformly modest subtask figures (Difference in 3D Geometry: 13.9%, Difference Relative Positioning: 21.03%); we interpret this as evidence that chain-of-thought or stepwise decoding elicits more stable open-ended responses. The weaker coupling between subtask-gap variance and overall gap within this family points in the same direction, indicating that instruction-tuned reasoning dampens format sensitivity. *Recommendation:* To reduce format sensitivity, we recommend adopting chain-of-thought or stepwise decoding strategies together with instruction-tuning for open-ended spatial tasks, which improve generative consistency and diminish MCQ dependence.
4. **Spatio-navigation stresses sequential grounding.** Spatial navigation gaps predict the overall MCQ→open-ended disparity almost perfectly (Pearson $r = 0.99$), rendering them the clearest window onto the underlying failure mode. Models with large navigation gaps carry commensurately large overall gaps—SpaceQwen2.5-VL-3B-Instruct pairs a Spatial Navigation gap of 35.87% with a 29.78% overall gap—which we read as evidence that multi-step referential reasoning is disproportionately fragile in open-ended settings. The plausible mechanism is that MCQ options collapse the reasoning chain, masking deficits in sequential grounding and relational computation that surface only under free generation. *Recommendation:* To better capture sequential reasoning ability, we recommend incorporating dedicated open-ended spatio-navigation tasks alongside targeted multi-step reasoning supervision, which sharpens grounding and curbs the overestimation introduced by MCQ.
5. **MCQ distractor artifacts.** The negative and near-zero gaps expose a second-order effect: distractor design can distort MCQ measurement in either direction. o4-mini registers a negative relative position gap (−3.18%) while remaining positive elsewhere, a profile that implicates poorly chosen distractors in artificially depressing its MCQ accuracy. The converse failure is equally plausible—Qwen-VL2.5-32B-Instruct’s pronounced gaps (overall = 29.85%) across multiple subtasks indicate that easy or leading distractors can inflate perceived competence in selection-based tasks. In either regime, the option set rather than the underlying spatial ability becomes the confound. *Recommendation:* We recommend systematic auditing and refinement of MCQ distractors, complemented by open-ended evaluation, to yield more accurate and less biased measures of model spatial reasoning ability.

G.4 PRACTICAL IMPLICATIONS

Taken together, these results deliver an unambiguous methodological mandate: MCQ and open-ended results must be reported jointly. Reliance on MCQ alone inflates estimates of practical spatial competence by roughly 23% on average—a bias far too large to leave unaccounted for. Subtask-level breakdowns are indispensable, since spatial navigation disproportionately drives the observed differences and would otherwise be obscured within aggregate scores. Diagnostic evaluations that separate classification from generative reasoning are the natural instrument for isolating these error modes. Beyond measurement, instruction-tuning and decoding strategies that promote stepwise reasoning emerge as promising levers for reducing MCQ dependence, and principled design and auditing of distractors is a prerequisite for fair cross-format comparison. In sum, the conjunction of dual-format reporting and per-subtask diagnostics is not an optional refinement but a necessary condition for trustworthy spatial-reasoning evaluation.

H DETAILS ON IMPROVING VISUAL REASONING CAPABILITIES

To determine how far spatial reasoning can be advanced on our diverse visual question benchmark, we mount a systematic investigation of four complementary intervention strategies: (i) exploiting the models’ inherent reasoning mechanisms, (ii) Chain-of-Thought (CoT) prompting together with its self-reflective extension, (iii) supervised fine-tuning (SFT), and (iv) a purpose-built multi-agent system. Considered jointly, these experiments map with precision where targeted interventions genuinely advance spatial competence and where the benchmark’s difficulty proves obstinately resilient.

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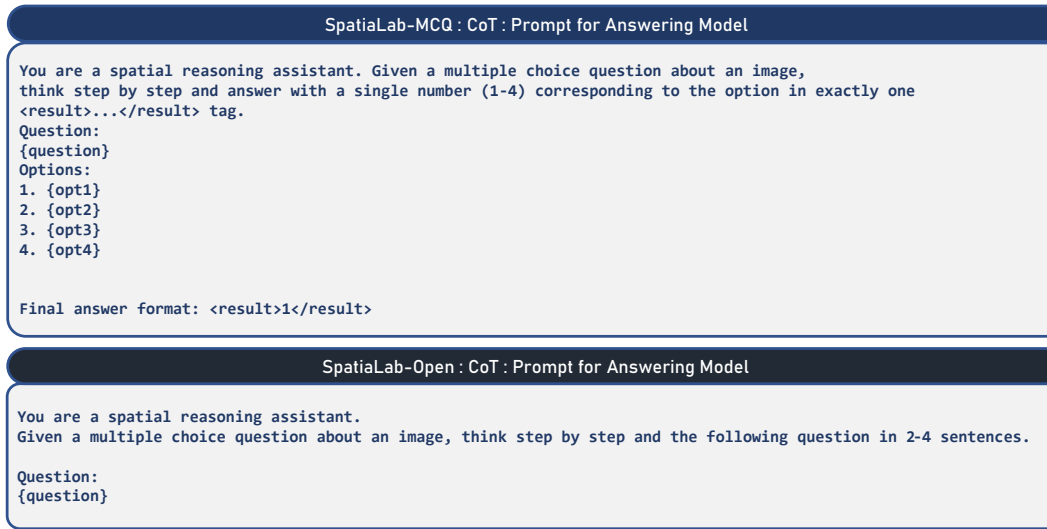


Figure 7: Chain-of-thought (CoT) Prompts.

H.1 INHERENT REASONING MECHANISMS

We begin by isolating the contribution of models’ built-in reasoning mechanisms, contrasting standard configurations against their reasoning-enabled counterparts (Table 2, Table 3). In the multiple-choice regime, activating reasoning confers a decisive and reproducible advantage, though its magnitude is unmistakably category-dependent. Gemini-2.5-Flash-Thinking, for instance, outstrips its non-reasoning variant by +4.64% overall (52.93% vs. 48.29%), propelled by pronounced gains in Relative Positioning (+13.1%) and Orientation (+8.6%). The identical trajectory recurs for o4-mini-medium, which attains the strongest overall MCQ accuracy recorded in our study (53.21%). Taken together, these figures establish that explicit reasoning scaffolds are at their most potent in structured, discrete-answer settings, where a bounded hypothesis space rewards deliberate elimination.

Open-ended evaluation yields a more textured picture. Reasoning-equipped models again lead their baselines, yet the margins are conspicuously uneven. Gemini-2.5-Flash-Thinking edges past Gemini-2.5-Flash (32.77% vs. 30.93%), while the most emphatic advance belongs to o4-mini-medium (37.86%), which comfortably eclipses every non-reasoning model. This evidence indicates that reasoning mechanisms attenuate the instability endemic to free-form generation, most visibly in demanding categories such as 3D Geometry and Spatial Navigation. The benefit is far from uniform, however: reasoning variants occasionally regress outright, as when Gemini-2.5-Flash-Thinking falls sharply in Size & Scale (21.74% vs. 26.59%)—a reversal that reprises our broader finding that generative settings amplify brittleness.

Read jointly, these comparisons crystallize a clear dual finding: explicit reasoning modules substantially elevate structured MCQ performance and can stabilize open-ended responses, even as their returns fluctuate across categories. Reasoning demonstrably sharpens logical consistency and relational chaining, and yet it does not, on its own, dissolve the harder obstacles of scale sensitivity, occlusion, and free-form linguistic grounding, which endure as the benchmark’s most stubborn frontiers.

H.2 CHAIN-OF-THOUGHT (CoT) PROMPTING

To adjudicate whether explicitly elicited step-by-step reasoning improves spatial performance, we evaluated models under Chain-of-Thought (CoT) prompting. From each of the 30 sub-categories in SPATIALAB we drew 5 items (seed fixed to 42), assembling a balanced test set that spans all major spatial phenomena while keeping evaluation computationally tractable. The step-by-step reasoning scaffolds supplied to the models are illustrated in Figure 7. Because model scale and training are held constant across conditions, this protocol cleanly isolates the marginal contribution attributable to CoT reasoning itself.

Table 19: Effect of chain-of-thought (CoT) prompting on per-category accuracy for InternVL-3-78b and Gemini-2.5-Flash, reported for both the MCQ and open-ended formats. All values are accuracies (%); the Gain column is the signed difference between the with-CoT and without-CoT conditions, so that positive values indicate an improvement attributable to CoT prompting and negative values a regression.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain
3D Geometry	52.00	52.00	0.00	28.00	28.00	0.00	48.00	20.00	-28.00	60.00	28.00	-32.00
Depth & Occlusion	60.00	60.00	0.00	16.00	12.00	-4.00	48.00	28.00	-20.00	40.00	16.00	-24.00
Orientation	60.00	64.00	4.00	24.00	28.00	4.00	56.00	24.00	-32.00	0.00	32.00	32.00
Relative Positioning	80.00	72.00	-8.00	44.00	56.00	12.00	60.00	36.00	-24.00	50.00	32.00	-18.00
Size & Scale	56.00	44.00	-12.00	28.00	20.00	-8.00	48.00	60.00	12.00	25.00	8.00	-17.00
Spatial Navigation	64.00	64.00	0.00	48.00	40.00	-8.00	68.00	52.00	-16.00	40.00	28.00	-12.00
Overall	62.00	59.33	-2.67	31.33	30.67	-0.66	54.67	36.67	-18.00	38.24	24.00	-14.24

Our analysis (Table 19) delivers a sharply delimited verdict: Orientation is the sole category in which CoT prompting reliably lifts performance, while the remaining categories stagnate or decline. The pattern is diagnostic. Reasoning pays dividends precisely when a task reduces to the logical alignment of directions, but it falters wherever richer perceptual priors—depth, scale, relative positioning—are prerequisite. We trace this asymmetry to a representational gap: current models do not encode 3D perception and geospatial relationships as intrinsic knowledge, and are therefore not “aware” of these concepts in the human sense. Under such conditions, chaining reasoning steps does nothing to repair the underlying misconceptions; it entrenches them, propagating and compounding errors across the deliberation.

This result underscores that SPATIALAB is genuinely non-trivial: it cannot be dispatched by merely bolting multi-step reasoning onto existing models. Where textual reasoning tasks reward CoT almost unconditionally, our benchmark lays bare deeper architectural limitations in perceptual and spatial grounding. In categories such as depth and size, faulty priors guarantee that each additional reasoning step carries the model further from the correct solution. The pronounced degradations we observe—most acute for Gemini in the MCQ setting—demonstrate just how fragile reasoning becomes once it is deprived of robust perceptual knowledge.

This resistance to CoT is, in fact, a deliberate strength of the benchmark’s design: rather than rewarding surface-level reasoning, SPATIALAB interrogates whether models genuinely understand geospatial and 3D relational concepts. Solving it demands far more than the chaining of logical steps; it calls for embeddings and training objectives that internalize core aspects of perception and spatial cognition. In exactly this manner, the benchmark stresses models at a fundamental level and pinpoints a consequential gap in contemporary architectures.

H.3 CHAIN-OF-THOUGHT (CoT) PROMPTING WITH SELF-REFLECTION

We extend the CoT setup with an explicit self-reflection stage, in which the model is prompted to review and revise its initial reasoning before committing to a final answer (see Figure 8). The evaluation protocol is held identical to the standard CoT setting—5 balanced samples per sub-category (seed 42)—so that the effect of self-reflection is cleanly isolated and rendered directly commensurable with baseline CoT performance.

Augmenting Chain-of-Thought (CoT) prompting with self-reflection yields outcomes that are strongly model-dependent, and the contrast between models is itself instructive. For InternVL-3-78b, overall accuracy drifts downward (−2.66% MCQ, −8.67% open-ended), with steep losses in depth, relative positioning, and navigation—signaling that the reflective pass more often entrenches errors than corrects them. Localized gains do surface, notably +12% in Size & Scale (MCQ), yet they remain inconsistent and are swamped by large degradations elsewhere (−28% in relative-positioning open-ended). Gemini-2.5-Flash, by contrast, posts robust MCQ gains (+20.66%), anchored by sizable improvements in 3D Geometry (+32%) and Depth & Occlusion (+24%), while its open-ended improvement stays negligible (+0.67%). Read jointly, these results expose a decisive asymmetry: reflection aids multiple-choice settings, where models can prune implausible options, but fails to transfer to free-form generation, where residual uncertainty bleeds into ungrounded rationales. Just as tellingly, the categories reflection most improves are those governed by geometric consistency (orientation, depth, size), whereas tasks demanding relational integration across multiple objects

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Figure 8: Chain-of-thought (CoT) with Self-Reflection Prompts.

Table 20: Chain-of-thought (CoT) versus CoT augmented with Self-Reflection (CoT-SR) on per-category accuracy for InternVL-3-78b and Gemini-2.5-Flash, reported for both the MCQ and open-ended formats. All values are accuracies (%); the Gain column is the signed difference between the CoT-SR and CoT conditions, so that positive values indicate that adding self-reflection helps and negative values that it hurts.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain
3D Geometry	52.00	56.00	4.00	28.00	20.00	-8.00	20.00	52.00	32.00	28.00	24.00	-4.00
Depth & Occlusion	60.00	44.00	-16.00	12.00	16.00	4.00	28.00	52.00	24.00	16.00	16.00	0.00
Orientation	64.00	64.00	0.00	28.00	32.00	4.00	24.00	48.00	24.00	32.00	36.00	4.00
Relative Positioning	72.00	64.00	-8.00	56.00	28.00	-28.00	36.00	72.00	36.00	32.00	32.00	0.00
Size & Scale	44.00	56.00	12.00	20.00	12.00	-8.00	60.00	56.00	-4.00	8.00	12.00	4.00
Spatial Navigation	64.00	56.00	-8.00	40.00	24.00	-16.00	52.00	64.00	12.00	28.00	28.00	0.00
Overall	59.33	56.67	-2.66	30.67	22.00	-8.67	36.67	57.33	20.66	24.00	24.67	0.67

(relative positioning, navigation) degrade. Reflection thus functions as a stabilizer within structured choice settings but does not reliably fortify naturalistic reasoning.

This inconsistency mirrors our earlier observations with plain CoT: reasoning enhancements succeed only when the model commands stable perceptual anchors on which to ground its steps. Self-reflection introduces an explicit verification pass, yet that mechanism can be no stronger than the internal priors it interrogates. When depth cues, occlusion ordering, or coordinate frames are poorly encoded, the second pass magnifies misconceptions rather than repairing them—which is precisely why Gemini,

with its firmer perceptual grounding, benefits under MCQ conditions while InternVL-3-78b degrades more frequently. The deeper cause is a pretraining mismatch: current VLMs inherit reasoning templates from language modeling yet lack the embedded 3D and geospatial representations that humans routinely exploit. Absent these anchors, reflection cannot perform genuine error correction; it operates instead as a linguistic filter that burnishes the syntax of reasoning without enriching its perceptual substrate. Reflective loops therefore afford modest utility in discrete-answer formats but collapse under open-ended demands, where no factual verification is available to adjudicate the revised rationale.

These findings reaffirm that SPATIALAB exposes a fundamental gap in current architectures: surface-level reasoning layers alone cannot substitute for absent spatial cognition, and meaningful progress will demand the deeper integration of geometric and perceptual structure. The disparity is compounded by the data distribution itself: because VLMs are predominantly trained on web-scale corpora, the concepts richly represented online are exactly those on which models excel, while under-represented spatial and 3D concepts languish—entrenching the very performance gaps our benchmark quantifies.

H.4 SUPERVISED FINE-TUNING (SFT)

H.4.1 SETUP AND IMPLEMENTATION

We adapted the Qwen 2.5 VL 3B instruction-following model (unsloth/Qwen2.5-VL-3B-Instruct) via a parameter-efficient fine-tuning procedure grounded in LoRA (Hu et al., 2021). To remain within a constrained GPU budget, we quantized the base model to 4-bit precision (`load_in_4bit=True`) and injected low-rank adapters into both the attention and the feed-forward projection layers, leaving the frozen backbone entirely intact. The adapter configuration—rank, scaling factor, dropout, and target modules—is enumerated in Table 21. Gradient checkpointing further curtailed memory consumption, while a fixed random seed guaranteed the reproducibility of adapter initialization.

Tokenization conformed to the Qwen-2.5 chat template, ensuring that the formatting encountered during training precisely matched that used at inference. Each example was serialized into a single text sequence, and dynamic padding was delegated to `DataCollatorForSeq2Seq`. We trained with an effective batch size of 4, realized through a per-device batch size of 1 coupled with gradient accumulation over 4 steps. The optimization schedule—learning rate, optimizer, weight decay, and scheduler—is specified in Table 22. The procedure unfolded as four sequential passes, each spanning a single epoch, so that one checkpoint was persisted per epoch for a total of four.

In aggregate, this configuration enabled efficient fine-tuning of a large vision-language model under limited GPU resources while preserving substantial adaptation capacity.

Table 21: LoRA Hyperparameters for SFT

Hyperparameter	Value	Notes
LoRA rank (r)	16	Low-rank update capacity
LoRA alpha	16	Scaling factor
LoRA dropout	0	No dropout applied
Gradient checkpointing	Enabled	Library-specific flag
LoRA random seed	3407	Adapter initialization

H.4.2 SUPERVISED FINE-TUNING (SFT) PERFORMANCE

The learning trajectories in Figure 4a expose pronounced divergences between the MCQ and Open-ended evaluation modes across training epochs. MCQ accuracy departs from a lower baseline of 25.63% yet climbs consistently, converging near 36.59% by the fourth epoch. This steady ascent indicates that the model benefits strongly from additional supervised fine-tuning, particularly on tasks that confine answers to a limited candidate space. Open-ended evaluation, by contrast, starts higher at 34.40% but plunges sharply by the second epoch, bottoming out as low as 12.62%. Only in the later epochs does it partially recover, reaching 35.48% at the end. This instability betrays a

Table 22: Trainer and Optimization Hyperparameters for SFT

Hyperparameter	Value
Base model	unsloth/Qwen2.5-VL-3B-Instruct
Max sequence length	2048
Load in 4-bit	True
Per-device train batch size	1
Gradient accumulation steps	4
Effective batch size	4
Learning rate	2e-4
Optimizer	paged_adamw_8bit
Weight decay	0.01
LR scheduler	linear
Logging steps	1
Seed	42
Tokenizer template	Qwen-2.5

sensitivity in open-ended generation that is absent from the MCQ setting, where the structure of multiple-choice questions furnishes more stable optimization signals. We read the fluctuations in open-ended performance as evidence that fine-tuning may overfit to MCQ-like reasoning patterns at the expense of free-form language generalization. The dotted baselines in the plot further accentuate how the MCQ mode secures gains well beyond its initial standing, whereas Open-ended remains approximately stagnant. On balance, the figure demonstrates that SFT decisively favors MCQ tasks in both stability and consistent accuracy gains.

The tabulated results in Table 4b complement the figure with a granular before-and-after breakdown across six spatial reasoning categories. On MCQ tasks, performance improved in every category, with gains spanning a modest 1.59% in Relative Positioning to a substantial 17.11% in Size and Scale. The Grand Total gain for MCQ reaches 10.97%, confirming that the fine-tuning procedure effectively strengthens structured-answer performance. Open-ended results, by contrast, are far more mixed, interleaving positive and negative changes. Categories such as 3D Geometry and Spatial Navigation register small but positive gains (3.50% and 2.82%, respectively), yet Size and Scale suffers a marked drop of -3.95% . Relative Positioning likewise declines by -0.79% , evidence that fine-tuning can inadvertently degrade certain reasoning modes in free-form contexts. The overall Open-ended gain is a mere 1.07%, statistically negligible beside the MCQ improvements. These results confirm that the efficacy of SFT is acutely sensitive to the evaluation paradigm. By isolating the categories, we pinpoint where training helps, where it fails, and how its benefits diverge between constrained and unconstrained outputs.

A deeper inspection surfaces striking disparities across categories and formats. For MCQ, Size and Scale and Spatial Navigation deliver the highest improvements, suggesting that models trained with multiple-choice prompts are especially adept at learning tasks demanding scale discrimination and navigational reasoning. In Open-ended form, however, Size and Scale actually declines, revealing that the same learned representations transfer poorly to generative contexts. Relative Positioning likewise barely improves in MCQ and worsens in Open-ended, exposing a persistent weakness in relational reasoning irrespective of format. Conversely, Orientation posts strong MCQ gains ($+13.33\%$) alongside small Open-ended improvements, indicating that certain categories generalize more robustly. We conjecture that the divergence between the two formats stems from the structural scaffolding MCQs provide, which relieves the model of the burden of generating precise linguistic formulations. Open-ended tasks, by contrast, demand simultaneous mastery of reasoning and natural language fluency, which amplifies latent weaknesses. The trends make plain that while fine-tuning markedly enhances pattern recognition in discrete-choice settings, it injects volatility and even degradation once free expression is required. This duality clarifies that training interventions optimized for MCQs need not generalize seamlessly to unconstrained reasoning.

The implications of these findings are twofold. First, we establish that SFT as currently applied confers strong benefits for structured tasks but limited or even negative transfer to open-ended reasoning. This limitation is not merely an artifact of fine-tuning; it is symptomatic of how contemporary

VLMs are pretrained. Existing pretraining pipelines lavish emphasis on broad image–text alignment, object recognition, and captioning, yet furnish scant explicit supervision on compositional spatial structures, occlusion layers, or geometric transformations. Consequently, models learn to excel at surface-level associations while failing to acquire robust representations of depth, orientation, or scale. The instability we observe in open-ended spatial reasoning suggests that the model’s learned priors are fragile: stripped of multiple-choice scaffolding, they regress to shallow pattern-matching in place of grounded inference. Categories such as Size and Scale and Relative Positioning, where declines are most pronounced, betray missing inductive biases in pretraining—biases that humans develop through embodied interaction with 3D environments. A further contributing factor is the reliance on synthetic or instruction-following datasets that flatten spatial diversity into puzzle-like abstractions, foreclosing generalization to real-world clutter, noise, and complexity. These findings underscore the imperative to rethink both pretraining and fine-tuning for spatial reasoning, incorporating datasets and objectives that mirror naturalistic visual–spatial challenges. Ultimately, we contend that absent targeted reforms, current VLMs risk entrenching an algorithmic straightjacket: they appear competent under constrained evaluations yet remain brittle and misaligned with the embodied, flexible reasoning that humans bring to real-world spatial understanding.

H.4.3 SFT DYNAMICS ANALYSIS

To interrogate the stability of Supervised Fine-Tuning (SFT) on spatial reasoning tasks, we conducted a multi-seed validation study ($N = 5$, seeds A–E) using the Qwen-2.5-VL-3B-Instruct model. The training trajectory, summarized in Table 23, reveals a stark divergence in performance dynamics between the Multiple-Choice Question (MCQ) and Open-Ended evaluation formats.

Divergent Learning Trajectories. As illustrated in Table 23 and Figure 9, performance on SPATIALAB-MCQ exhibits monotonic improvement, ascending from a baseline of 25.63% to an average of 35.74% at Epoch 4. This confirms that the model readily adapts to the discriminative nature of the task, learning to map visual features onto constrained output options with mounting reliability.

In sharp contrast, SPATIALAB-OPEN traces a pronounced *U-shaped* trajectory across all seeds. Far from steady improvement, the model sustains an immediate and severe performance collapse in the early epochs (plunging from 34.40% to an average of 19.26% at Epoch 1). Although performance rebounds in later epochs (converging to $\approx 34.82\%$ at Epoch 4), the final gain over the zero-shot baseline is negligible ($+0.42\%$).

Table 23: Multi-seed validation of SFT dynamics ($N = 5$). While MCQ performance improves steadily, Open-Ended performance exhibits a systematic “U-shaped” collapse and recovery, indicating representational instability in generative spatial tasks.

Epoch	Type	Seed A	Seed B	Seed C	Seed D	Seed E
0 (Base)	MCQ	25.63%	25.63%	25.63%	25.63%	25.63%
	Open	34.40%	34.40%	34.40%	34.40%	34.40%
1	MCQ	29.56%	35.83%	24.13%	27.63%	26.53%
	Open	22.50%	23.20%	12.50%	14.00%	24.10%
2	MCQ	34.33%	37.76%	27.33%	31.83%	34.43%
	Open	12.62%	25.30%	28.80%	23.80%	16.65%
3	MCQ	34.68%	37.33%	32.86%	32.23%	34.73%
	Open	28.69%	22.22%	30.34%	32.90%	28.30%
4	MCQ	36.59%	38.48%	34.43%	34.36%	34.83%
	Open	35.48%	34.59%	33.92%	34.70%	35.40%

Representational Brittleness and Alignment Tax. We ascribe this phenomenon to the fragility of spatial representations in current VLMs when subjected to fine-tuning. The initial collapse in open-ended accuracy is consistent with *catastrophic forgetting* of the linguistic priors required for fluent, grounded generation. As the model optimizes for the specific instruction-following format of

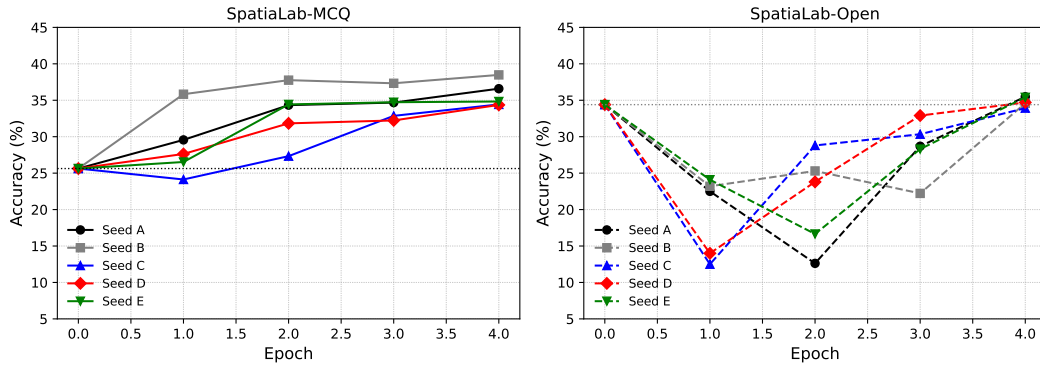


Figure 9: SFT Dynamics Analysis.

the training data, it overfits to the structural logic of the tasks (benefiting MCQ) at the expense of the flexible, generative spatial grounding demanded by open-ended descriptions.

The ensuing recovery phase reflects the model re-learning to articulate spatial concepts within the new distribution, yet it fails to surpass the initial baseline in any meaningful degree. This dynamic crystallizes a critical finding: SFT is effective for teaching models to *select* correct spatial answers, but it does not fundamentally enrich the underlying *generative spatial representation* required for robust open-ended reasoning.

H.4.4 PERFORMANCE GAIN ON EXTERNAL BENCHMARKS

To determine whether the representations learned from SPATIALAB generalize beyond the benchmark’s own format, we assessed the transferability of our fine-tuned model to three external spatial reasoning datasets: OMNISPATIAL, SPACE (Multimodal), and MIND THE GAP. We took the Qwen2.5-VL-3B-Instruct model, fine-tuned on SPATIALAB via the SFT protocol detailed in Appendix H.4.1, and evaluated it on these external benchmarks in a strictly zero-shot setting.

Table 24: Transfer learning performance of Qwen2.5-VL-3B-Instruct on external benchmarks after fine-tuning on SPATIALAB. The consistent gains indicate that the model learns generalizable spatial representations rather than merely overfitting to the source dataset’s format.

Benchmark	Base Accuracy	After SFT	Improvement
OmniSpatial	40.30%	47.35%	+7.05%
SPACE (Multimodal)	23.43%	28.67%	+5.24%
Mind The Gap	35.86%	47.56%	+11.70%

As Table 24 shows, fine-tuning on SPATIALAB yields substantial and uniformly positive performance improvements across all three evaluated benchmarks, ranging from +5.24% to +11.70%. The largest gain arises on MIND THE GAP (+11.70%), indicating that our benchmark’s emphasis on occlusion and spatial continuity directly redresses the reasoning gaps that dataset was built to probe. In the same vein, the improvements on OMNISPATIAL (+7.05%) and SPACE (+5.24%) confirm that the geometric and relational concepts encoded in our taxonomy align squarely with broader definitions of spatial intelligence in the literature. These results establish the quality of the SPATIALAB data. Although the primary purpose of this work is to serve as a rigorous diagnostic benchmark, the consistent transfer across OMNISPATIAL, SPACE, and MIND THE GAP demonstrates decisively that the dataset does not merely invite overfitting to a specific question template. Rather, the diversity of its 30 sub-categories compels the model to acquire robust, transferable spatial representations that generalize to unseen distributions and task formats.

H.5 AI AGENTS FOR SPATIAL REASONING

We developed SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), to perform structured spatial reasoning on images within the SPATIOLAB evaluation. The system decomposes complex visual reasoning into specialized sub-tasks, each delegated to a dedicated agent, thereby enabling focused processing of objects, attributes, relations, symmetries, and transformations. This modular design advances both accuracy and interpretability while sustaining multi-step reasoning, allowing the system to propagate low-level perceptual cues upward through higher-order relational and transformational inferences. The architecture, agent implementation, and their coordinated interaction are elaborated in the sections that follow, with the full complement of prompts provided in Figures 10, 11, and 12.

H.5.1 AGENTS

Base Visual Analysis Agent. The Base Visual Language Model (VLM) agent initiates the pipeline by generating a detailed textual description of the input image, capturing object types—including geometric shapes, natural elements, and artificial icons—together with attributes such as size, orientation, color, and shading. Relative positions such as leftmost, center, and top are encoded explicitly, and the prompt insists on exhaustive detail without summarization. Images are converted to data URIs and dispatched with the prompt to a Gemini-2.5-based model, whose output is parsed into a structured bullet-list via regex-based extraction. This structured description constitutes the foundation for all downstream agents, furnishing a comprehensive perceptual anchor that encodes both explicit and implicit spatial cues. The approach ensures that even subtle visual features—partial occlusions or minor shape variations—are preserved for subsequent relational and transformational reasoning.

Object Segmentation Agent. The Object Segmentation agent converts the verbose textual description into a discrete list of uniquely identified objects, disambiguating visually similar instances. Each object is assigned a stable identifier (e.g., Obj1, Obj2) to preserve consistent references across the ensuing reasoning steps. The agent issues carefully formatted prompts and invokes the same Gemini-2.5 model to structure the description, while post-processing splits lines by object ID to enforce consistency. This structured representation curtails ambiguity in downstream attribute and relation extraction, enabling precise mapping of complex scenes. By explicitly isolating objects, the agent further facilitates the detection of overlaps, repetitions, and hierarchical groupings that may remain latent in raw descriptions.

Attribute Extraction Agent. The Attribute Extraction agent translates each object into a structured JSON record capturing core properties such as shape, size, color, shading, orientation, and positional descriptors. Prompts enforce strict JSON output, and responses are parsed using `json.loads()`, with fallback defaults that safeguard robustness against model inconsistencies. This structured encoding permits both quantitative and qualitative comparisons of objects, underpinning the later stages of spatial reasoning. By retaining both absolute and relative measures, the agent maintains fidelity to the original perceptual content while normalizing descriptors for computational consistency. The resulting JSON structure enables programmatic downstream operations, such as computing symmetries or detecting transformations.

Spatial Relation Agent. The Spatial Relation agent encodes object interactions as triples (ObjectA,Relation,ObjectB), capturing directional relations (e.g., left_of, above, inside) as well as higher-order alignments such as rows, grids, or repetition patterns. Consistent object IDs are enforced to forestall relational ambiguity, and parsing isolates output lines beginning with parentheses, followed by automated consistency checks. This step transmutes static object descriptions into a relational graph that encodes spatial topology and hierarchy. The resulting structure supports reasoning over both pairwise interactions and more elaborate scene configurations—an essential prerequisite for tasks such as symmetry detection, grouping, and multi-step navigation of spatial layouts.

Grouping and Symmetry Agent. The Grouping and Symmetry agent abstracts higher-order patterns from the object lists and spatial relations, identifying clusters, rows, grids, and symmetries—vertical, horizontal, rotational, and translational alike. Outputs are requested in structured JSON keyed by `groups` and `symmetries`, with empty structures as fallbacks to preserve downstream compatibility. This agent captures emergent scene regularities that remain imperceptible at the

object or pairwise-relation level, supplying critical context for pattern recognition and reasoning. By formalizing structural redundancies and symmetries, the agent underpins both perceptual inference and anticipatory reasoning about dynamic transformations.

Transformation Tracking Agent. For multi-frame sequences, the Transformation Tracking agent logs object-level changes—translations, scaling, rotations, color/shading shifts, and appearance or disappearance events. Structured logs are extracted line-by-line via regex patterns that capture directional arrows, yielding a temporal representation of the scene. This encoding allows the system to track not merely static spatial configurations but also their dynamic evolution, supporting reasoning about motion, causality, and object interactions over time. The agent binds each transformation to consistent object IDs, enabling coherent cross-frame reasoning and predictive modeling.

Representation Standardization Agent. Finally, the Representation Standardization agent consolidates every upstream output into a unified JSON format comprising objects, relations, groups, symmetries, and transformations. Prompts enforce consistent object IDs and structured formatting, while JSON parsing failures revert to empty dictionaries to safeguard pipeline robustness. By centralizing all perceptual, relational, and temporal information, this agent synthesizes a comprehensive scene representation that is at once human-interpretable and machine-readable. The resulting unified structure constitutes a versatile foundation for downstream tasks, including reasoning evaluation, spatial pattern analysis, and multi-frame prediction.

Open-Ended Spatial Reasoning Agent. The standardized representation feeds into the reasoning agent, which answers multiple-choice or pattern-recognition questions. Its inputs comprise objects, attributes, relations, groups, symmetries, and transformations. The model weighs geometric patterns, relational alignment, and dynamic transformations to generate or select the correct answer.

H.5.2 TECHNICAL IMPLEMENTATION DETAILS

For the technical implementation, we employ Gemini-2.5-Flash at a low temperature (0.1) to secure near-deterministic outputs, and we perform multiple iterations for refinement. API calls are routed through OpenRouter, which accommodates both text and image inputs, while the parsing of model responses relies on regex and JSON-based extraction to yield structured outputs. All interactions are logged in JSON format for reproducibility, and the terminal output is a unified per-image JSON encompassing comprehensive information on objects, relations, symmetries, and transformations. This multi-agent decomposition permits each specialized model to concentrate on a constrained sub-task, thereby elevating overall accuracy, interpretability, and robustness in structured spatial reasoning.

H.5.3 EVALUATION AND ANALYSIS

The evaluation protocol remains identical to the standard CoT and CoT-SR setting, drawing 5 balanced samples per sub-category (seed 42) for a total of 150 samples.

Table 25: Agent Performance Analysis

Category	MCQ		Gain	Open Evaluation		Gain
	Normal	With Agents		Normal	Agent Results	
3D Geometry	48.00%	44.00%	-4.00%	60.00%	48.00%	-12.00%
Depth & Occlusion	48.00%	44.00%	-4.00%	40.00%	16.00%	-24.00%
Orientation	56.00%	64.00%	+8.00%	0.00%	36.00%	+36.00%
Relative Positioning	60.00%	60.00%	0.00%	50.00%	36.00%	-14.00%
Size & Scale	48.00%	52.00%	+4.00%	25.00%	24.00%	-1.00%
Spatial Navigation	68.00%	64.00%	-4.00%	40.00%	28.00%	-12.00%
Overall	54.67 %	57.33%	+2.66%	35.83 %	31.33%	-4.50%

The evaluation of Gemini-2.5-Flash with agents throws into relief both the strengths and the limitations of agentic reasoning on perceptual–geometric tasks. Orientation is the sole category in which agents deliver consistent and substantial gains, improving by +8.00% in MCQ and an emphatic +36.00% in open evaluation. This pattern indicates that agent-based multi-step reasoning is effective precisely when a task can be decomposed into the sequential alignment of directional cues, where structured deliberation confers a clear advantage over single-step outputs.

By contrast, most other categories register negative or neutral gains, revealing that agent reasoning does not universally enhance performance. Depth & occlusion, for instance, suffers a severe -24.00% drop in open evaluation, while spatial navigation falls by -12.00%. These tasks demand deeper perceptual grounding in three-dimensional space, and the absence of such priors means that reasoning steps tend to reinforce misconceptions rather than rectify them. Relative positioning likewise stagnates in MCQ and declines by -14.00% in open evaluation, indicating that agent loops fail to instill robustness where relational understanding is inherently weak.

A handful of categories show marginal improvements, such as size & scale (+4.00% in MCQ), but these gains prove unstable and fail to carry over to open evaluation. Even in 3D geometry, agents furnish no boost, incurring -4.00% and -12.00% drops in MCQ and open evaluation, respectively. These results underscore that our benchmark is genuinely complex and cannot be solved merely by layering multi-step agentic reasoning atop existing models. It demands instead a deeper integration of geospatial understanding and perceptual representations—precisely what current architectures lack.

On balance, the mixed results make clear that agentic workflows alone are insufficient to advance performance in this domain. While orientation tasks profit from structured reasoning, the majority of categories expose the models’ want of inherent awareness of perceptual concepts. Errors therefore propagate through the multi-step reasoning, amplifying the performance drops. Our benchmark consequently functions as a rigorous stress test, making evident that future progress will hinge on embedding genuine perceptual and geospatial priors rather than leaning solely on agent-based reasoning.

I QUALITATIVE ERROR ANALYSIS AND ERROR PATTERNS

This section advances a rigorous qualitative account of the failure modes elicited across our visual reasoning benchmark. For each error class we specify the observable behavior, derive a mechanistic root-cause diagnosis, and isolate the recurring structure that unifies these failures, situating the discussion within the verification protocols we adopt. This is not a catalogue of incidental slips: the analysis demonstrates conclusively that model errors are systematic, interpretable, and diagnostic. They lay bare the shared inductive biases of contemporary vision–language pipelines and, precisely because they are structured, translate directly into concrete and actionable prescriptions for future architectural and evaluation design.

I.1 OVERVIEW

Across the sample set, models exhibit robust coarse perception: they reliably resolve object categories, colors, and simple binary relations. They nonetheless falter with striking regularity the moment multiple cues must be integrated into a single judgment. These breakdowns are not stochastic noise. They condense into a compact and fully interpretable taxonomy of five recurring classes: spatial mislocalization and reference confusion, perspective and scale mistakes, occlusion and ordering failures, attribute confusion and semantic swap, and ungrounded open-ended rationalization. That failures organize into so few and so legible categories is itself a strength of the benchmark, which is deliberately calibrated to expose these structural weaknesses rather than dilute them. The classes recur invariantly across model architectures, across parameter scales, and across prompting styles, which establishes that their origin lies in inductive biases endemic to current vision–language pipelines rather than in isolated training artifacts. Tellingly, an identical error signature surfaces in large-scale proprietary systems and in small open-source baselines alike, decisive evidence of a shared structural bias that transcends any single model family. In what follows we dissect each error class through representative examples and precise root-cause hypotheses.

I.2 ERROR TAXONOMY AND ANALYSIS

I.2.1 SPATIAL MISLOCALIZATION AND REFERENCE CONFUSION.

■ **Observation.** Whenever a query requires that a single object be individuated from a set of visually similar candidates, models misidentify the target with pronounced regularity. Prompted to indicate “the leftmost red cube among three stacked cubes,” a model routinely returns the central cube instead, captured by a lighting highlight or by sheer positional centrality. The difficulty intensifies sharply in cluttered environments such as kitchen counters or crowded table scenes, where multiple overlapping objects share identical color and shape. The resulting mistakes are anything but arbitrary: the model consistently privileges whichever object is most perceptually salient, by size, brightness, or centrality, even when that salience stands in direct contradiction to the explicit spatial instruction in the query. The stability of this pattern is exactly what identifies the failure as systematic rather than accidental.

★ **Analysis.** This recurring pattern indicates the absence of a robust object-centric representation, what cognitive science terms a persistent “index” that endures across successive reasoning steps. Rather than sustaining a stable pointer to the designated candidate throughout the reasoning chain, the model drifts, reallocating attention at intermediate layers toward whatever entity is most visually conspicuous. The consequence is a characteristic slippage whereby the chain initializes on the leftmost cube yet terminates on the brightest or largest one. Because the spatial referent cannot be held invariant, relational queries such as “the ball behind the second chair” grow acutely fragile. In effect, the model operates not as a referential reasoner but as a salience detector.

⊗ **Root cause.** At the architectural level, feature-pooling operations, whether instantiated through attention or convolution, blend contextual information across neighboring regions. Advantageous as this is for aggregating global context, it concurrently dissolves the fine-grained, object-specific identity needed to separate similar candidates. Once pooled, the representations of two adjacent red cubes become all but indistinguishable in latent space. Absent an explicit binding mechanism, such as slot-based representations or pointer tokens, the network collapses distinct entities into a single blended embedding. This structural limitation precludes reliable referent tracking across reasoning steps and fully accounts for the consistent confusion of dense, overlapping objects.

I.2.2 PERSPECTIVE AND SCALE MISTAKES.

■ **Observation.** Tasks that demand metric reasoning about relative size, fit, or scale are handled with conspicuous inconsistency. Coarse comparisons such as “is a car larger than a bicycle” are dispatched effortlessly, because the relation coincides with a strong statistical prior. Near-threshold decisions, by contrast, such as “will the small sphere fit inside the slightly larger cylinder,” provoke frequent and erratic errors. A recurring signature is uncritical reliance on canonical object sizes: the model insists that a door must exceed a truck in height even in a rendering where the truck has been deliberately scaled down. A parallel failure arises in scenes that intentionally invert familiar proportions, such as toy miniatures or architectural models, where the model adheres to its priors and disregards the geometric evidence plainly present in the image.

★ **Analysis.** These behaviors expose a heavy dependence on the distributional statistics of object categories in lieu of genuine geometric inference. Rather than reasoning over projection geometry, relative bounding-box extents, or depth cues, the model substitutes memorized world knowledge of typical scales. The heuristic serves reliably in ordinary contexts yet fails catastrophically in edge cases that depart from the training distribution: presented with an anomalously small airplane rendered beside a person, the model nonetheless declares the airplane the larger object. Such errors reveal both the missing grounding in pixel-level metric cues and the failure to exploit perspective structure, such as vanishing points, in any consistent manner.

⊗ **Root cause.** Current architectures embed no explicit geometric reasoning module capable of preserving projection-aware relationships. Because supervision for scale and fit is scarce, models acquire correlations between category labels and typical size but never internalize the geometry that governs them. Compounding this, training objectives reward aggregate accuracy rather than fine-grained metric consistency, so scale-sensitive errors go systematically under-penalized. Lacking either an inductive bias for 3D reasoning or targeted supervision, the model retreats to memorized priors about canonical sizes instead of interpreting the perspective geometry actually present in the image.

I.2.3 OCCLUSION AND ORDERING FAILURES.

■ **Observation.** Adjudicating which object stands in front, which behind, and which is partially hidden remains a core weakness. Models routinely misreport the foreground element when two animals overlap or when semi-transparent barriers such as glass or fences intervene. In an image where a cat sits partially behind a curtain, for example, the model may invert the relation and describe the curtain as occluded by the cat. Thin or fragile structures, such as wires, tree branches, or railings, prove especially error-prone: they are frequently omitted altogether, yielding wrong answers to questions such as “what object lies behind the fence.” The mistakes recur with marked consistency, most acutely where occlusion cues are subtle or the depth separation is slight.

★ **Analysis.** Such failures indicate that depth and occlusion are encoded as weak, secondary signals. Although the pixel data furnish unambiguous occlusion cues, such as edge boundaries and T-junctions, the intermediate features the model learns fail to retain them at adequate resolution. Aggregation layers smear boundaries and down-weight thin structures, in effect discarding the very cues on which ordering depends. This accounts for why models resolve gross depth distinctions between foreground and background yet collapse in the fine-grained regime where partial occlusion must be inferred. In practice they guess rather than infer, exhibiting a systematic bias toward conferring foreground status on whichever object is visually larger or brighter.

⊗ **Root cause.** The architecture affords no explicit mechanism for multi-scale feature retention and no dedicated supervision for depth or occlusion reasoning. Thin or partially occluded signals are readily extinguished once features are pooled across large receptive fields. The training data compound the problem by underrepresenting subtle occlusion cases, particularly those involving transparency or fine boundaries. The learned representations therefore flatten layered scenes into 2D appearance maps, foreclosing the robust construction of depth-aware relational reasoning.

I.2.4 ATTRIBUTE CONFUSION AND SEMANTIC SWAP.

■ **Observation.** Models frequently conflate perceptual attributes, such as color, texture, or material, with functional or semantic roles. A canonical instance is judging whether a crosswalk is safe: the model may rest its verdict entirely on the presence of painted stripes while disregarding the state of the traffic light and the approaching vehicles. In the same vein, a surface with a metallic sheen may be classified as “steel” despite being painted plastic, and in tool-use settings the mere presence of a handle or a familiar silhouette can induce the model to ascribe a function the object does not possess. Such confusions arise reliably wherever a task requires fine-grained visual grounding of attributes.

★ **Analysis.** These errors demonstrate that language priors exert a dominant bias over prediction whenever the visual input is ambiguous. The decoder gravitates toward plausible narratives underwritten by training-data co-occurrence, such as “crosswalks are safe when stripes are visible” or “shiny textures correspond to metal.” This shortcut lets the model emit semantically coherent answers without ever interrogating the pertinent visual evidence, producing reasoning that appears correct in familiar contexts yet fails systematically in atypical or adversarial ones. In effect, the model overfits to textual stereotypes rather than grounding its judgments in perceptual detail.

⊗ **Root cause.** The governing training objective, maximizing the likelihood of fluent and plausible text, imposes no requirement of visual grounding whatsoever. So long as the generated answer is semantically reasonable, the loss is minimized irrespective of whether the underlying reasoning is correct. The model is thereby incentivized to substitute common-sense priors for genuine perceptual inference. Absent auxiliary losses or verification steps that bind textual claims to specific image regions, the interchange of perceptual and semantic attributes will persist.

I.2.5 OPEN-ENDED RATIONALIZATION WITHOUT VERIFICATION.

■ **Observation.** On explanatory or why-style questions, models routinely generate coherent yet unsupported narratives. Asked “why is the object leaning,” a model may respond “because the surface is slippery,” even though slipperiness admits no visual inference; a tilted shelf may be explained “due to an earthquake,” a claim lying wholly beyond the evidence in the image. Such errors betray a disposition to fill evidential gaps with imaginative but ungrounded rationales, producing fluent explanations that misinform users about what the image actually supports.

★ **Analysis.** The issue originates in a decoupling of text generation from visual grounding. The model produces high-quality language that reads convincingly, yet no mechanism tethers these rationales to the actual visual features. Poor confidence calibration aggravates the problem: fabricated

explanations are frequently issued with high certainty, lending them a spurious air of reliability. Open-ended outputs may thus present as authoritative while remaining factually baseless, a failure mode of acute concern in safety-critical domains, where a plausible but false rationale could misdirect a downstream decision.

⊗ **Root cause.** Training is dominated by language-modeling objectives that reward fluency and coherence while remaining indifferent to factual grounding. The decoder is optimized to emit the most probable continuation of text given the prompt, under no constraint that anchors its explanation to specific image evidence. Lacking any explicit verification layer or grounding requirement, the model defaults to linguistic plausibility over perceptual truth. Until additional architectural or training constraints are imposed, open-ended rationalization will continue to privilege surface coherence at the expense of evidential accuracy.

I.3 ERROR PATTERNS AND CROSS-MODEL SIGNALS

Comparing failures across models throws several consistent patterns into relief, each illuminating the mechanisms that generate the errors. The first is that inter-model disagreement is itself systematic rather than random: certain models lean consistently on geometric cues such as relative size, depth ordering, and occlusion, whereas others retreat to linguistic priors and common-sense assumptions distilled from training text. The upshot is predictable divergence, as when two models return opposite verdicts on which object lies “in front” in a shared scene. Errors likewise spike whenever several cues must be integrated at once; tasks that couple occlusion, relative scale, and perspective are the most demanding, and judging whether a larger object that partially occludes a smaller one lies in front or behind reliably elicits coordinated mispredictions. Calibration constitutes a further recurring signal: high-confidence predictions correspond disproportionately to incorrect answers, most conspicuously in open-ended explanations and complex spatial queries. Taken together, these patterns localize the source of error in how each model arbitrates between perceptual grounding and linguistic plausibility. Distinct architectures manifest distinct tendencies, some over-relying on visual geometry, others overfitting to language correlations, so that failures are not isolated mistakes but expressions of a deeper trade-off between competing reasoning strategies. The corollary is decisive: any durable improvement must confront both sources of bias jointly rather than treat errors as independent events.

I.4 MITIGATION STRATEGIES

Because the failures are systematic, they admit systematic remedy, and the taxonomy above maps directly and one-to-one onto a set of targeted interventions that curtail error without demanding a wholesale architectural redesign. *Geometry-aware supervision* introduces auxiliary tasks such as depth prediction, occlusion ordering, and overlap estimation, compelling the model to respect spatial constraints and sharpening the consistency of its predictions. *Multi-scale feature retention*, realized through edge-aware pooling or attention, preserves information from thin or partially occluded structures that would otherwise be annihilated under standard downsampling. *Coordinate anchoring* dissolves the confusion between viewer-centric and object-centric frames by representing reference points for relative positioning explicitly. *Counterfactual augmentation* systematically swaps or perturbs distractor objects, forcing the model to depend on genuine referent cues rather than salience heuristics. And *lightweight verification modules* can be layered atop existing outputs, cross-checking each claim against visual or geometric constraints and filtering out implausible predictions. Each intervention is incrementally deployable, furnishing a clear and low-cost path to suppressing systematic failure while leaving the core model largely intact.

Complementing these architectural and training augmentations, *supervised fine-tuning* (SFT) offers a practical route to instilling more reliable spatial reasoning. By curating datasets that foreground the failure cases themselves, near-threshold scale comparisons, occlusion ambiguities, and reference-resolution traps, SFT confronts the model directly with the edge scenarios in which it would otherwise default to heuristics. Instruction-style fine-tuning, judiciously designed, further elicits step-by-step reasoning, prompting the model to verbalize intermediate spatial relations rather than leap to a guess; this simultaneously strengthens grounding and improves interpretability, since errors become traceable to specific reasoning steps. SFT can moreover be coupled with counterfactual or adversarial augmentation, so that the model acquires invariances to distractors and context shifts instead of merely memorizing. Critically, none of this demands wholesale architectural change: these interventions adapt existing models by reshaping their inductive biases through supervision alone. In this respect

SFT operates as a lightweight yet potent complement to the geometry-aware and verification-based strategies above.

To substantiate this claim empirically, we performed SFT on 40% of the training data and observed clear performance gains, most pronounced on multiple-choice questions. The pattern establishes that targeted supervision helps the model internalize spatial cues and resolve distractor confusions when the answer options are well defined. Gains on open-ended responses were more limited, however: the model persists in producing fluent but weakly grounded explanations, a direct reminder that SFT alone cannot fully redress rationalization in the absence of verification. These findings corroborate the central prescription of our analysis, that SFT must be combined with geometry-aware supervision, counterfactual augmentation, and lightweight verifiers to secure balanced progress across both structured and free-form tasks.

Our analysis establishes decisively that these failures are not arbitrary but structured and interpretable, arising from recurring patterns rather than random noise. VLM errors follow systematic trajectories dictated by the inductive biases of their architectures and training regimes. Geometric reasoning collapses predictably when occlusion cues are weak, whereupon language priors seize control and return answers that sound plausible yet want visual grounding; scale inference drifts, no less predictably, whenever canonical size priors override pixel-level metric cues, producing consistent misjudgments in near-threshold scenarios. The through-line is unambiguous: the gap is one not of capacity but of representation, since current models lack explicit mechanisms for anchoring visual features to grounded geometric relations.

Because the diagnosis is precise, the remedy can be too. Adopting the verification protocols and targeted mitigation strategies proposed here confronts these weaknesses at their source. Geometry-aware supervision and counterfactual augmentation blunt the model’s reliance on shortcuts, while multi-scale feature retention guarantees that thin structures and partial occlusions are not discarded before they can be reasoned over. SFT, judiciously applied, further consolidates structured reasoning by aligning predictions with task constraints, provided it is combined with grounding objectives so as not to overfit to discrete-choice formats. Verification modules, for their part, supply a lightweight means of enforcing consistency between claimed relations and image evidence, directly counteracting the impulse to rationalize without checking. Together these steps chart a practical pathway toward systems that reason more nearly as humans do: robust to clutter, interpretable in their failures, and able to explain not merely what they predict but why.

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Base VLM Agent Prompt

You are a Visual Scene Describer. Produce a raw, exhaustive text description of the given image.

Input image:
{image_input}

Your focus:

- Object Identification:
 - Shapes (circle, square, triangle, polygon, etc.)
 - Natural elements (tree, sun, water, etc.)
 - Artificial objects (arrows, patterns, icons)
- Attribute Noticing:
 - Size (small, medium, large, relative comparison)
 - Orientation (upright, rotated, tilted)
 - Shading or texture (solid, striped, shaded, transparent)
 - Color (if available)
- Positioning:
 - Relative location (leftmost, rightmost, top, bottom, center)
 - Proximity to other objects
- Level of Detail:
 - Do not summarize
 - Be verbose and explicit about every visible detail

Object Segmentation Agent Prompt

You are an Object Segmentation Specialist. Convert a verbose image description into a structured list of distinct objects.

Input description:
{visual_description}

Your focus:

- Object Isolation:
 - Treat each unique instance as a separate object
 - If multiple objects are similar, assign separate IDs
- Labeling:
 - Assign IDs: Obj1, Obj2, Obj3...
 - Keep descriptions short but complete
- Example Output:
 - Obj1: Small shaded circle (leftmost)
 - Obj2: Large unshaded square (center)
 - Obj3: Triangle rotated 45° (rightmost)

Attribute Extraction Agent Prompt

You are an Attribute Extraction Specialist. For each object, extract its properties into structured JSON.

Your focus:

- Attributes to Capture:
 - Shape
 - Size (relative if necessary)
 - Color or shading
 - Orientation or rotation
 - Position (leftmost, rightmost, center, top, bottom, etc.)
- Output Format:
 - JSON array, e.g.:
[
 {{ "ID": "Obj1", "Shape": "Circle", "Size": "Small", "Shading": "Yes", "Rotation": "0°", "Position": "Leftmost" }},
 {{ "ID": "Obj2", "Shape": "Square", "Size": "Large", "Shading": "No", "Rotation": "45°", "Position": "Center" }}
]

Figure 10: Prompts for SPATIOXOLVER (Part 1).

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Spatial Relation Agent Prompt

You are a Spatial Relation Specialist. Extract spatial relationships among objects.

Your focus:

- Core Relations:
 - left_of, right_of, above, below
 - inside, surrounding
 - touching, overlapping
 - aligned, centered
- Precision:
 - Always specify directionality (ObjA left_of ObjB is not the same as ObjB left_of ObjA)
 - Use consistent object IDs
- Structural Patterns:
 - Detect repeated relations (e.g., "each circle is above a square")
 - Note spacing and alignment
- Output Format:

Triples in form: (ObjectA, Relation, ObjectB)

Example:

(Obj1, left_of, Obj2)

(Obj3, inside, Obj4)

Grouping Symmetry Agent Prompt

You are a Grouping and Symmetry Specialist. Identify higher-order structures from objects and relations.

Relations:
{relation_list}

Your focus:

- Grouping:
 - Clusters of objects (rows, columns, triangles, grids)
 - Repeated patterns
- Symmetry:
 - Vertical, horizontal, rotational
 - Mirror or reflective symmetry
 - Translational repetition
- Output Format:

JSON with two fields: "groups" and "symmetries"

Example:

```
{{  
  "groups": ["Obj1, Obj2, Obj3 form a horizontal row"],  
  "symmetries": ["Vertical symmetry across Obj2"]  
}}
```

Transformation Tracking Prompt

You are a Transformation Tracker. Analyze changes across multiple frames of an image sequence.

Your focus:

- Types of Transformations:
 - Movement (left, right, up, down)
 - Scaling (larger, smaller)
 - Rotation (degrees)
 - Shading or color change
 - Shape change (square → circle)
 - Appearance/disappearance
- Output Format:

Per-object transformation logs:

Obj1: Frame1 → Frame2 (rotated 90° clockwise)

Obj2: Frame2 → Frame3 (moved right)

Figure 11: Prompts for SPATIOXOLVER (Part 2).

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Representation Standardization Agent Prompt

You are a Symbolic Abstraction Specialist. Consolidate all perception results into a standardized structured format.

Inputs:

- Objects: {object_list}
- Attributes: {attributes_json}
- Relations: {relation_list}
- Groups & Symmetries: {grouping_symmetry}
- Transformations: {transformations}

Your focus:

1. Create a unified JSON representation containing:
 - Objects with attributes
 - Relations as triples
 - Groups and symmetries
 - Transformations across frames
2. Ensure consistency in IDs and formatting
3. Example Output:

```
{
  "objects": [...],
  "relations": [...],
  "groups": [...],
  "symmetries": [...],
  "transformations": [...]
}
```

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and select the correct option for the multiple-choice question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Options: {options_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.
- Select only the correct option Only (1, 2, 3, 4).
- Do NOT provide explanations, reasoning, or extra text. Only the option letter.

Output format: "<option_number>"

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and give correct answer for the question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.

Figure 12: Prompts for SPATIOXOLVER (Part 3).

J REPRESENTATIVE BENCHMARK SAMPLES ACROSS ALL SUB-CATEGORIES

To convey the breadth of SPATIALAB concretely, we present representative benchmark samples spanning all 30 sub-categories, each instantiating a distinct spatial reasoning challenge. The selection foregrounds the full spectrum of visual phenomena the benchmark probes, from 3D geometry and occlusion through orientation, relational positioning, size and scale, and navigation. Taken together, these examples afford a tangible view of the reasoning each model must execute to succeed across the benchmark.

3D Geometry: Gravity Effects



Question: From which direction would the wind cause the liquid in the glass which is in the front to spill onto the ground first?

Options

1. From the left side
2. From the right side
3. From the front
4. From the back

Gold MCQ Answer: 1

Gold Open ended Answer: From the left side

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** A gust blowing from the back of the patio (the top of the image, toward the viewer) would push the liquid forward over the near edge and make it spill first. Winds from the sides would tend to slosh it sideways rather than directly over the front rim.
- **Gemini-2.5-Flash:** The glass in the front is the one on the right side of the coffee table. Given its position, a wind blowing from the left (from the direction of the wooden cabinet and tall plant) would likely cause the liquid to spill first. This is because the glass is positioned with its opening facing somewhat towards the left, making it more susceptible to a gust from that direction.
- **InternVL3.5-72B:** The wind would need to come from the upper-right direction to cause the liquid in the glass on the front right of the table to spill first. This is due to the positioning of the glass towards the front edge of the table, closest to the right side of the image.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about the direction the wind would have to come from to spill the liquid in the front glass. Looking at the position of the glass: it's on the left front edge of the coffee table. The table seems to have a flat surface. If the wind blows from the left side (from the viewer's left, which would be the front of the image if the viewer.

Figure 13: Representative benchmark items probing how objects settle, balance, and respond under gravity, paired with model responses and evaluation outcomes (3D Geometry: Gravity Effects).

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3D Geometry: Shape Projection



Question: If the hanging light casts a shadow on the wall, which outline might appear?

Options

1. A rectangle narrowing at the bottom From the right side
2. A vertical cylinder
3. A circle with fuzzy edges
4. A triangle pointing downward

Gold MCQ Answer: 2

Gold Open ended Answer: A vertical cylinder

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would see the pendant's dome-shaped profile: a rounded semicircle or oval shadow with a thin vertical line or narrow neck above where the cord hangs. The shadow would fall on the wall to the right of the window, roughly over the nightstand.
- Gemini-2.5-Flash: The hanging light is a simple, cylindrical pendant. If it casts a shadow on the wall, the outline would likely be a circular or elliptical shape, depending on the angle of the light source relative to the wall. Given its position near the window, the shadow might also be influenced by ambient light or other objects in the room.
- InternVL3.5-72B: The hanging light fixture is likely to cast a shadow resembling its own shape onto the wall. Given its cylindrical form, the shadow would appear as a soft-edged, elongated circle or oval, depending on the angle of the light source relative to the wall.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking which outline a hanging light would cast on the wall if it casts a shadow. The wall near the window is light-colored, so the shadow would be visible. Since the shade is cylindrical, the shadow might be a cylinder. But maybe the outline is of the lampshade's perimeter. Alternatively, if the light is hanging from a chain, the shadow could be more elongated. But based on typical light fixtures, a cylindrical shade would cast a circular or oval shadow. The answer should be the outline of the hanging light's shade.

Figure 14: Representative benchmark items requiring inference of the 2D silhouette a 3D solid casts under a given viewpoint, with model responses and evaluation outcomes (3D Geometry: Shape Projection).

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3D Geometry: Spatial Containment



Question: Look at the tall pot in the background with the wide mouth (top left). could the small jug from the center of the photo fit completely inside it?

Options

1. Yes, easily with room to spare
2. No, the small jug hand will stuck
3. No, the small jug is too tall.
4. Yes, but it would be a perfect fit with no room left

Gold MCQ Answer: 2

Gold Open ended Answer: No, the small jug hand will stuck

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** No — the tall pot's mouth appears too narrow compared with the small jug's widest part and its handle. The jug wouldn't pass through the opening to sit completely inside.
- **Gemini-2.5-Flash:** No, the small jug from the center of the photo could not fit completely inside the tall pot in the background. While the tall pot has a wide mouth, the jug's overall height and width appear to be too large to be fully contained within the pot's interior. The jug's handle would also likely prevent it from fitting snugly.
- **InternVL3.5-72B:** The tall pot in the background with the wide mouth appears large enough to potentially accommodate the small jug from the center of the photo. Considering the proportions and the wide opening of the tall pot, it is likely that the small jug could fit completely inside it.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about whether the small jug in the center of the photo could fit completely inside the tall pot in the background with the wide mouth (top left). However, without explicit scale references, this conclusion relies on typical proportional relationships between similar pottery forms. Based on typical proportions, the small jug's height is likely shorter than the tall pot's, and its width seems narrow enough to fit.

Figure 15: Representative benchmark items assessing whether one object fits within or is enclosed by another, with model responses and evaluation outcomes (3D Geometry: Spatial Containment).

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3D Geometry: Volume Comparison



Question: Compare the internal cooking volume of the microwave with the internal cooking volume of the larger oven below the stove.

Options

1. They both have the same volume A vertical cylinder
2. The microwave has slightly more volume
3. The oven below the stove has much more volume
4. The oven has more width, but the microwave has more height, so they are equal

Gold MCQ Answer: 3

Gold Open ended Answer: The oven below the stove has much more volume

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 3
- **InternVL3.5-72B:** 3
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** The microwave’s cooking cavity is much smaller than the larger oven below the cooktop — it’s only a small fraction of the oven’s internal volume. The countertop microwave is sized for single plates or small dishes, whereas the built-in oven can hold full baking trays and multiple racks for larger meals.
- **Gemini-2.5-Flash:** The internal cooking volume of the microwave appears to be significantly smaller than that of the larger oven below the stove. The microwave is a compact countertop model, suitable for reheating or cooking smaller items..
- **InternVL3.5-72B:** The internal cooking volume of the microwave on the right is likely smaller than that of the larger oven located below the stove. This is because microwaves typically have more compact dimensions to fit on countertops, while ovens have a greater internal capacity to accommodate larger dishes and multiple items. The size and design visible in the image suggest the oven has a deeper and wider cooking area compared to the microwave.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants me to compare the internal cooking volumes of the microwave and the larger oven below the stove in the provided image. So, the internal cooking volume of the oven below the stove is larger than that of the microwave. Based on their relative sizes in the image, the internal cooking volume of the oven below the stove is substantially larger than that of the microwave

Figure 16: Representative benchmark items requiring comparison of the volumes of competing 3D objects, with model responses and evaluation outcomes (3D Geometry: Volume Comparison).

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3D Geometry: Shape Projection



Question: If you look at the seating arrangement from above, what shape does it form?

Options

1. A perfect square
2. A round shape
3. A rectangle
4. An irregular shape

Gold MCQ Answer: 4

Gold Open ended Answer: An irregular shape

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Viewed from above, the seating is arranged in a U-shape (a horseshoe) around the central fire pit. The L-shaped sectional plus the two facing chairs create that U configuration.
- Gemini-2.5-Flash: The seating arrangement, when viewed from above, primarily forms an L-shape with the large sectional sofa. This L-shape is complemented by two individual chairs placed opposite the open ends of the sectional. Together, the furniture creates a U-shaped configuration around the central fire pit, designed for communal gathering.
- InternVL3.5-72B: From above, the seating arrangement forms a U-shape around the central fire pit. The sectional sofa curves around the pit, leaving an open side opposite the tall fence, creating a semi-enclosed space.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user wants to know the shape formed by the seating arrangement when viewed from above. When viewed from above, the configuration clearly forms a right-angled L-shape, combining two parallel sections at a 90-degree angle. The symmetry of the layout and the placement of furniture around the fire pit reinforce this geometric form. The seating arrangement forms a right-angled L-shape when viewed from above.

Figure 17: Further benchmark items testing recovery of the projected outline of a 3D form across differing orientations, with model responses and evaluation outcomes (3D Geometry: Shape Projection).

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Spatial Navigation: Accessibility Constraints



Question: The notebook at the leftmost corner of the table is inaccessible because of which object?

Options

1. Chair
2. Vase
3. Table
4. Mug with pens and pencil in it

Gold MCQ Answer: 4

Gold Open ended Answer: Mug with pens and pencil in it

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** The leftmost corner notebook is blocked by a wooden chair. The chair sits directly in front of the desk, preventing easy access to the notebook.
- **Gemini-2.5-Flash:** The notebook at the leftmost corner of the table is inaccessible because of the chair. The chair is positioned directly in front of the notebook, blocking direct access to it. To reach the notebook, one would first need to move the chair out of the way.
- **InternVL3.5-72B:** The notebook at the leftmost corner of the table is inaccessible because of the chair, which is positioned in front of it. The chair's backrest and seat block direct access to the notebook.
- **Kimi-VL-A3B-Thinking-2506:** Alright, let's tackle this spatial reasoning question. So, the question is asking why the notebook at the leftmost corner of the table is inaccessible. The chair is positioned near the leftmost corner of the table, and given its placement, it's reasonable to conclude that the chair is blocking access to the notebook. The chair's presence in that space would make it difficult to reach the notebook from the left side without moving the chair first. The chair is the object that's making the notebook at the leftmost corner inaccessible.

Figure 18: Representative benchmark items testing whether a target location can be reached given physical accessibility constraints, with model responses and evaluation outcomes (Spatial Navigation: Accessibility Constraints).

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Spatial Navigation: Obstacle Avoidance



Question: There is a clear direct path to the end of the hallway without colliding with any object.

Options

1. TRUE
2. FALSE
3. Maybe
4. Unable to determine

Gold MCQ Answer: 1

Gold Open ended Answer: TRUE

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

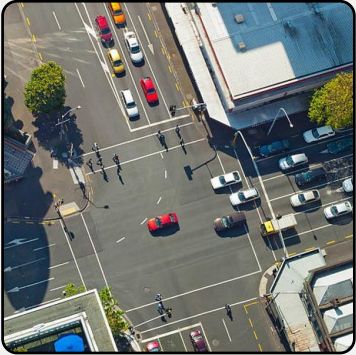
Open-ended Answers:

- **GPT-5-mini:** Yes. The central aisle is unobstructed from the foreground to the far end, with shelving stacked along both sides and no items blocking the floor. You can walk straight down the hallway without colliding with any objects.
- **Gemini-2.5-Flash:** Yes, there is a clear direct path to the end of the hallway without colliding with any object. The image shows a wide, unobstructed aisle between two rows of tall shelving units filled with boxes. The floor is clear, indicating a safe passage for movement.
- **InternVL3.5-72B:** The image shows a large warehouse with high shelving units full of boxes on both sides of a central aisle. The aisle appears to be clear and unobstructed, leading directly to the end of the hallway. There are no objects or obstacles in the pathway, ensuring a direct and clear route to the end of the aisle.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The user says there's a clear direct path to the end of the hallway without colliding with any object. A clear direct path exists because the shelving units are positioned symmetrically, leaving a central aisle free of objects. There are no visible hazards like hanging items, loose debris, or protruding elements in the visible portion of the hallway, allowing safe, uninterrupted movement to the end. The path is clear because the shelving units are symmetrically arranged, leaving the central walkway obstacle-free, with no visible hazards in the visible hallway.

Figure 19: Representative benchmark items requiring routes that circumvent intervening obstacles, with model responses and evaluation outcomes (Spatial Navigation: Obstacle Avoidance).

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Spatial Navigation: Pathway Existence



Question: Which walkways are the safest to cross according to the picture?

Options

1. Top and bottom
2. Top and left
3. Left and right
4. Bottom and right

Gold MCQ Answer: 1

Gold Open ended Answer: Top and bottom

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

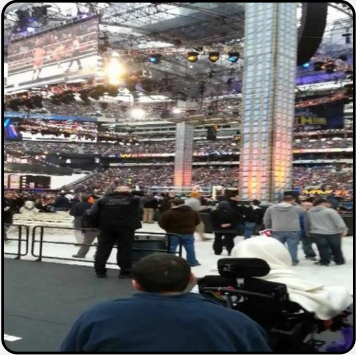
Open-ended Answers:

- GPT-5-mini: The marked white crosswalks at each corner of the intersection (the painted lines across the road) are the safest to use. They are signalized and where pedestrians are waiting, unlike cutting across mid-block or diagonally through the intersection.
- Gemini-2.5-Flash: The walkways marked with white lines, commonly known as crosswalks, are the safest to cross. These designated areas indicate where pedestrians have the right-of-way or are expected to cross the street. In the image, several people are visible using these marked crosswalks, suggesting they are the intended and therefore safest routes for pedestrians.
- InternVL3.5-72B: The crosswalks aligned with the pedestrian signals at the corners are the safest places to cross, as they are equipped with traffic signals ensuring vehicles are stopped. The designated pedestrian areas with painted lines across the streets are clearly marked for safety. Additionally, sidewalks provide a safe space for walking along the roads.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user wants to know which walkways are the safest to cross based on the image provided. The safest walkways to cross, based on the image, are the marked crosswalks equipped with pedestrian signals. Specifically, the crosswalk at the bottom-left corner of the intersection is safest because it features a visible pedestrian signal indicating walk, reducing ambiguity with traffic flow.

Figure 20: Representative benchmark items probing whether a traversable path exists between two points, with model responses and evaluation outcomes (Spatial Navigation: Pathway Existence).

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Spatial Navigation: Viewpoint Visibility



Question: The human standing the closest (blue jacket) has clear view of the stadium, how?

Options

1. Seeing directly
2. Watching on the big screen on the left side
3. There is no stadium
4. Unable to determine

Gold MCQ Answer: 2

Gold Open ended Answer: Watching on the big screen on the left side

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- Gemini-2.5-Flash: The person in the blue jacket is seated in a wheelchair, which places them at a lower vantage point than the standing crowd. However, they have a clear view of the stadium because they are positioned at the very front of a section, likely an accessible seating area. This front-row placement ensures an unobstructed line of sight over the heads of any standing individuals further back.
- InternVL3.5-72B: The person in the blue jacket is standing near the front of the arena, which allows them a clear, unobstructed view of the stadium. Their position is close to the barrier separating the audience from the performance area, providing a direct line of sight towards the stage and the large screens overhead. This proximity minimizes visual obstructions from other spectators or structures.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking how the person in the blue jacket (who is standing the closest) has a clear view of the stadium. The person's elevated seating, minimized obstructions from nearby spectators, and potential alignment with stadium features (e.g., screens or lighting) enable their clear view despite being relatively close to the ground.

Figure 21: Representative benchmark items testing what is visible from a specified vantage point, with model responses and evaluation outcomes (Spatial Navigation: Viewpoint Visibility).

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Spatial Navigation: Spatial Sequence



Question: Which path is the easiest for the human sitting in the bench(reading paper) to follow to get on the train?

Options

1. Go straight to the first red colored door
2. To make a turn and go to the second floor
3. Go through the door at the end of the train
4. No viable option

Gold MCQ Answer: 1

Gold Open ended Answer: Go straight to the first red colored door

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The simplest way for the person to board the train is to rise from the bench, turn right, and walk a short distance along the platform. They would then cross the yellow safety line and step through one of the train doors. This route is straightforward and involves minimal effort.
- **InternVL3.5-72B:** The person sitting on the bench can follow a straightforward path to board the train by moving forward along the platform, parallel to the train tracks, while keeping to the right of the yellow safety line. This path will lead them directly to the nearest open train door. They should avoid crossing the yellow line until they are ready to enter the train to ensure safety.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about determining the easiest path for the person sitting on the bench to get on the train. First, I need to look at the image provided. The person is sitting on the bench which is alongside the platform walls. The train is on the opposite side. To get to the train, the person would..

Figure 22: Representative benchmark items requiring the correct ordering of locations along a route, with model responses and evaluation outcomes (Spatial Navigation: Spatial Sequence).

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Size and Scale: Scale Consistency



Question: With the electric pole's capacity can all this wires be redesigned to carry properly in aligned structured design?

Options

1. The size of wires are too big
2. The wires need to be move into another pole
3. Wires can be set there following a layout
4. No, the bulk size of the wires are too much here for one pole

Gold MCQ Answer: 4

Gold Open ended Answer: No, the bulk size of the wires are too much here for one pole

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- **GPT-5-mini:** Not safely in its current state — the pole is clearly overloaded and the wires are tangled, so simply re-bundling them on the same pole would exceed mechanical, clearance and safety limits. A proper redesign is possible but requires removing redundant lines, separating circuits (high-voltage vs. telecom) onto organized crossarms, trays or underground ducts, and likely installing stronger or additional poles and anchors.
- **Gemini-2.5-Flash:** While it's technically possible to redesign the wiring for better alignment, the sheer volume and entanglement of wires in the image suggest that the current electric pole's capacity is severely overtaxed. Simply reorganizing the existing wires on the same pole would not address the underlying capacity issues or the safety hazards presented by such a dense and disorganized network.
- **InternVL3.5-72B:** Redesigning the extensive web of wires around the electric pole into a properly aligned and structured design is feasible but would likely require significant effort and resources. Proper planning and coordination with multiple service providers would be essential. Implementing such changes would enhance safety and aesthetics, but it involves logistical challenges and coordination to minimize service disruptions.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking if all the wires on the electric pole can be redesigned to be properly aligned in a structured way. A redesigned system is feasible but dependent on evaluating regulatory, technical, and safety constraints. Prioritizing organized routing for essential cables while removing redundant ones could create.

Figure 23: Representative benchmark items testing whether object sizes remain mutually consistent across a scene, with model responses and evaluation outcomes (Size and Scale: Scale Consistency).

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Size and Scale: Shadow-Size Projection



Question: If one light is rotated so that they are perpendicular to each other, how will the shadow look over the wall?

Options

1. It will cover almost all of the wall
2. The shadow will only take the size of the light frame as it's actual size
3. There won't be any shadow
4. Shadow will smaller than the actual light body

Gold MCQ Answer: 1

Gold Open ended Answer: It will cover almost all of the wall

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** If one lamp is rotated 90° so the two beams are perpendicular, you'll get two elongated oval patches of light on the wall at right angles to each other. Where the ovals overlap it will be brightest; outside them the wall will be darker, producing an L-shaped darker region and two perpendicular shadow lobes cast by the fixture.
- **Gemini-2.5-Flash:** If one light is rotated to be perpendicular to the other, the shadow cast on the wall would change significantly. The current shadows are likely elongated and somewhat parallel due to the lights' current orientation. With one light rotated, the shadows would intersect at a more acute angle, creating a more complex and possibly darker overlapping shadow where the light beams cross.
- **InternVL3.5-72B:** If one light is rotated so that the two lights are perpendicular to each other, the shadows on the wall will intersect at a right angle, creating intersecting patterns. The light beams will cast shadows in perpendicular directions, resulting in a cross-like effect where the shadows overlap. The intensity and shape of the shadows will depend on the orientation and angle of the lights relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The user is asking about rotating a light so that they're perpendicular to each other and how the shadow on the wall would look. If they're further away, the shadow would be smaller.

Figure 24: Representative benchmark items inferring object size from cast-shadow geometry, with model responses and evaluation outcomes (Size and Scale: Shadow-Size Projection).

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Size and Scale: Perspective Distortion



Question: If similar shaped pipes are stacked together separately will it take up less space?

Options

1. Yes
2. No
3. Take more
4. Similar

Gold MCQ Answer: 1

Gold Open ended Answer: Yes

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Yes - when like-shaped pipes are grouped you can pack them in their most efficient pattern ,which reduces wasted voids and the total footprint. The exact savings depend on orientation and dimensions; in some cases clever mixing or nesting of different shapes can beat separate stacks, but as a general rule same-shape grouping is more space-efficient.
- **Gemini-2.5-Flash:** Yes, if similarly shaped pipes are stacked together separately, it would likely take up less space. This is because organizing them by shape allows for more efficient packing and reduces wasted space between different forms. For example, square pipes can be stacked flush against each other, and circular pipes can be nested or arranged in a hexagonal pattern for optimal density.
- **InternVL3.5-72B:** Yes, stacking similar shaped pipes together separately will generally take up less space. Organizing them by shape allows for more efficient packing, reducing gaps and empty spaces between different sizes and types. This method optimizes the use of available storage area and improves accessibility.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is asking if stacking similar shaped pipes separately would take up less space. Stacking similar-shaped pipes separately can potentially optimize space usage. When pipes of the same diameter and shape (e.g., all round or rectangular) are grouped together, they can be arranged more tightly, minimizing gaps between

Figure 25: Representative benchmark items requiring size judgments to be corrected for perspective foreshortening, with model responses and evaluation outcomes (Size and Scale: Perspective Distortion).

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Size and Scale: Relative Size Comparison



Question: The ash-colored, thicker coiled spring will be surpassed by which coiled springs if all are kept in the standing position?

Options

1. Right bottom corner
2. Right top corner
3. Left top corner
4. Bottom mid spring

Gold MCQ Answer: 1

Gold Open ended Answer: Right bottom corner

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 2

Open-ended Answers:

- **GPT-5-mini:** The ash-colored thicker spring in the center will be shorter than the thin, taller silver spring just behind it (top-center) and the long, dark-gray spring at the right edge when all are stood upright. Those two coils extend above the ash-colored spring.
- **Gemini-2.5-Flash:** The ash-colored, thicker coiled spring, located near the center-right of the image, would be surpassed by the longer, thinner coiled spring positioned in the bottom right corner. Additionally, the two smaller, conical springs in the upper left quadrant would also be taller if fully extended and standing. These springs vary in height and coil density, indicating different potential applications.
- **InternVL3.5-72B:** The ash-colored, thicker coiled spring in the image will be surpassed in height by the two taller springs on its right. One of these taller springs is also ash-colored, while the other has a slightly darker hue. Both of these springs have more coils and are visibly longer when standing upright.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The image shows a bunch of different coiled springs. The question is asking which coiled springs would be surpassed by the ash-colored, thicker one if all are kept in the standing position The thinner, tightly coiled springs (smaller in diameter or fewer coils) would be surpassed in height by the ash-colored, thicker spring when all are kept upright.

Figure 26: Representative benchmark items requiring direct comparison of the relative sizes of two objects, with model responses and evaluation outcomes (Size and Scale: Relative Size Comparison).

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Size and Scale: Relative Size Comparison



Question: If the red bow is taken out of it's place and kept far away in the behind how will it look and what will be it's capacity comparing with it's other similar clips at front?

Options

1. Large, same
2. Same, less
3. Small, same
4. Small, large

Gold MCQ Answer: 3

Gold Open ended Answer: Small, same

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** If you move the red bow to the back it will appear smaller and more obscured by the bows in front because of perspective and overlap, and its red color will look less vivid from that angle. Physically nothing about the bow changes — its size, shape and clasp strength remain the same as the other similar clips in front, so its holding capacity is unchanged.
- **Gemini-2.5-Flash:** If the red bow were taken out and placed far behind, it would appear smaller and less distinct due to perspective. Its capacity, referring to its visual prominence or ability to hold hair, would remain the same as the other similar clips. However, its perceived capacity might diminish as it would no longer be in the foreground, drawing less attention.
- **InternVL3.5-72B:** If the red bow is moved to the back, it will be partially obscured by the other bows in the foreground, making it less prominent. Its size and capacity to hold hair will remain the same as the others, but visually, it won't stand out as much as it does currently when placed in the front.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about removing the red bow and placing it behind, then comparing its appearance and capacity to the other similar clips in front. Removing the red bow would reduce its central visual impact, making the remaining front-facing bows appear denser but less balanced. Its capacity (physical or visual space) in the front area would decrease, potentially limiting the number of similar bows that can be displayed

Figure 27: Further benchmark items contrasting object magnitudes under varied placements and scales, with model responses and evaluation outcomes (Size and Scale: Relative Size Comparison).

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Depth & Occlusion : Complete Occlusion Inference



Question: What is creating the striped shadow on the wall?

Options

1. The sunflower
2. Blinds
3. Window
4. Not applicable

Gold MCQ Answer: 2

Gold Open ended Answer: Blinds

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: Sunlight streaming through window blinds (slatted shades) is casting the diagonal striped shadow on the wall. The slats break the light into parallel bands, producing the striped pattern.
- Gemini-2.5-Flash: The striped shadow on the wall is created by sunlight passing through a window with horizontal blinds. The blinds block portions of the light, casting distinct parallel lines onto the wall. This effect is common when direct sunlight enters a room through slatted window coverings.
- InternVL3.5-72B: The striped shadow on the wall is created by blinds or window slats. The sunlight is passing through the gaps between the horizontal blinds, casting parallel lines of light and shadow. This pattern is often seen when light is filtered through window coverings with narrow openings. The alignment and spacing of the lines suggest that the blinds are partially open, allowing the sun to project the shadow pattern onto the wall.
- Kimi-VL-A3B-Thinking-2506: The striped shadow on the wall is created by sunlight passing through a set of window blinds or shutters with narrow, evenly spaced slats. The parallel lines of the shadow result from the angular interaction between the light source (likely the sun) and the rigid, horizontal slats of the blinds, casting consistent geometric patterns onto the wall. The uniformity and sharpness of the shadows suggest the light is direct and possibly filtered through the gaps between the slats.

Figure 28: Representative benchmark items requiring inference about objects that are fully hidden from view, with model responses and evaluation outcomes (Depth and Occlusion: Complete Occlusion Inference).

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Depth & Occlusion : Layering Order



Question: What is the frontmost object in the picture?

Options

1. Trolley
2. Box
3. Man
4. Woman

Gold MCQ Answer: 1

Gold Open ended Answer: Trolley

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- **Gemini-2.5-Flash:** The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- **InternVL3.5-72B:** The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- **Kimi-VL-A3B-Thinking-2506:** The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 29: Representative benchmark items requiring recovery of the front-to-back layering of overlapping objects, with model responses and evaluation outcomes (Depth and Occlusion: Layering Order).

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Depth & Occlusion : Partial Occlusion



Question: What is the color of the ball behind the white colored square shaped device?

Options

1. Cyan
2. Yellow
3. Red
4. Not applicable

Gold MCQ Answer: 1

Gold Open ended Answer: Cyan

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The ball directly behind the white, square-shaped device is blue (a cyan/teal colored ball). It sits slightly to the left and behind the device.
- Gemini-2.5-Flash: The ball directly behind the white-colored square-shaped device (GameBoy) is teal. There is also a yellow-green ball further to the right of the GameBoy.
- InternVL3.5-72B: The ball behind the white, square shaped device is blue.
- Kimi-VL-A3B-Thinking-2506: The color of the ball behind the white square-shaped device is blue.

Figure 30: Representative benchmark items reasoning about partially hidden objects from their visible fragments, with model responses and evaluation outcomes (Depth and Occlusion: Partial Occlusion).

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Depth & Occlusion : Reflective Surfaces



Question: Which object is the shiniest among all the objects in the image?

Options

1. Kettle
2. Tea glass
3. Table clock
4. Lamp chimney

Gold MCQ Answer: 4

Gold Open ended Answer: Lamp chimney

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 31: Representative benchmark items disentangling real objects from their reflections on mirrored surfaces, with model responses and evaluation outcomes (Depth and Occlusion: Reflective Surfaces).

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Depth & Occlusion : Transparency Effects



Question: How many objects has filaments inside them in the image?

Options

- 7
- 6
- 5
- 4

Gold MCQ Answer: 4

Gold Open ended Answer: 4

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Three. The glowing bulb being held clearly shows a filament, and two more filament bulbs are visible inside the round box on the left.
- Gemini-2.5-Flash: There are three objects with filaments inside them. One is the illuminated light bulb being held, and the other two are unlit light bulbs inside the "Sunny Day Kit" box. All three are incandescent light bulbs, which contain a filament that glows when heated.
- InternVL3.5-72B: In the image, there are four objects with visible filaments. These include the lit light bulb being held, and three unlit bulbs inside the container labeled "Sunny Day Kit."
- Kimi-VL-A3B-Thinking-2506: There are 3 objects with filaments in the image - one held in hand and two in the "Sunny Day Kit" container.

Figure 32: Representative benchmark items reasoning about objects seen through transparent or translucent media, with model responses and evaluation outcomes (Depth and Occlusion: Transparency Effects).

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Orientation : Cardinal Direction

Question: If you go straight following route 6, which direction are you traveling?

Options

1. North
2. South
3. East
4. West

Gold MCQ Answer: 3

Gold Open ended Answer: East

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** You would be traveling east. The Route 6 shield is labeled "EAST" and the straight-ahead arrow beneath it shows that going straight follows eastbound Route 6.
- **Gemini-2.5-Flash:** If you go straight following route 6, you will be traveling East. The sign for Route 6 is directly below the "EAST" sign, and there is an arrow pointing straight up beneath the Route 6 shield, indicating that going straight leads to Route 6 East
- **InternVL3.5-72B:** If you go straight following Route 6, you are traveling east. This is indicated by the sign with the number "6" and the arrow pointing forward.
- **Kimi-VL-A3B-Thinking-2506:** Standard highway signs usually have the path of the route. So if it's route 11 East, the arrow would indicate the direction. But in the image, the left arrow is under the NORTH 11 EAST sign. Maybe that's indicating that route 11 East goes left? But the sign says East. Alternatively, maybe the arrows are indicating the direction of the route numbers. Like, route 11 North is going straight, route 11 East is turning left, etc. But then the route 6 sign has an upward arrow. So if you take route 6 straight, which direction is that? The arrow on the route 6 shield is pointing straight up. But the Question: is about direction—like North, South, East, West. Wait, the signs here are for routes, not cardinal directions. So maybe the answer is that going straight on route 6 would be in the direction indicated by the upward arrow. But the upward arrow might be part of the route 6 sign.

Figure 33: Representative benchmark items requiring assignment of cardinal directions within a scene, with model responses and evaluation outcomes (Orientation: Cardinal Direction).

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Orientation : Facing Direction



Question: Suppose, Rabindranath Tagore's portrait is hung on the left wall and it faces directly into the room, what is the direction of the portrait with reference to the wall?

Options

- Up
- Down
- Away
- Toward

Gold MCQ Answer: 3

Gold Open ended Answer: Away

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini:** If the portrait is on the left wall and faces directly into the room, it is oriented perpendicular to that wall and pointing inward. In other words, it faces away from the wall toward the center of the room.
- Gemini-2.5-Flash:** The portrait of Rabindranath Tagore is located on the wall to the left, as seen in the image. If it faces directly into the room, its direction with reference to the wall would be perpendicular to the wall. This means the portrait is facing straight out from the wall into the space of the room.
- InternVL3.5-72B:** If Rabindranath Tagore's portrait is hung on the left wall and faces directly into the room, it would be positioned perpendicular to the wall. This means the portrait is oriented so that viewers inside the room can face it directly when looking toward the wall on which it is hung.
- Kimi-VL-A3B-Thinking-2506:** The portrait is hung on the left wall and faces directly into the room, meaning it is oriented inward toward the interior space.

Figure 34: Representative benchmark items determining which way an object or agent is facing, with model responses and evaluation outcomes (Orientation: Facing Direction).

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Orientation : Object Rotation



Question: What is the approximate angle of the cases of the pocket watch?

Options

- 90
- 45
- 180
- 120

Gold MCQ Answer: 3

Gold Open ended Answer: 180

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: About 180°. The two covers are opened in opposite directions and lie nearly flat, forming an almost straight line.
- Gemini-2.5-Flash: The cases of the pocket watch are open, revealing the intricate mechanism inside. The front cover is open to approximately a 120-degree angle relative to the watch face. The back case is also open, but its angle is less clear, appearing to be around 90 degrees or slightly more.
- InternVL3.5-72B: The pocket watch in the image appears to be partially open, with the angle between its two cases estimated to be around 60 degrees. The case opens hinge-wise, and the visible gap suggests that it is made to display the internal mechanical components of the watch. This angle provides a good view of the watch face and some of its intricate gears..
- Kimi-VL-A3B-Thinking-2506: When the watch is open, the cases form an angle. If the covers are both open to 180 degrees relative to each other, that would be a straight line. But in this image, the back cover isn't fully open. It looks like it's at around 90 degrees from the front cover. So the angle between the cases would be approximately 90 degrees.

Figure 35: Representative benchmark items reasoning about how an object appears after rotation, with model responses and evaluation outcomes (Orientation: Object Rotation).

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Orientation : Stacking Orientation



Question: What is the relationship between the road and the other objects in the image?

Options

1. Behind the building and plants
2. In front of the plants and leads towards the building
3. The road is beside the building, with no clear direction
4. The road is stacked above the building and plants

Gold MCQ Answer: 2

Gold Open ended Answer: In front of the plants and leads towards the building

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The road runs in front of the large white building, curving around a central planted island and seating area. It separates and encircles the entrance and decorative planters, forming a driveway/roundabout in the courtyard.
- Gemini-2.5-Flash: The road in the image serves as an access path, leading towards the grand white building. It curves around a central island feature, which contains a decorative fountain and some potted plants. This suggests the road is a driveway or entrance loop designed to facilitate vehicle access to the building.
- InternVL3.5-72B: The road in the image forms a semi-circular driveway in front of a large white building with columns. It encircles a landscaped garden area with plants, a stone sculpture, and a fountain at the center, leading up to the building's entrance. The road is bordered by a curb that separates it from the garden and pedestrian space. Lamp posts are situated alongside the road, enhancing the building's stately appearance.
- Kimi-VL-A3B-Thinking-2506: The road curves around the front of the grand white building, acting as a transitional pathway that guides movement and attention toward the entrance while dividing space between the main structure and other elements like the pergola. It establishes scale and perspective while integrating with the surrounding landscape.

Figure 36: Representative benchmark items reasoning about the orientation of stacked objects, with model responses and evaluation outcomes (Orientation: Stacking Orientation).

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Orientation : Tool Handedness



Question: Considering the likely process of creating the mural, which tool's handedness would be most evident in the smooth, curved lines of the bengali script?

Options

1. Paint roller
2. Paint brush
3. Tape measure
4. Stencil

Gold MCQ Answer: 2

Gold Open ended Answer: Paint brush

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Those flowing, tapered curves are most characteristic of a paintbrush (or broad paint marker) rather than a spray can. The handedness of the brush user — most likely right-handed — would be most evident, since right-handed strokes tend to pull and taper in predictable left-to-right sweeps visible in the Bengali script.
- **Gemini-2.5-Flash:** The smooth, curved lines of the Bengali script suggest the use of a brush, likely a fine-tipped one, to achieve such fluidity. The artist's handedness would be most evident in the consistent direction and pressure applied to the brushstrokes.
- **InternVL3.5-72B:** In creating the mural, the smooth, curved lines of the Bengali script would most likely show the artist's handedness, particularly if the lines consistently curve more naturally in one direction. If the curves are more fluid and precise on one side, it could suggest that the artist is either right-handed or left-handed, depending on the direction of the dominant curves.
- **Kimi-VL-A3B-Thinking-2506:** If the artist is left-handed, they might use the brush in their left hand, leading to strokes that curve more naturally. Alternatively, the tool itself (like a specific brush tip) might be designed for left or right handed use.

Figure 37: Representative benchmark items judging the left- or right-handed configuration of tools, with model responses and evaluation outcomes (Orientation: Tool Handedness).

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Relative Positioning : Alignment Patterns



Question: Which of the objects is the closest to the centerline of the image?

Options

1. Rice
2. Nut
3. Avocado
4. Berries

Gold MCQ Answer: 4

Gold Open ended Answer: Berries

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- GPT-5-mini: The red bowl of berries (raspberries and blueberries) is closest to the image's centerline. It sits almost directly in the middle of the composition, slightly left of the exact center..
- Gemini-2.5-Flash: The red bowl containing raspberries and blueberries is the closest object to the centerline of the image. Its position is almost perfectly centered horizontally, and it is located in the upper-middle section of the image vertically. While other objects like the avocado halves are also quite central, the red bowl's overall placement makes it the most aligned with the image's central axis.
- InternVL3.5-72B: In the image, the avocado halves are the closest to the centerline. One avocado is placed near the top center, and the other is slightly below it to the right, straddling the vertical centerline of the image.
- Kimi-VL-A3B-Thinking-2506: The bowl of mixed berries is closest to the centerline.

Figure 38: Representative benchmark items detecting how objects align along shared axes or rows, with model responses and evaluation outcomes (Relative Positioning: Alignment Patterns).

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Relative Positioning : Betweenness Relationships



Question: What is the object between the building and the street lamp?

Options

1. A monument of various statues
2. Statue
3. Tree
4. Humans

Gold MCQ Answer: 1

Gold Open ended Answer: A monument of various statues

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Between the building and the street lamp stands a stone statue of a person. It appears to be one of the sculpted figures that are part of the fountain in the foreground.
- **Gemini-2.5-Flash:** The object located between the building and the street lamp is a large, ornate monument. This monument features multiple levels adorned with various statues and intricate carvings. At its very top, there is a prominent statue of a person.
- **InternVL3.5-72B:** The object between the building and the street lamp is a detailed stone monument or fountain. It features several statues, including a prominent figure standing on top and additional figures around its structure. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.
- **Kimi-VL-A3B-Thinking-2506:** The object between the building and the street lamp is a detailed stone monument or fountain. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.

Figure 39: Representative benchmark items judging whether one object lies between two others, with model responses and evaluation outcomes (Relative Positioning: Betweenness Relationships).

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Relative Positioning : Corner/Angle Positioning



Question: What is in the top left corner of the plant?

Options

1. A lobster
2. A crab
3. An octopus
4. Nothing

Gold MCQ Answer: 2

Gold Open ended Answer: A crab

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: B

Open-ended Answers:

- GPT-5-mini: A crab is in the top left corner of the plant, perched on the rim of the fishbowl. It's hanging over the water with its claws visible.
- Gemini-2.5-Flash: The plant in the image is a type of seaweed or coral, and it is contained within a glass bowl. In the top left corner of this plant, there is a branch of the plant extending upwards and slightly to the left. This branch is a dark, greenish-brown color.
- InternVL3.5-72B: In the top left corner of the plant, there is a small crab that is perched on edge of the bowl, outside the water. It is gripping the rim with its legs, and its body is partially inside the bowl. The background shows the open water, providing contrast to the scene.
- Kimi-VL-A3B-Thinking-2506: The answer is the crab

Figure 40: Representative benchmark items locating objects relative to corners and angular regions of a scene, with model responses and evaluation outcomes (Relative Positioning: Corner/Angle Positioning).

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Relative Positioning : Directional Relations



Question: What is located directly behind the cheetah?

Options

1. Person
2. Tree
3. Safari vehicle
4. Cheetah

Gold MCQ Answer: 3

Gold Open ended Answer: Saffari vehicle

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Directly behind the cheetah is a white open-sided tour vehicle or tram. Several people are seated inside the tram, watching and taking photos.
- Gemini-2.5-Flash: Directly behind the cheetah, there is a white open-sided vehicle, likely a safari or tour cart. Inside this vehicle, several people are seated, observing the cheetah. Some of these individuals are holding up cameras or phones, seemingly taking pictures or videos.
- InternVL3.5-72B: Directly behind the cheetah, there is a group of people sitting and standing around, appearing to watch or interact with the cheetah. They are positioned near a small covered vehicle, which looks like a safari cart or shuttle. The area is a grassy patch, and there are trees and a hill in the further background.
- Kimi-VL-A3B-Thinking-2506: People seated on benches under a canopy are located directly behind the cheetah.

Figure 41: Representative benchmark items resolving directional relations such as left-of, above, or behind, with model responses and evaluation outcomes (Relative Positioning: Directional Relations).

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Relative Positioning : Proximity Gradients



Question: Which book is closer to the book "american marriage" - "atonement" or "cloud atlas"?

Options

1. Cloud Atlas
2. Atonement
3. Crossed
4. nan

Gold MCQ Answer: 2

Gold Open ended Answer: Atonement

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: "Atonement" is closer to "american marriage" than "Cloud Atlas." "Atonement" sits just to the right with only one small book between them, while "Cloud Atlas" is further to the right on the shelf.
- Gemini-2.5-Flash: "Atonement" is closer to "An American Marriage" than "Cloud Atlas." "Atonement" is directly adjacent to "An American Marriage" on its right side. "Cloud Atlas" is further to the right, with "Americanah" separating it from "Atonement."
- InternVL3.5-72B: The book "atonement" is closer to "american marriage" than "cloud atlas." In the image, "atonement" is directly next to "american marriage," while "cloud atlas" is located further to the left..
- Kimi-VL-A3B-Thinking-2506: Cloud Atlas is closer to "American Marriage" than "Atonement."

Figure 42: Representative benchmark items ranking objects by relative nearness along a proximity gradient, with model responses and evaluation outcomes (Relative Positioning: Proximity Gradients).

K STATISTICAL ROBUSTNESS AND DATASET STABILITY

A foundational commitment of this work is that the reported scores measure genuine differences in model capability, not stochastic artifacts of random initialization or dataset noise. We substantiate this commitment directly, subjecting both the evaluated models and the benchmark instrument itself to a comprehensive battery of statistical tests that jointly quantify model robustness across independent runs and dataset stability across repeated resampling trials. Such scrutiny is not optional for small-to medium-scale benchmarks, where even minor fluctuations can corrupt fine-grained conclusions (Drummond, 2009; Drummond & Japkowicz, 2010); the analyses that follow demonstrate that our evaluation withstands it decisively, and they do so uniformly across two contrasting models, both response formats, and both sampling regimes.

Throughout, let N denote the number of benchmark questions and $R = 3$ the number of independent runs of a given model. Each run is executed under a distinct random seed while every other hyperparameter is held fixed, so that any observed variation is attributable exclusively to stochastic effects during inference.

K.1 MODEL ROBUSTNESS

To characterize the robustness of model evaluation, we assemble a sequence of complementary statistical instruments, each formalized below, that together probe reproducibility from both an aggregate and an item-resolved perspective.

K.1.1 MULTIPLE RUN AVERAGES AND DEVIATIONS

Let $a_{i,r} \in 0, 1$ denote the correctness of item $i \in 1, \dots, N$ on run $r \in 1, 2, 3$. The per-item mean accuracy is then

$$\bar{a} * i = \frac{1}{R} \sum_{r=1}^R a_{i,r} \quad (1)$$

with its corresponding standard deviation

$$\sigma_i = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (a_{i,r} - \bar{a}_i)^2} \quad (2)$$

The aggregate accuracy attained on a single run is

$$A_r = \frac{1}{N} \sum_{i=1}^N a_{i,r} \quad (3)$$

and the dispersion of this quantity across runs is captured by its standard deviation

$$\sigma_A = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_r - \bar{A})^2} \quad (4)$$

where $\bar{A} = \frac{1}{R} \sum_{r=1}^R A_r$ denotes the accuracy aggregated over all runs. The magnitude of σ_A therefore constitutes a direct index of run-to-run reproducibility: the smaller it is, the more precisely a single run recovers a model’s underlying performance.

K.1.2 INTRA-CLASS CORRELATION (ICC)

To quantify inter-run agreement beyond aggregate accuracy alone, we compute the Intra-Class Correlation coefficient (Shrout & Fleiss, 1979) under a two-way mixed-effects model:

$$\text{ICC}(3, k) = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \frac{\sigma_{\text{within}}^2}{k}} \quad (5)$$

where $\sigma_{\text{between}}^2$ denotes the variance attributable to genuine differences between items and σ_{within}^2 the residual variance across runs. This formulation isolates the fidelity with which the item-level difficulty ordering is preserved from one run to the next. Following established interpretive thresholds, ICC values above 0.75 signal good reliability, whereas values above 0.9 denote excellent, effectively deterministic consistency (Koo & Li, 2016).

K.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Beyond the behavior of any individual model, we interrogate the stability of the dataset itself, asking whether the difficulty structure of its questions remains invariant across runs and across sampling regimes.

K.2.1 RESAMPLING STUDY

To probe **robustness** from a distributional standpoint, we conduct a resampling analysis. From the full complement of $C = 30$ subcategories, we draw random subsets of size $S \in 20, 25$ within each subcategory, repeating the procedure for each of the $R = 3$ runs and across both models and both evaluation types (MCQ and open-ended). This design directly tests whether accuracy estimates remain stable when the effective sample per subcategory is deliberately curtailed.

For each run r and subset size S , the accuracy is

$$A_{r,S} = \frac{1}{S} \sum_{i=1}^S a_{i,r,S}, \quad (6)$$

where $a_{i,r,S}$ denotes the accuracy for subcategory i on run r under subset size S .

Averaging over the R runs yields the mean accuracy for each subset size,

$$\bar{A} * S = \frac{1}{R} \sum_{r=1}^R A_{r,S}, \quad (7)$$

together with its standard deviation,

$$\sigma_S = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_{r,S} - \bar{A}_S)^2}. \quad (8)$$

As a complementary, distribution-free safeguard, we apply the Wilcoxon signed-rank test to detect any systematic difference in the median accuracy between paired samples. Let $d_r = A_{r,20} - A_{r,25}$ denote the accuracy difference for run r , with ties excluded ($d_r \neq 0$). The test statistic is then

$$W = \min(W^+, W^-), \quad (9)$$

where W^+ and W^- are the sums of the signed ranks of the positive and negative differences, respectively. Its standardized z -score is

$$z = \frac{W - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}, \quad (10)$$

where n is the number of nonzero pairs.

Whenever the resulting p -value satisfies $p > 0.05$, we fail to reject the null hypothesis H_0 , and we conclude that the accuracies obtained from the 20- and 25-sample subsets are statistically indistinguishable.

K.2.2 ITEM-LEVEL CONSISTENCY

For each item i , we define its difficulty as $d_i = 1 - \bar{a} * i$. If this difficulty structure is stable across runs, the correlation between the per-item outcomes of any two runs should be correspondingly high:

$$\rho * r_1, r_2 = \frac{\text{Cov}(a_{\cdot, r_1}, a_{\cdot, r_2})}{\sigma_{r_1} \sigma_{r_2}} \quad (11)$$

A high mean pairwise correlation ($\bar{\rho} > 0.8$) is thus diagnostic of stable item difficulty and, by extension, of dependable dataset behavior.

K.2.3 CRONBACH’S ALPHA

To summarize the overall internal consistency of the measurement instrument in a single coefficient, we compute Cronbach’s alpha (Cronbach, 1951):

$$\alpha = \frac{R}{R-1} \left(1 - \frac{\sum_{r=1}^R s_r^2}{s_T^2} \right) \quad (12)$$

where s_r^2 is the variance of run r and s_T^2 the variance of the total scores aggregated across runs. Intuitively, α measures the proportion of total score variance that reflects true, shared signal rather than run-specific noise. Values of $\alpha \geq 0.9$ are conventionally taken as evidence of excellent internal consistency (Tavakol & Dennick, 2011).

K.3 EMPIRICAL RESULTS

We instantiate the analysis on two deliberately contrasting systems: one large proprietary model, **Gemini-2.5-Flash**, and one small open-source model, **Qwen-2.5-vl-7b-Instruct**. This pairing straddles both the large- and small-scale regimes and the proprietary and open-source paradigms, so that whatever robustness we establish is exhibited across the breadth of realistic deployment settings rather than for a single idiosyncratic system. Each test is further replicated under both the multiple-choice (MCQ) and open-ended evaluation formats (the latter scored as binary: correct or incorrect), guaranteeing that our conclusions hold uniformly across response modalities. The statistical computations rest on the following Python packages and functions: `scipy.stats`² (Virtanen et al., 2020) for `wilcoxon` test; `pingouin`³ (Vallat, 2018) for `intraclass` correlation and `cronbach` alpha; and `numpy`⁴’s (Harris et al., 2020) `numpy.corrcoef` for correlations.

K.3.1 MODEL EVALUATION ROBUSTNESS

Multiple Run Averages and Deviations

We first examine the stability of aggregate model accuracy across the $R = 3$ independent runs. For every model and format, the cross-run standard deviation of accuracy (σ_A) is negligible ($< 0.6\%$), establishing at the outset that the reported scores are precise point estimates rather than realizations perturbed by random-seeding noise.

Table 26: Mean accuracy and cross-run dispersion over three independent runs, reported for both models under each evaluation format. The standard deviation σ_A remains below 0.6% in every case.

Model	Format	Mean Accuracy (A)	Std Dev (σ_A)
Qwen-2.5-vl-7b	MCQ	0.4079	0.0021
Gemini-2.5-Flash	MCQ	0.5193	0.0054
Qwen-2.5-vl-7b	Open	0.1936	0.0047
Gemini-2.5-Flash	Open	0.3124	0.0042

These uniformly minute standard deviations decisively establish that our performance estimates are stable and reproducible, displaying negligible sensitivity to stochastic run-to-run variation. The tightness of the estimates is striking: under MCQ, Qwen-2.5-vl-7b attains a mean accuracy of 0.4079 with a deviation of only 0.0021, and Gemini-2.5-Flash reaches 0.5193 at a deviation of 0.0054; under the open-ended format, Qwen-2.5-vl-7b scores 0.1936 ($\sigma_A = 0.0047$) and Gemini-2.5-Flash 0.3124 ($\sigma_A = 0.0042$). In every configuration the between-model gap eclipses the run-to-run noise by roughly two orders of magnitude, so the observed accuracy differences cannot plausibly be ascribed to random initialization or decoding and must instead reflect genuine underlying capability. The open-ended setting registers marginally larger variance than MCQ, yet its absolute magnitude remains vanishingly small, confirming that free-form generation is assessed with the same fidelity under our protocol. These figures carry a further implication: the dataset size and scoring methodology already deliver convergent accuracy estimates without recourse to extensive repetition, so single-run

²<https://scipy.org/>

³<https://pingouin-stats.org/build/html/index.html>

⁴<https://numpy.org/>

evaluation suffices for large-scale comparison while multi-run analysis serves chiefly to certify that sufficiency. Taken together, the evidence substantially reinforces the empirical credibility of the reported results and licenses confident cross-model and cross-format interpretation.

Intra-Class Correlation (ICC)

We next compute ICC(3,k) to quantify the reliability of the scoring mechanism at the finer resolution of preserved item ordering.

Table 27: Intra-Class Correlation ICC(3,k) reliability scores across the three runs. Every value falls within the “excellent” band, exceeding 0.98 in each model–format combination.

Model	Format	ICC(3,k)	Interpretation
Qwen-2.5-vl-7b	MCQ	0.988	Excellent
Gemini-2.5-Flash	MCQ	0.990	Excellent
Qwen-2.5-vl-7b	Open	0.983	Excellent
Gemini-2.5-Flash	Open	0.981	Excellent

The uniformly high ICC(3,k) scores demonstrate near-perfect reliability across independent runs, irrespective of model scale or response format. The coefficients cluster tightly at the ceiling of the interpretive scale (0.988 for Qwen-2.5-vl-7b and 0.990 for Gemini-2.5-Flash under MCQ; 0.983 and 0.981, respectively, under the open-ended format), situating every configuration well inside the “excellent” band. Such values establish that the item-level difficulty ordering is faithfully preserved from run to run and that stochastic fluctuations leave the relative ranking of samples untouched. This degree of agreement confirms that the scoring pipeline is internally coherent and impervious to output-level randomness, and it decisively rules out random guessing as a dominant factor even in open-ended generation, the very setting where variability is ordinarily most pronounced. From an evaluation standpoint, the benchmark therefore furnishes a dependable signal of model capability, allowing longitudinal comparisons and fine-grained ablations to be conducted with full confidence in measurement stability.

Taken jointly, the evidence from multiple-run variance analysis and ICC reliability demonstrates that the evaluation is highly robust to stochastic effects. Both absolute performance (low standard deviation) and relative performance (near-perfect ICC) remain stable across independent runs, confirming that the results are not driven by random seeding or decoding noise. This robustness guarantees that the observed performance gaps across models and formats reflect genuine capability differences, and it thereby furnishes a reliable, reproducible foundation for comparative analysis and benchmarking.

K.3.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Resampling Study To secure statistical reliability, the resampling procedure is repeated over **1,000 independent trials**, each drawing fresh random subsets of size $S = 20$ and $S = 25$ from the available subcategories. Repeating the test at this scale stabilizes the empirical distribution of accuracy differences and p -values, so that no conclusion can hinge on a single fortunate draw. The uniform failure to reject H_0 across all trials therefore constitutes strong evidence that the observed robustness is systematic rather than incidental.

Table 28: Resampling Study (1,000 Trials): Wilcoxon Signed-Rank Test ($S = 20$ vs. $S = 25$).

Model	Format	Mean p -value	% Trials $p > 0.05$	Conclusion
Qwen-2.5-vl-7b	MCQ	0.290	100%	No Sig. Diff.
Gemini-2.5-Flash	MCQ	0.285	100%	No Sig. Diff.
Qwen-2.5-vl-7b	Open	0.294	100%	No Sig. Diff.
Gemini-2.5-Flash	Open	0.294	100%	No Sig. Diff.

The resampling results supply strong empirical evidence that accuracy estimates have already converged at relatively small subcategory sizes. The consistently high mean p -values and the complete absence of statistically significant differences across all 1,000 trials establish that our performance metrics are insensitive to moderate reductions in sample size. This stability holds uniformly across both models and both evaluation formats, indicating that neither model scale nor output structure

injects additional sampling uncertainty. Crucially, these findings imply that the variance in accuracy is governed by model behavior rather than by data sparsity at the subcategory level. The current subcategory size (≈ 50) can accordingly be regarded as statistically generous, affording a substantial safety margin beyond the minimum required for reliable estimation. This corroborates the soundness of the benchmark design and indicates that future extensions could reallocate samples across categories without compromising statistical robustness.

Item-Level Consistency and Cronbach’s Alpha We measure the mean pairwise Pearson correlation ($\bar{\rho}$) between runs together with Cronbach’s Alpha to assess internal consistency.

Table 29: Item-Level Consistency Metrics.

Model	Format	Mean Pairwise $\bar{\rho}$	Cronbach’s α
Qwen-2.5-vl-7b	MCQ	0.966	0.988
Gemini-2.5-Flash	MCQ	0.972	0.990
Qwen-2.5-vl-7b	Open	0.950	0.983
Gemini-2.5-Flash	Open	0.946	0.981

The exceptionally high mean pairwise correlations (> 0.94) demonstrate strong item-level stability, establishing that the relative difficulty of individual questions is preserved across independent runs. This consistency shows that model responses are systematically structured rather than randomly variable, even under stochastic decoding. The similarly high Cronbach’s α values (> 0.9) further confirm that the dataset possesses excellent internal coherence, with items contributing reliably to the aggregate accuracy measure. The slight reductions observed in the open-ended format are the expected consequence of its greater expressive flexibility, yet they remain comfortably within the range of excellent reliability. Together, these results establish that the benchmark operates as a unified and internally consistent measurement instrument, so that model performance differences can be confidently attributed to true capability variation rather than to instability at the item level.

K.3.3 OVERALL DISCUSSION

Our expanded statistical testing establishes that both the evaluation and the dataset are robust: **ICC and Cronbach’s α scores consistently exceed 0.98**, and the **standard deviation of accuracy across runs is negligible**. Moreover, the **resampling study** confirms that even smaller subsets yield results statistically indistinguishable from the full set, proving that our subcategory sizes (~ 50) are more than sufficient for robust conclusions. These results demonstrate that the granular analysis reported in the paper reflects genuine differences in model capability, not statistical noise.

In sum, our expanded statistical testing establishes that the evaluation protocol and dataset are highly robust, with **ICC and Cronbach’s α scores consistently exceeding 0.98**, evidencing near-perfect reliability and internal consistency. The **standard deviation of accuracy across runs remains negligible** ($< 0.6\%$), confirming that reported model performance is stable and unmoved by stochastic effects such as random seeding or decoding variability. The **resampling study** further demonstrates that even substantially smaller subsets ($N = 20$ and $N = 25$) yield accuracy estimates statistically indistinguishable from the full set, establishing that the chosen subcategory size (~ 25) lies well beyond the threshold required for convergence. Collectively, these results show that both absolute scores and relative performance rankings remain stable across runs and across sampling regimes. This statistical robustness guarantees that fine-grained comparisons across models, formats, and subcategories are meaningful, and it ensures that the granular analyses reported in the paper reflect genuine differences in model capability rather than artifacts of sampling variance or evaluation noise.